Chapter 1
Research Design

Research design is the organization of data collection so that the data collected will support unambiguous conclusions about the problem being studied. The perspective of this book is that empirical research studies are designed to support arguments. The data collected are linked to the conclusions by a warrant that gives a logic explaining why the empirical evidence collected supports the validity of those conclusions. An important aspect of the research argument is the scope of applicability of the research findings and the logic supporting the applicability of the research findings to that scope. The research design is therefore an important part of that warrant, supporting why the data collection supports the conclusions within the scope to which they are intended to apply.

A crucial objective of empirical research designs is to ensure transparency of the research process. That is, research designs help make explicit what research data are considered relevant, the process by which that evidence is collected, and the relation between that evidence and how it is organized in the analysis to support research conclusions. Transparency helps other scholars understand one's work, enables it to be subjected to public scrutiny, and enables future research to build on that work.

Sound research design proceeds from an understanding of the problem to be addressed and the relevant context. This book focuses on the problem of the evaluation of education interventions, products, or services during the development process. One might argue that this is a rather broad problem space, but it does imply important constraints. For example, it mandates that the research is likely to be carried out in schools rather than laboratory settings (which has implications for sampling) and that it will involve relatively small sample sizes of individuals and schools (compared to large scale field trials that occur after an intervention has been fully developed and may have no connection to the development team). Although this book was inspired by the problem of educational research and development, the research designs have far broader applicability and we hope they may be useful in many other contexts.

A. The Research and Development Process

Educational research and development involves a series of steps ranging from basic research on teaching and learning to the development of interventions products or services (hereafter we will use the term "intervention" to cover all three) to small scale testing in schools and large scale efficacy or effectiveness studies. These steps often iterate as trials reveal the need for more development and then more trials of the modifications. An example of research that followed these steps was the development of the University of Chicago secondary school curricula in the 1980s (Stodolsky and Hedges, 1990). Each year-long curriculum was developed by a team of authors who used materials in their own classrooms as part of the development process. Revisions of these curricula were then used in small scale field tests in a handful of schools with other teachers and the results were compared with those from matched classrooms taught by other teachers. Finally, each curriculum was tested in a national field trial using random assignment of classes to treatments (whenever possible).

Similar steps are embedded in funding programs. For example, these steps were embedded in the US Institute of Education Sciences (IES) goal structure (which has now been slightly modified). Goal 1 studies were intended to be exploration and hypothesis generating studies. Goal 2 studies were intended to be development studies: Studies that developed interventions and included small scale initial field tests to evaluate them. Goal 3 studies were intended to be large scale efficacy studies, while Goal 4 studies were intended to be even larger scale effectiveness studies to test scale up and generalizability.

There is a well-developed literature on laboratory and design experiments (which may involve a handful of students) on the one hand and efficacy and effectiveness trials (which may involve 50 – 100 schools and thousands of students) on the other. This literature characterizes the research design and analysis strategies in first and last steps in the development process (see, e.g., Collins, 1992; or Collins, Joseph, and Bieraczyc, 2004). However, the literature describing small scale field testing during development (which corresponds to what IES would call development studies and much of the STEM education research and development work supported by the National Science Foundation) is less well developed.

One of the difficulties that plagues small scale testing arises because of the structure of schooling. Students are embedded in a complex nested structure of classrooms, schools, and school districts. These structures are typically exploited in sampling and can result in statistical artifacts (statistical clustering) that make research designs less efficient than they would be if these structures did not exist (that is if simple random sampling of students were possible). These statistical artifacts are the reason that efficacy and effectiveness trials require large samples (typically large numbers of schools). For example, it is not uncommon for an efficacy trial to involve 50 or more schools.

Large sample sizes are not feasible, and perhaps not even desirable in initial testing of an intervention. They are not feasible because most of the budget in a development project rightly should be allocated to development, not evaluation. Large sample sizes are arguably not desirable not only because of cost and complexity, but because of an ethical issue. Every research study involves some cost to the participants (both students and school personnel), which needs to be weighed against the magnitude and certainty of the benefits. During initial testing of a intervention, it is difficult to be sure of the benefits of the intervention and therefore whether they outweigh the costs—that is why there is an evaluation. Therefore, it is important to involve only the smallest sample of participants necessary to obtain the evidence needed. The same argument would also imply that there is an ethical imperative to assure that an evaluation study is well enough designed that it will provide unambiguous evidence about the benefits (if any) of the intervention. This ethical imperative is more compelling when vulnerable populations such as disadvantaged students are involved.

The problem facing developers in designing initial field trials is therefore how to design a study that is as small as possible, yet big enough to yield relatively unambiguous evidence about an intervention's effects. The terms big and small, sample sizes are purposefully ambiguous, but exact numbers of schools and students that might be involved will be clarified in subsequent sections of this paper. We begin this way because we are mindful that sample sizes that may look small in some situations, look rather lager in others. As a crude guideline, we will focus on research designs where a dozen or fewer schools and at most a few hundred students might be involved in a study.

B. Logic of Inquiry

While it is essential that research designs be well suited to the research problem, they must also be well suited to the way in which the empirical evidence will be used to draw conclusions – to the logic of enquiry used in the research. The logic of enquiry for a particular problem will be shaped by the intellectual traditions in which the researchers are working and by how they conceive the particular problem being studied. This will determine what kinds of evidence might be relevant in supporting the conclusions of the research, as well as how they might be organized in analysis.

Many logics of enquiry involve implicit or explicit comparisons of units being studied (which may be individuals or collectives such as classrooms, schools or even countries). Some logics of enquiry involve comparisons of a unit or units with external standards, conventions or ways of understanding. This is most obvious in studies like experiments or quasiexperiments that compare one group of individuals with another as an essential feature of their logic of enquiry. Such studies compare groups that are intended to be the same in every important respect except the intervention or treatment they are evaluating, arguing that any difference observed must be caused by the treatment or intervention.

A similar logic of enquiry involving comparisons motivates some kinds of research that do not manipulate or intend to draw causal conclusions. For example, survey research that attempts to describe populations often involves comparisons of one part of the population to another to give meaning to both or to define new comparative concepts, like gaps between societal groups in academic achievement. Another kind of comparison that gives meaning in logic of research arises in studies of a single unit, where the comparison is with the unit itself at a different point in time, for example a single subject time series that traces behavior over time using comparison of the unit with itself over time to gauge progress or decline in desired behavior. The same logic applies to time series representing the performance of aggregate units like schools over time.

Even studies that investigate a single unit (whether individual or aggregate) like intensive case studies use comparisons to give meaning to the evidence about that unit. Such studies often use internal comparisons as part of their logic of enquiry. They may also use comparisons to external standards or broader experience to give meaning to the evidence even if it is never explicitly compared to another unit. Such use of comparisons with external standards is also part of the logic of other research designs, such as surveys (e.g. assessments) that include measures with externally referenced performance standards.

In many quantitative research traditions, including those described in this book, the logic of enquiry involves highly specific logics involving statistical methods for data analysis and hypothesis testing. For studies in these traditions, important considerations in research design involve ensuring that the data collected will support the validity of inferences drawn using the statistical analyses, for example statistical power analysis to ensure that the sample size is adequate for the hypothesis tests to have a high likelihood of detecting effects if they are present. In survey research, methods for drawing probability samples are highly specified to ensure that findings will be generalizable to the appropriate population (See for example, McGrath, 2011).

In many qualitative research traditions, the logics of enquiry are less specifically technical but are none the less highly evolved. For example, some logics of enquiry used in ethnography involve the idea that the process of developing descriptions, claims and interpretations must be based on evidence, subject to searches for alternative interpretations, and attempts at disconfirmation. **[Needed?]**

C. Types of Quantitative Research Designs

There are many kinds of research designs, but they differ on at least two dimensions. One dimension is whether the research design involves manipulation of putative independent variables or passive observation. The other dimension is whether the research design involves a single unit (which may be a person or an aggregate unit such as a classroom or school) or multiple units. While these dimensions do not adequately describe all of the research designs that are used in education, they are useful as principles to suggest the breadth of research designs that are most frequently used in educational evaluation. This classification of research designs is summarized in Table 1, which gives examples of research designs for each cross classification of these two dimensions.

Table 1 A typology of types of research design and some prominent examples of each type

	Focus of study			
	Single unit		**Multiple units**	
	Type of design	**Objective**	**Type of design**	**Objective**
	Case study	Describe	Survey	Describe
Passive observation	Time series	Describe	Cross sectional comparative Longitudinal/panel study	Describe Describe
			Natural experiments	Infer causal effect
Manipulated treatment	Single case designs	Infer causal effect	Randomized experiments Difference in difference designs	Infer causal effect
	Interrupted time series	Infer causal effect	Assignment by covariate (RDD) Nonequivalent control group designs	Infer causal effect Infer causal effect

Some research designs are organized to study single units. Sometimes the purpose of the study is simply to describe the unit at one point in time. More often, designs study single units over time. Intensive case studies, both contemporaneous and retrospective, fall into this category of designs. Quantitative studies such as time series designs that observe one or more variables over time for a single unit also fall into this category. In such designs, the units are typically not people, but aggregates (such as schools). While aggregate units like schools my be composed of individuals, they are *social* units that retain their identity even though the particular people within the unit may change over times (as they do in a series of mean achievement test scores for a school over several years).

Some research designs focus on a single unit but involve an attempt to determine if manipulating one (or more) putative independent variables will have a causal effect on another variable. For example, classic studies of single subject behavior (often grounded in behavioral psychology) observe the behavior of a single individual over time, but do so as different interventions are applied to try to influence that behavior. The timing of the application of different interventions may be arbitrary or it may be highly formalized, even determined under a random allocation scheme.

Interrupted time series where outcome variables are measured over time both before and after an intervention is introduced fall into this category. The simplest of these interrupted time series designs is when there is only one observation before and one observation after the intervention is introduced (this is often called the pre-test/post-test design).

Several research designs focus on *multiple* units but involve no attempt to determine if manipulating one (or more) putative independent variables will have a causal effect on another variable. These designs can be described as using passive observation, although the data collection itself may involve questionnaires, interviews, the collection of administrative or even biologic data (such as blood or saliva samples). However, the more intrusive the measurement process, the more likely it is to have effects of its own and the notion that these designs do not involve manipulation becomes more questionable (see, for example, Hill, 2012).

Cross sectional surveys focus on the measurement of a collection of units at one point in time, supporting logics of enquiry that focus on comparisons between individuals at one point in time. Longitudinal surveys (also known as panel studies) collect data on the same units at several points in time to allow comparisons not only between different units at one point in time, but also the same units across time.

While surveys and panel studies are among the most common research designs in this category, there are other important designs. Natural experiments are one of the most important. A natural experiment arises when existing differences in policies or practices create a situation in which naturally occurring groups that are quite similar on many variables except a key policy or practice (then labelled the 'treatment') that is the focus of the research problem. Such natural experiments often occur when similar administrative entities adopt different policies or practices, such as when adjacent schools or school districts that are similar in other ways adopt different curricula or programs. Natural experiments are so named because they resemble artificial experiments that (randomly) assign units to different interventions, policies, or practices.

Natural experiments are difficult to distinguish from quasi-experiments using a comparison group that did not arise from manipulation but was created or identified by the researcher (sometimes called the non-equivalent control group design or an observational study). In both designs, the logic of enquiry is to try to deduce whether the causal effect of an intervention differs from that of another (the comparison) condition by comparing the groups on one or more outcome variables.

Research designs in which the researcher actively manipulates the value of an independent variable so that different units have different values of that independent variable can provide strong evidence about the causal effects of independent variables on outcomes. There are principally two kinds of designs that rely on manipulation of independent variables by the researcher. The most prominent designs are experiments with random assignment. A lesser known alternative is the class of designs in which treatment assignment is based strictly on the value of a covariate, but where there is no random assignment.

Randomized experiments have a distinctive role in scientific enquiry because, in principle, they can provide valid inferences about the causal effect of treatments without requiring any statistical modelling assumptions. Natural experiments and other study designs that involve matching provide valid causal inferences only if all of the covariates that affect outcome have been measured and properly matched between the treatment and comparisons groups. Randomization achieves (on average) matching on all the covariates that a researcher may have imagined, and all the covariates the researcher *has not* imagined as well. Consequently, randomization is a powerful tool for assuring the validity of causal inferences.

This book focuses on four types of designs whose objective is to infer causal effects. We start with several specific types of designs involving random assignment of units to treatments—experimental designs. We begin with experiments because they are the simplest designs to implement and tend to produce the least ambiguous results. Then we turn to three quasi-experimental designs: difference-in-difference designs, interrupted time series designs, and nonequivalent control group designs. In each case, we discuss validity concerns for that type of design. We discuss statistical analysis of data collected from the design, design sensitivity, and what factors influence design sensitivity. Then we turn to planning studies using the design and how to interpret the results of statistical analyses of the design.

D. Relevant Concepts and Terminology

There are several primitive concepts in research design. One of them is the concept of units, which are the elemental units of study. Often units are people (students, teachers, or administrators), but units can be aggregates such as classrooms, schools, or school districts. The key point is that the units being studied are defined by the problem and the logic of inquiry.

The concepts of independent and dependent variable (outcome) are likewise defined by the logic of inquiry. Independent variables can, in principle, be continuous or discrete, but in this book we focus on treatments that are discrete: Treatments that may be education interventions, products or services. Thus, the independent variable is exposure to the treatment versus some comparison condition. Similarly, we focus on dependent variables that are continuous: typically measures of academic achievement, attitude, or socio-emotional competence that are measured on a continuous scale.

A covariate is any variable that cannot be affected by the treatment that units experience. Covariates that are useful in research design are variables that are also correlated with the outcome variable and so can be used to control variation by matching or statistical adjustment methods. Covariates are typically measured before the intervention occurs (such as pretest of other baseline information). Sometimes variables measured after the intervention has begun can be considered covariates if the characteristics measured are unlikely to change over time (like gender, race, or social class).

An important concept in research design is whether a particular variable has values that are manipulated by the investigator or not. In experimental design, variables that are manipulated by randomly assigning their values are called *factors*, and their values are called *levels*. Treatments (e.g., variables having treatment group or control group as their levels) are the most obvious factors, but factors can be associated with quantitative values as well (e.g., amount of a cash payment given to individuals to motivate test performance). In contrast, variables whose values are not manipulated by the investigator are called blocks. There are many kinds of blocks, but they include any preexisting characteristics of units (e.g., gender, race/ethnicity, social class, previous test scores). Blocks may also be defined by membership in an aggregate group (e.g., classroom or school). The terminology for describing the values of variables that are blocks is not standardized. Sometimes particular values are called blocks (e.g., the classrooms to which students belong are called blocks). Other times the variable whose values are blocks is referred to as the blocking variable, or the term block is used as a verb (e.g., we blocked by classroom). The blocks usually used in experimental design are chosen because they are correlated with the outcome and the blocks are used to reduce variability in the design.

Two other concepts are often used in experimental design are crossing and nesting. Both terms arose as experimental design was being developed in agricultural research bit are useful in creating unambiguous descriptions of the relation between factors and blocks in research designs. Two factors are said to be *crossed* if every value of one variable (factor or block) occurs in conjunction with every other values of the other variable (factor or block).

Imagine a plot of ground divided into vertical strips that will be assigned one treatment, and further subdivided by a set of horizonal strips crossing the vertical strips so that the resulting pattern is a cross hatched set of small squares or rectangles. One can imagine two treatment factors (e.g., seed varieties and fertilizer types) assigning a different seed variety to each vertical strip and a different type of fertilizer to each horizontal strip so that each small square or rectangle has a unique combination of seed variety and fertilizer type. The crossed variables need not both be treatments. Blocks can be crossed with treatments too (e.g., if the plot of ground were sloped so that the horizontal strips had different drainage or microclimate, these strips might be used to control for the effects of differences in drainage or microclimate.

Crossed factors or crossed blocks and factors play a major role in defining designs used in agriculture and laboratory experiments. For example, designs in which all factors are crossed with one another are called *factorial designs* and these designs have some special properties. Designs that have factors crossed with blocks (usually classrooms or schools) can be very useful role in evaluation research.

Another important concept is that of nesting. Variable B is said to be nested within variable A if each value of variable B occurs only within a single value of variable A. One can imagine several plots of ground separated geographically from one another and that each of he separate plots is in turn subdivided. One can imagine a different treatment applied to each of the subdivisions of the separate plots. In this case the variable defined by the plots (which is a block, since we cannot assign the properties of that plot) are nested with the variable treatments (which is a factors since the treatments can be assigned). For educational purposes a clearer example may be schools. In general, students are nested within schools because each student occurs only within a single school. If a design assigns the same treatment to all individuals within that school, then schools are nested within treatments because each school occurs in conjunction with only one treatment.

E. Principles of Research Design

In its most general formulation, empirical research is about understanding the relation between variables of interest: How are variables of interest (like variation in treatments) related to other variables (outcomes of interest). Because empirical research is an attempt to understand *relations* between variables, it is always comparative. That is, the concept of relatedness, implies that a change in the independent variable corresponds to (in causal research, *causes*) a change in another variable (the dependent variable). In correlational research, the implied comparison is between the values of the dependent variable corresponding to different values of the independent variable, quantified, for example as a regression or correlation coefficient. In experimental research, the comparison is between the values of the dependent variable corresponding to different values of the treatment (e.g., treatment or control).

Treatment effects are always comparative: They describe what the outcome is when units receive the treatment compared to the outcome when units receive something else. In fact, the often-used term "treatment effect" is actually conceptually underspecified. Without specifying what the treatment is being compared with, the term "treatment effect" is meaningless. The fully qualified term should be "treatment effect compared to *X*," where *X* is the control or counterfactual condition to which the treatment is being compared. The treatment effect compared to a placebo is unlikely to be the same as the treatment effect compared to another active treatment. To put it in educational terms, the effect of a new math curriculum versus no math instruction at all is unlikely to be the same as the effect of the new math curriculum compared to the old math curriculum. This point may seem obvious but is extremely important. A great deal of confusion arises in interpretation of studies comparing the same treatment to somewhat different control conditions.

Empirical research always evaluates the relations by comparing the strength of the empirical relation of interest to the variation in outcomes that is presumably a consequence of variation in other variables that are not directly of interest in the research. The former is often called systematic variation while the latter is often called noise or error variation. This basic idea that systematic variation is evaluated by comparing its magnitude to that of error variation (roughly a signal to noise ratio) is a primitive logic that is the basis of the technical apparatus of statistical hypothesis testing. Sound research design begins with the idea of controlling variation, so that irrelevant sources of variation are controlled in order that systematic variation will loom large enough to be observed.

What counts as systematic and what counts as irrelevant or error variation depends on the research problem. Lee Cronbach (1957) articulated this at the broadest level in his famous paper describing the two disciplines of scientific psychology. Individual difference psychologists are interested in differences among individuals. Thus, for them, individual differences are systematic but groups differences (of the kind experiments intent to create) are error. On the other hand, experimental psychologists use experimental manipulations to try to create differences between groups. For experimental psychologists, group difference are systematic effects in which they are interested, and individual differences are error variation. Thus, the systematic effects of one group of psychologists are the error variation of the other.

Three basic principles of research design are employed to create research designs that can lead to relatively unambiguous conclusions. All of them start with the idea of controlling variation. The first principle is control by matching. The second principle is control by randomization. The third principle is control by statistical adjustment.

E.1. Control by Matching

If we can make all the units we are studying (students, teachers, or schools), then we have controlled the variation associated with the ways in which units could be different. For example, if we match the students or teachers that we study on gender, then any variation associated with average differences between genders is controlled. Similarly, if we match students or teachers by selecting students or teachers from the same schools, we have controlled for variation across schools (on the average).

The condition "on the average" is important here. All women or all schools are not identical in their individual outcomes. Matching on these characteristics does not control for variation within the group of women or the variation among individuals within a school. Matching on gender only controls for the variation between genders on the average (that is between the average of males and the average of females). Similarly, matching on school does not control for the variation within schools, only the variation between school averages. While it may not be obvious, systematic group differences can have a very powerful cumulative effect, dwarfing that of much larger individual differences. We will show in the discussion of statistical clustering how uncontrolled group effects can dramatically reduce the sensitivity of research designs.

The implementation of matching on discrete variables such as gender or school or classroom is straightforward. Matching can obviously be used to control variation on variables that are continuous. The implementation of matching on continuous variables raises the question of how close do the values have to be for units to constitute a "match?" There are various ways to implement such matching, the most obvious being "caliper matching" in which two units re declared to match if the difference on the matching variable s sufficiently small. For example, we define a number δ as being sufficiently small and say that a unit with value x_i on the matching variable matches a unit with value x_j on the matching variable if $|x_i - x_j| < \delta$.

Caliper matching on a single matching variable is straightforward, but often insufficient to ensure good control. Matching on many variables at once is much more complex and raises practical an methodological issues like “Is it better to get good matching on one variable even if that leads to worse matching on another matching variable?” The problem of multivariate matching has become a subdiscipline of statistics and has a large literature (see, e.g., Rubin, 2006). Some very practical approaches to multivariate matching rely on reducing the multivariate problem to a unidimensional one. This can be done by computing a summary of the multivariate “distance” between units to be matched using multivariate distance measures such as Mahalanobis distance (Rubin, 1980) or propensity scores (Rosenbaum and Rubin, 1983).

In studies where different treatment groups are compared, matching can be used to control variation between groups that are assigned to different treatment conditions. It can also be used to control variation within treatment groups. Used this way, matching can reduce the error variation and increase sensitivity

While control for differences on the average may seem a crude instrument, matching is actually the most reliable control procedure in research design. While it is limited, it is easy to implement and it absolutely controls the variation that is within its scope to control. No other method of control can make this claim.

E.2. Control by Randomization

The second major method of control is control by randomization, which is the basis of randomized experiments. Control by matching requires the investigator to be aware of and measure the relevant matching variables. Control by randomization does not require the investigator to know what the important matching variables might be. In fact, it controls for all the matching variables that are known, and also all the potential matching variables that are unknown, but only in the average. The fact that randomization can control for variation due to variables that are not even known to the investigator is the profound advantage of control by randomization.

The logic of random assignment is that if we assign individual units to two (or more) groups at random, any difference between the two groups must have occurred due to chance. Therefore, we would not expect any difference between groups to be large than would be expected due to chance. Because we can evaluate (using statistical methods) how large a difference we might expect to see due to chance, we can determine if a difference we do observe (e.g., in an outcome) is larger than we would expect as a consequence of chance. The apparatus of statistical hypothesis testing is designed to identify those differences in outcomes between groups (e.g., between treatment and control groups) that are bigger than can be attributed to chance. We call these statistically significant differences.

Unlike matching, which can be used to control variation between treatment groups or within treatment groups (e.g., variation between individuals), randomization controls only variation between treatment groups. Random assignment controls variation only between the groups that have been created by random assignment—it cannot be used to reduce error variation (“noise”). Because randomization alone cannot increase sensitivity by controlling error variation, the only way to make a randomized comparison more sensitive is to increase sample size (which effectively amplifies the “signal”) or to use it in conjunction with other principles such as matching.

Moreover, random assignment is not as efficient as matching in controlling for differences between groups. For example, matching on gender assures absolute balance on gender between treatment groups. Random assignment assures only balance that is as good as chance. The gender balance in a randomized study is 50/50 in the long run, and it will be very close once the sample size is sufficiently large, but it may be far from 50/50 if the number of units randomized is small (because in that case, chance can produce larger imbalances)

E.3. Control by Statistical Adjustment

Covariates are variable that could not have been affected by treatment assignment. Typically, covariates are variables measured before treatment assignment of are variables that are theoretically or practically unlikely to be influenced by treatment assignment. Control by statistical adjustment uses the empirical relation between covariates and outcomes to estimate what the outcome would have been in each group if the groups had had the same average value of covariate. Thus statistical adjustment is a kind of pseudo matching. The analysis of covariance and various regression strategies ae the most common kinds of statistical adjustment.

Statistical adjustment is the weakest of the control strategies for several reasons. First, statistical adjustment depends on a statistical model, (for example a linear regression model). If the model is wrong, so is the statistical adjustment. Models can be incorrect in subtle ways that can be difficult to detect. For example, the correct model may be nonlinear, but be a reasonably good fit to the center of the data but quite poor for the extremes. Second, even if the model is correct, if the covariate is poorly measured (if it has lots of measurement error), the adjustment will be incorrect. For example, adjustments based on an unreliable covariate tend to be too small. Third, statistical adjustments are about counterfactuals, they make predictions about situations that are not actually observed (i.e., the situation in which treatment and control groups have the same mean on the covariate). Such counterfactuals may be meaningful, but they may not be or may not address the question of interest to the investigator (see Lord, ????). For example, consider a study that examines differences in academic achievement between disadvantaged students given remedial assistance (the treatment) using a control group of students who initially had higher achievement scores. We might attempt to use the latter group as a comparison group by using prior achievement as a covariate. In essence we would be asking how the group assigned to remedial intervention would have performed after the intervention *if they had not needed the intervention*. But if they had not needed the intervention, they would never have gotten it.

Most statisticians distrust statistical adjustment.

F. History of Research Design

The basic principles of research design were codified by R. A. Fisher in the early 20th century, but their origins are much earlier. A much-lauded example is the work of Sir James Lind on scurvy. In the 16th century, the disease of scurvy ravaged the crews ships during long sea voyages. The disease was not entirely unknown on land, and its cause was believed to be associated with the lack of fresh fruits and vegetables in the sailors diets at time, but the precise cause and cure were unknown. Lind served as a ships surgeon who tried to discover cures for scurvy. In one experiment he reports that

> I selected twelve patients in the scurvy. On board the *Salisbury* at sea. Their cases were as similar as I could have them. They all in general had putrified gums, the spots and lassitude, with weakness in the knees. They all lay together in one place. Being a proper apartment in for the sick in the forehold; and had one diet common to all, …

He then goes on to describe how the group was divided into six groups and a different treatment was given to each group. One of the treatments was to give each man two oranges and a lemon each day. The results were striking in that the two sailors who received the oranges and lemons were significantly better after six days. Perhaps unknown to Fisher and Lind, many of the ideas that they used had their origins much earlier.

F.1 Comparison Groups

The idea of evaluating a treatment by comparing a treated group to a control group has is origins in antiquity. For example, in the bible the book of Daniel (Chapter 1, verses 12 – 14), Daniel demonstrates that a vegetarian diet will make he children of Israel just as healthy as one including meat, by giving one group the vegetarian diet for 10 days while another group ate a meat based diet. Explicit reference of the need for an *untreated* control group to evaluate the effects of medical treatments was made by Persian physicians in the 10th century (al-Razi, 10th century; Tibi, 2005). Other comparative studies appear later in the history of medicine, for example, in 11th century Chinese medical texts (Ben Cao Tu Jing, *Atlas of Materia Medica*) which described a study to evaluate the effect of genuine Shangdang ginseng.

F.2 Control by Matching

The idea that simply comparing individuals who receive different treatments is not necessarily a valid way to evaluate a treatment—the idea that matching is necessary to ensure a valid comparison—also has a long history. Control by matching is explicitly mentioned by Petrarch (1501) in a proposal for a study to determine whether people would be better off avoiding rather than seeking medical care. Donaldson (2016) translates a portion this letter as follows

> *if one hundred or one thousand men, of the same age and character and* [eating] *the same diet, one and all affected by the same disease, one half shall turn to the advice of Doctors of the kind that there are in our time, and the other* [half] *without any Doctors shall follow natural instinct and their own discernment then I have no doubt that of the former* [half] *many shall die and of the latter* [half] *many shall escape*.

Studies actually using matching were carried out by physicians as early as 1575, who used two different treatments on different portions of the same patient's burns, effectively using the patient as a matching variable (Paré, 1575).

F.3 Control by Randomization

Control by randomization is in some ways a more complex idea than matching and therefore has a more complex history. Jean-Baptiste von Helmont (a Flemish physician) proposed an evaluation of the efficacy bleeding and purging (the conventional treatment or standard of care at the time) for fevers by a method that involved randomization. He proposed dividing a collection of patients into two groups, then "casting lots" (randomization) to see which group would be treated with bleeding and purging and comparing the number of deaths in each group (von Helmont, 1648). It is not clear if this experiment was ever carried out, nor if a reiteration of the proposal by the English physician Starkey in 1657 (who seems not to have mentioned the casting of lots) was ever carried out. Note that von Helmont's proposal differs from the modern use of randomization in that the groups were assembled first, then one randomization was used to decide which group was assigned to which treatment. In modern language, von Helmont proposed a cluster randomized trial with two clusters (the preassembled groups of patients), which would be a very poor design from the modern point of view because it would have very poor statistical sensitivity.

The idea that casting lots was used to control variability is implicit (but not elaborated) in von Helmont's proposal. Something much closer to a modern randomized trial was proposed by Franz Mesmer (1781) to evaluate the efficacy of treatment based on the then new method of "animal magnetism"(p. 112). Mesmer proposed a "comparative trial of the new method against the old one" (p. 113). He proposes that 24 patients be selected by the Faculté de Médecine de Paris, 12 to be treated by the then conventional methods, the others by the new method. He then goes on to say that

> *In order to avoid any later argument and all the questions that could be raised about differences in age, in temperament, in diseases, in their symptoms etc. the assignment of the patients shall be made by the method of lots* (p. 113).

While it is not completely clear that the casting of lots (randomization) is applied to each individual, this description could be interpreted that way. If this interpretation is correct, this could be the earliest description of an individually randomized experiment.

The Society of Truth-loving Men in Nürnberg reported a trial to determine if homeopathic provings could be distinguished from 'snow water', by placing one or the other in numbered vials, which were then shuffled and given to participants. The participants later reported their observations of the effects. One innovative feature of this trial is that it was blinded from the participants: A list of the contents of each numbered vial was placed in a sealed envelope and a list of the vial number each participant received was placed in another list, and both envelopes were opened only *after* the patients reported their observations, (Löhner 1835; Stolberg 2006). This trial seems to be unambiguously a randomized trial in the modern sense of the word.

Although the idea of controlling for differences between treatment and comparison groups gained acceptance in medicine in the 19th century, control by randomization did not. Instead, alternation (e.g., assigning every second patient to the treatment) or if there were more than two treatment, rotation were the methods of choice. For example, Lesassier Hamilton reported using alternation to assign patients among three physicians in his doctoral thesis (1816). Some years later Thomas Graham Balfour used alternation to allocate children to receive either a treatment for scarlet fever (belladonna) or a control, and was explicit that he did so "to avoid the imputation of selection" (Balfour, 1854; Chalmers and Toth, 2009). The actual term "alternation" as a method seems not have been used until the early 20th century when it was deemed "die alternative Method" by Polverini (1903), '*the alternate case method*' and '*rational alternation*' by Choksy (1908), '*la méthode alternante*' in France (Cousin, 1905; Netter, 1906) and just 'alternation' in the United States (Bullowa, 1928). Although some early medical trials claimed to use random assignment (e.g., Doull, et al., 1931; Theobald, 1937; Bell 1941), the word random was often used at the time to refer to alternation and there is considerable evidence that thee trials actually used alternation, not random assignment (Armitage, 2002). This is true even in the work of Sir Austin Bradford Hill, who became the most prominent advocate of randomization in Britain (Hill, 1937; Chalmers, 2010). The British streptomycin trial marked the first truly modern randomized trial that revolutionized medical trial design (Medical Research Council, 1948).

Control by randomization appears to have been embraced by at least some psychologists and education researchers much earlier than it was in medicine. Charles Pierce and Joseph Jastrow did experiments asking subjects to compare weights, but used true random assignment (based on shuffling a special deck of cards) to determine the order in which the different weights were given to the subject (Pierce and Jastrow, 1885; see also Stigler, 1978). In other writings, Pierce makes clear that he understood the role of random assignment in assuring control of variation between groups on the long run average (see Stigler, 1992).

Control by randomization continued to be used in some areas of psychology (e.g., psychophysics), but the understanding of randomization did not appear to be widespread in education and psychology, even after R. A. Fisher's foundational work on experimental design in the early 1920's. For example, a 1923 textbook (McCall, 1923) on the design of education experiments was chronologically contemporary with R. A. Fisher's early writings on experimental design, but clearly used the word "experiment" in a more general way than we do today, to refer to any comparative study. For example, although McCall appreciated the need for control of variation, he (unlike Pierce and Jastrow or Fisher) did not appreciate the special role of random assignment in controlling variation. For example, McCall's book mentions equating groups 'by chance' (i.e., random assignment) but described random assignment and alternation both as (essentially equal) methods to accomplish this. Moreover, equating groups by chance is only one of the methods mentioned for assuring equality of experimental groups and seems to imply that matching was a superior method for equating groups. Other educational studies before 1940 seems to have shared the preference of matching over randomization for controlling variation between treatment groups (see Hedges and Schauer, 2018).

A remarkable book by educational psychologist E. F. Linquist in 1940, *Statistical Analysis in Educational Research*, helped introduce Fisher's ideas about experimental design (including control by randomization) to American education researchers. Lindquist illustrated a variety of statistical methods and designs that are useful in educational research, including cluster randomized trials (his Design III). Linquist's book and a subsequent edition of it in 1953 were hugely influential in promoting randomized experiments in educational psychology and decades later (in the 1960's and 1970s) in promoting randomized field trials as part of large scale policy evaluations (see Hedges and Schauer, 2018).

F.4 Control by Statistical Adjustment

The basic methodology for statistical control has its origins in one of two ideas: Stratification (cross classification) and linear regression. Stratification by cross classification is essentially a form of matching. By cross classifying a treatment variable (e.g., treatment versus control) with other control variables (covariates), the treatment-control comparisons within strata of the control variable are matched on the control variable. Thus, any statistical aggregate of those stratum-specific treatment-control comparisons has implicitly adjusted for the control variable. The idea of stratification in population statistics has its origins in the beginning of statistical tabulations for the state. It seems remarkable today that stratification as an analytic too had very little use in the context of large census data collections which were the norm prior to 1900. The reason is very straightforward: Until the advent of mechanical (and later electronic) sorting machines, stratification of large datasets (such as that from the US census) was practically infeasible (see Hollerith, 1894). However, stratification of small datasets such as those collected in research studies was feasible and we see it used as early at the mid 1800's in medical studies. For example, Tessier (1852) reports a study of homeopathic versus allopathic treatments, stratified by sex of patient for each treatment-control comparison.

The linear regression approach to statistical control has its origins in Galton and Edgeworth's formulation of regression in the late 19th century. This is the method Fisher incorporated into his development of analysis of covariance (Fisher, 1923?). Later such methods have become a staple of economic and social research. Yet early in the 20th century, and except for balanced experimental designs, the computations required for these methods were quite difficult to carry out with the existing computational technology. It noteworthy that Fisher's early work at the Rothamsted agricultural station involved attempting to use regression methods to address confounding (that is, apply statistical control) to better understand data that had been collected earlier. It was this work that led him to invent the orthogonal polynomial contrast approach to simplify computations in regression (a method still taught today in courses on experimental design) (Fisher, 1921). It also led him to recognize that the confounding factors that needed to be controlled in agricultural studies often had larger effects that the treatments of interest, which led him to develop the idea of randomized experiments and systematic experimental design to address that problem (Fisher, 1923).

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Chapter 2
Validity of Research Designs

The principal function of research design is to support validity claims about the findings of research studies. The dominant perspective on validity of research designs in quantitative social science grew out of an important paper (later published as a small book) by Campbell and Stanley (1963) that has been elaborated since then (see Shadish, Cook, and Campbell, 2002). The key to this framework is the idea that conclusions reached from research design are subject to potential threats to their validity. If we think of the research design as part of the process of developing an argument supporting conclusions, these threats to the validity of the design are like counter-arguments. While their framework was developed in the quantitative domain, the general thrust of their analysis is quite general.

In the Campbell-Stanley-Cook-Shadish framework, there are four classes of threats to the validity of research designs, which can be described generically in terms of the process of deciding if a particular independent variable has a causal effect on a dependent variable. Data analysis validity is about whether the design permits an accurate assessment of whether there is a relation between the independent variable and the dependent variables as measured. Internal validity concerns whether the design permits an accurate assessment of whether the association is causal. External validity is about whether the design permits an accurate assessment of whether the causal relation found will generalize to other situations. Construct validity of explanation is about whether the design permits accurate identification of the components of the independent variable that actually cause the relation with the outcome. We discuss slightly generalized versions of these types of validity in the sections of this chapter that follow.

A. Data analysis validity

Cook and Campbell (1978) originally called this class of threats to validity 'statistical conclusion validity', but the idea is clearly more general. All empirical research designs involve the collection of empirical materials which are organized by the analysis in order to draw conclusions. If the design does not assure that the empirical materials used in the analysis are adequate to draw conclusions or if the analysis is not organized to permit drawing valid conclusions even if the materials collected are sufficient to do so, then the conclusions drawn from the research design will be invalid.

In quantitative studies the analysis of the data collected is statistical, hence the name statistical conclusion validity. Several specific threats to data analysis validity are described in the sections below.

A.1 Unreliable data Elements

Unreliable data elements might be measurements, notations of occurrences from field notes, incorrect observations of participants. Often this will occur due to too few observations, too little time spent observing or poor choices of what to observe (e.g. intrinsically variable as opposed to more stable characteristics). In quantitative studies where the individual observations are in the form of tests, rating scales, or attitude scales, there is a highly developed set of psychometric theory and methods to assess reliability of measurements of various kinds. One aspect of this theory is a highly developed theory of reliability of measurements and how to assess it (e.g., methods like Cronbach's alpha coefficient). In quantitative studies where the observations are systematic observations of behavior (e.g., classroom observations of instruction or group processes) there is also a highly developed psychometric theory of the dependability of behavioral measurement and how to assess it (e.g., generalizability theory). Both of these methodologies include quantitative concepts (reliability and generalizability coefficients, respectively) that are explicitly linked to the sensitivity of statistical tests.

A.2 Incorrect analysis

Incorrect analyses occur when the analysis is an invalid summary of data elements. In qualitative work this could occur when the analysis relied on impressions from memory when more verifiable means are available, such as counting instances in field notes. In quantitative studies incorrect analyses can occur when an incorrect statistical method is used (e.g., one that makes assumptions that are untenable) or when it is applied incorrectly.

A.3 Incorrect data elements

Incorrect data elements include those using invalid measurements. In qualitative studies, this includes focusing data collection on the wrong participants or being deceived by informants. In quantitative studies, this can happen when data is coded incorrectly (e.g., there are errors in tabulation, matching, or aggregation). For example, when attitude questionnaire phrases some questions positively (e.g., I like science) and some negatively (e.g., I don't like science) but the ratings of the negatively phrased questions are not reversed before aggregations. Incorrect data elements can also arise when data is transformed in ways that undermine or exaggerate its properties. For example, when ranks are treated as independent ratings and aggregated as if they were test scores.

B. Internal validity

This was one of the two original classes of threats to validity mentioned by Campbell and Stanley (1963). It refers to the issue of whether the relation observed between putative independent and dependent variables is a causal relation as opposed to just an association that is not causal. Obviously, this class of threat applies only to research designs that are seeking to identify causal relations. Research designs or research traditions that seek just to identify associations are not subject to threats to internal validity. For research traditions that do make claims about cause and effect, achieving internal validity is a fundamental (and often daunting) validity challenge.

There is a long list of threats to internal validity in quantitative studies. Some of these threats apply only to specific types of designs, others apply more generally. For example, consider the threat of maturation given below. The threat is that individuals may have changed as part of a natural process over the period of the study and this might undermine the validity of causal claims made based on the design. Clearly this threat is more serious in a design such as the pre-test/post-test design (where it is completely indistinguishable from a change caused by the treatment) than in a design with a comparison group (where the comparison groups would also be subject to maturation and so could provide an indication of any maturation effects). Moreover, these threats are more important when a design is used to study some types of problem than when it is used to study others. For example, consider the threat of maturation again. Clearly this threat is more serious over a year-long study than a two-week study and more serious when the outcome being measured is one that is changing rapidly in the natural development of individuals like those in the study. Some important threats to internal validity are listed in the following sections, but a fuller discussion see chapter ??? of Shadish, Cook, and Campbell (2002).

B.1 Ambiguous temporal precedence.

Because causes must precede their effects, any claims of causality in which the putative cause does not unambiguously precede the putative effect is suspect. One of the less obvious implications of this principle arises when some form of control (often statistical control) but the control variable might have been influenced by the putative independent variable. For example, when a study uses a pretest that was administered after a part of the treatment had already been administered and therefore the value of the pretest reflects part of the treatment effect. Temporal precedence is the reason that covariates (which by definition must not be able to be influenced by treatment assignment) are usually measured *before* the treatment is administered.

B.2 Testing or observation effects

The act of observing (or measuring or interviewing) may change the phenomenon being observed in substantial ways. For example, attention to the ways students go about planning an investigation or interviews about their strategies in an investigation of scientific problem solving might change the students insights about problem solving. This presents a threat to internal validity when observation effects might be confused with relations between putative independent and dependent variables.

B.3 Instrumentation

The same instrument or observation protocol might not measure the same construct at different points in time. For example, a test of arithmetic might represent mathematical ability when children are very young but represent diligence or clerical ability when students are older (and every child has essentially mastered basic arithmetic). Over a shorter time period, some students might remember questions or answers between tests, so that performance on later tests measures both the intended construct and memory or reflection on the content of prior tests.

B.4 Maturation

Individuals grow older, wiser and more experienced over time for reasons that have nothing to do with interventions. For example, many competencies measured by achievement tests in early grades are learned eventually by virtually every student. Maturation presents a threat to internal validity when maturation effects might be confused with relations between putative independent and dependent variables.

B.5 Selection

When groups are being compared that are not randomly assigned it is possible that the groups differ in ways other than the putative independent variable which is the presumed cause of group differences in outcomes. This presents a threat to internal validity when the pre-existing differences between groups induced by selection could cause one of the groups to have higher outcome scores irrespective of the treatment. In studies that do not use random assignment, selection is often the most serious threat to the internal validity of evaluation studies and the most difficult to control.

B.6 Mortality

Often, not every unit (e.g. person, classroom, or school) that begins the study persists throughout the study. This presents a threat to the internal validity of the study if the units that drop out do so in ways that are related to both the putative independent and dependent variables. For example, in studies comparing two groups receiving different treatments, if the more able individuals tend to drop out of one group more frequently, and those who drop out differ from those who persist, this is a threat to the internal validity. Importantly, if mortality is unrelated to the outcome variables (that is if units that drop out do not have scores that are systematically different than those who do not drop out) then mortality is not a threat to internal validity. Note that it is not enough to know that the mortality (dropout) rate is the same in treatment and control groups. It is necessary to know that the units that drop out would have tended to have the same outcome scores in each group. These conditions are very difficult to verify empirically because we do not observe the outcomes of units that drop out of the study.

B.7 Regression

Regression to the mean was first discovered in genetics, where Galton called it "regression to mediocrity," where he noted that particularly tall (or particularly short) parents tended to have children that were taller (or shorter) than average, but closer to the average than their parents. The principle is far more general than the genetics of height, applying to any situation in which an individual or group is selected because it deviates from the mean on a measured variable.

We can even quantify this effect. If μ_X and μ_Y are the means of X and Y, the variances of X and Y are the same, and ρ is the correlation coefficient between X and Y in the reservoir from which the comparison group is composed, then the regression equation for predicting Y from X is

$$\hat{Y} = \mu_Y + \rho\left(X - \mu_X\right) .$$

This equation shows that if $\rho = 0$ there is no regression effect, but the stronger the correlation between X and Y, the stronger the regression effect.

For example, if individuals are selected for an evaluation study because they are above or below their group mean on one measure (like a pretest), then if these individuals are measured again (e.g., on a posttest) we would expect them to have scores that are closer to the mean on that second measure. This does not have implications for internal validity if all of the individuals in the study are from the same group or if they are selected from reservoirs of individuals that have the same means, but it can be a problem in designs that compare a treatment group to a comparison group and the individuals in the comparison group are selected from a reservoir of individuals who have a mean that is different from that of the treatment group.

For example, an evaluation study of a treatment that was intended for struggling students might treat a group of such students and compare their performance on a posttest to that of a comparison group of who were matched on a pretest but did not receive the intervention. Because the comparison group was selected from a group that presumably had higher pretest scores, we can predict that the comparison group would improve (be closer to their mean) on the posttest than they were on the pretest. This would tend to reduce the magnitude of the difference between treatment and comparisons group means, making the treatment look less effective or even harmful (see, e.g., Campbell and Erlebacher, 1970).

C. Generalizability

This was one of the two original classes of threats to validity mentioned by Campbell and Stanley (1963), where it was described as external validity. It refers to the issue of whether the relation observed between independent and dependent variables can be generalized from the settings, persons and contexts studied to those that are part of the scope of application intended by the researcher. Representative (that is probability or random) sampling can ensure external validity, but this is seldom a viable option outside survey research. Probability sampling requires the precise specification of the sampling frame that is the intended scope of generalization.

It is important to recognize that, while external validity or generalizability is often discussed as if it was a property of a study (like internal validity), this a terribly misleading. In fact, the generalizability of a study's results is an ambiguous concept without specifying the target of generalization—the supposed population or scope of applicability to which the study's results are supposed to apply (Hedges, 2013). We strongly discourage the use of the term 'generalizability 'or 'generalizable' without specifying the target of generalization in terms as precise as is possible.

One important generalizability consideration that arises in evaluation studies is whether the design is intended to support generalization as part of the statistical inference or not. Some designs formally support statistical inference to a larger collection of units than those actually present in the experiment. For example, an experiment involving a random sample of individuals is intended to support generalization to the population from which the individuals in the experiment were sampled. The statistical analysis of that experiment takes into account the sampling uncertainty relevant to that generalization.

More complex research designs may be subject to different analyses, which may or my not take into account all of the sampling uncertainty relevant to generalization to larger populations. For example, a multi-site study conducted in several schools may assign some students to treatment and some to the control condition in each school. One possible analysis of this study (the schools random model) is intended to generalize the treatment effects to a larger population of schools. Another analysis of this study (the schools fixed model) is intended to understand only the effects in these particular schools. The first (schools random) analysis incorporates the sampling uncertainty of sampling schools in determining the uncertainty of the treatment effect, the other (schools fixed) does not. Either analysis can be defended as appropriate depending on which generalization is intended. As one might guess, which analysis is intended has an influence on the details of the design. For example, if generalizations to a larger population of schools is intended to be incorporated in the statistical analysis, a design with a greater number of schools is better (will be more conclusive).

Note that the issue is not whether generalizations to broader population of schools is possible, but whether that generalization (and assessment of its uncertainty) is part of the statistical inference process or is relegated to conceptual reasoning (that is, is extra-statistical). The extra-statistical reasoning process about generalization to a large set of schools would consider how similar the target schools were to the sample in the study, which factors to consider in deciding about similarity, etc.

D. Construct validity of explanation

Cook and Campbell (1978) called this threat to validity 'construct validity of cause and effect' because they focused on empirical studies of causal effects, but the idea is clearly more general. This validity threat concerns whether the explanation of the study has correctly identified the constructs in the interpretation. This concept of validity draws its inspiration from the measurement concept of construct validity and in some it ways resembles the measurement concept of consequential validity (whether the interpretations of a measurement 'as used' are correct, see Messick, 1995). For example, a research design might identify a generalizable causal relation between an intervention and an outcome, but misinterpret the construct causing the effect. Hawthorne effects or novelty effects where the actual cause of the effects is attention or a change in routine rather than the attributed type of attention or the particular attributed change in routine are examples of how explanations can lack construct validity. Construct validity of explanation is not a property of designs, but of interpretation of evidence from designs, which inevitably involve the theoretical framework in which the research is embedded. However, certain design features can support the construct validity of explanations. For example, having several variations of the treatment in the design can make it possible to determine whether the effects of particular variations of the treatment are consistent with an explanation in terms of treatment constructs. While construct validity of explanation is crucial in the broader interpretation of evaluation research, it is difficult for a single evaluation study, particularly a small-scale study conducted in the development process, to address it.

E. Causal inference

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Chapter 3
Statistics

A research project is like an argument, linking evidence to a conclusion via a warrant that is describes how the evidence supports the conclusion. Research design organizes the production of evidence (data collection), but *statistical reasoning* provides the warrant that describes how the evidence supports the conclusion. This book focuses primarily on research design, but research design is inextricably linked to the analysis that will be used to support conclusions from that design. In this chapter we will focus on statistical analysis at a very broad level but will discuss the analysis of particular designs at greater length in chapters about specific research designs.

Populations and Samples

One of the most fundamental distinctions in statistics is between populations and samples. A *population* is the conceivable (and possibly infinite) set of *possible* observations. Statistical populations are sets of numbers (like test scores) but because the numbers in statistical populations are often associated with individuals, statistical populations are often associated with the sets of people that correspond to those observations. Because populations are sets of possible observations, all of the observations in a population are not usually observed.

A *sample* is a subset of a populations that can be observed. Samples can be small or large and can be selected from the population in any possible way. One but by no means the only, way to select a sample is random sampling. The simplest type of random sampling is simple random sampling. In a simple *random sampling*, every observation in the population is equally likely to appear in the sample.

Populations and samples are just collections of numbers, but we often wish to describe those collections by using descriptive characteristics like their average (mean) or their variability. We are often interested in the same descriptive characteristics of both populations and samples, thus different terms are used to denote whether a particular characteristic is that of a population or a sample. The characteristics of populations are called *parameters* while characteristics of samples are called *statistics*. Thus, there are population and sample means and there is population and sample variation (quantified, for example, by population and sample variances).

To help distinguish when a descriptive characteristic describes a sample (is a parameter) versus a population (is a statistic) a conventional notation is to use Greek letters to refer to population parameters and Roman letters to refer to sample statistics. For example, we often use the Greek letter μ (mu) to refer to the population mean and the corresponding Roman letter m (or the not so obvious correspondence $\bar{x}$) for the sample mean, the Greek letter σ (sigma) for the population standard deviation and the Roman letter S for the sample standard deviation.

Populations are, in principle, fixed sets of numbers even if they are unobservable. Therefore, population parameters (like the mean and standard deviation) are *fixed* quantities. Statistical samples are typically random samples of some type (usually, but not always simple random samples). Because which elements of the population happen to be in each simple random sample is a matter of chance, any description of the numbers in a random sample is itself random. Thus, statistics describing samples are *random* quantities.

While sample statistics describing random samples are in principle random, the set of all possible sample statistics itself describes a population of statistics. That population is called the *sampling distribution* of that statistic for the population. Distribution is just another term for a set of numbers and is synonymous with the term population. While the sampling distribution of any statistic based on samples from a population is, like the population itself, unobservable, mathematical methods can be used to deduce relations between the population and sampling distributions of statistics based on samples from that population. One example of such a deduction is that the mean of the sampling distribution of the sample mean for samples of any size is the same as the population mean. Another example is that if the population standard deviation is σ, the standard deviation of the sampling distribution of the mean of samples of size n from the population is $\sigma/\sqrt{n}$.

The concept of sampling distribution is central to statistical inference. We shall use mathematical results about sampling distributions based on statistical models in subsequent chapters without proof but provide references to more detailed treatments of the statistical inference methods we use.

One of the most fundamental (but surprisingly tricky) distinctions in statistics is between quantities that are *fixed* and those that are *random*. One way to characterize this distinction is that fixed quantities (e.g., population parameters) remain the same regardless of what random sample is chosen, while random quantities can (and usually do) change depending on which random sample is chosen. Thus, the population mean is a fixed quantity because it does not change over samples, but the sample mean is random because the exact value of the sample mean can (and usually does) changes from one random sample to the next. Of course, in research we usually only collect one sample, but from a statistical perspective, that sample is one of many that might have been collected.

Models and Notations

Although it may seem pedantic, well-defined notation is essential for clarity about concepts. The most basic statistical notation is that for describing observations and sample sizes. If that are some number (call that number n) of observations (call the observations Y's), then it is conventional to put the observations in a list (from first to n^{th}) and denote particular observations on the list by a subscript indicating the order of their occurrence on the list. Thus, Y_1 is the first observation, Y_2 is the second observation, etc., and more generally, Y_i is the i^{th} observation. It is also conventional to set letters that are used for algebraic variables in italic type (thus the Y in Y_1, both Y and i in Y_i, and n are set here in italic type). This notation helps distinguish between the use of a Roman letter as a word or part of a word and the use of the same letter as an algebraic symbol.

Statistical models describe the mechanism that generated the data at a very abstract level. The most general statistical model for a sample of n observations is something like

$$Y_i = \mu + e_i\,,\; e_i \sim \mathrm{N}(0, \sigma^2),\; i = 1, \ldots, n.$$

This model says that the observations $Y_1, \ldots, Y_n$ are all composed of a mean value (denoted by the Greek letter μ) plus a deviation from the mean (e_i) and that those deviations from the mean are normally distributed with mean zero and variance σ^2. This, like all statistical models, has two parts: A structural model (in this case: $Y_i = \mu + e_i$) and a stochastic specification [in this case: $e_i \sim \mathrm{N}(0, \sigma^2)$]. The structural model is simply a tautology because if we *define* e_i to be $Y_i - \mu$, the model will always be correct algebraically.

The structural model says more than its tautological nature might suggest. First it introduces two unobservable quantities, the mean μ, and the error e_i. They are unobservable because, while we observe Y, we cannot deduce either of the components from a single observation of Y.

The stochastic specification tells us that unlike the mean μ which is a fixed quantity, e_i is a random quantity. We know μ is fixed because, unlike e_i, μ is not mentioned in the stochastic specification. The stochastic specification tells us that the e_i arise from sampling from a normally distributed population. While the structural specification is a tautology, the stochastic specification is not—it can be incorrect. However, for many reasons (including mathematically deep ones like the laws of infinitesimal errors and the central limit theorems), errors of continuous measurements like academic achievement tests are very often well described by a normally distribution.

The model above is extended to deal with evaluation studies by incorporating a parameter to represent the treatment effect. For example, a model like

$$Y_i = \mu + \beta Treatment_i + e_i\,,\ e_i \sim \mathrm{N}(0, \sigma^2),\ i = 1, \ldots, n,$$

where $Treatment_i$ is a dummy variable taking the value 1 if the i^{th} individual in the sample is in the treatment group and zero otherwise. Examining the model, you see that μ is the mean value of observations in the control group (those for whom $Treatment_i = 0$) and that β is difference between the mean of the treatment group (those for whom $Treatment_i = 1$) and the control group. In other words, β is the treatment effect.

It is important that models do not have to be completely veridical to be useful. In this vein, John Tukey, one of the greatest statisticians of the 20th century, remarked that all models are wrong, but some are useful. By this he meant that a model that is approximately correct and simple enough to understand, can offer important insights into scientific phenomena under investigation that a completely veridical description could not.

Statistical Inference

Statistical inference involves inferring values of population parameters from observed data. The structure of inference (we infer about parameters from statistics) is important. Population parameters are fixed quantities, but sample data and statistics derived from them) are random. It makes little sense to infer the value of something that is not stable (with a few rare and complex exceptions). Often the population parameters that are the topic of statistical inference are parameters in the structural part of a statistical model, but sometimes part of the stochastic specification (such as variances) are also targets of statistical inference.

One form of statistical inference is called *estimation*. The object of estimation is to infer the value of a population parameter. In evaluation studies, there is almost always a desire to estimate the parameter that represents the treatment effect. Because estimates are statistics (they are random), they have a sampling distribution. The sampling distribution of a statistical estimate describes the set of values the statistic might take. While the exact value of the mean of the sampling distribution is often unknown, but its standard deviation is known (that is can be mathematically deduced). The standard deviation of the sampling distribution of a statistic is called its *standard error* and is used to quantify the uncertainty of the estimate.

The intuition often used to interpret uncertainty of estimates is that approximately 95% of the estimates should lie within ±2 standard errors of the parameter that is the target of estimation. This intuition is sometimes formalized into computation of a *confidence interval*, which is approximately and interval that is the value of the estimate ±2 standard errors of that estimate.

Another form of statistical inference is *hypothesis testing*. The object of hypothesis testing is to make a qualitative (not quantitative) inference about population parameters (usually parameters that represent a relation between variables of a treatment effect). Hypothesis testing makes use of a statistic computed from a sample to carry out the inference. Statistics used for hypothesis testing are called *test statistics*. The selection of test statistics and the derivation of their properties (such as their sampling distribution) is a major topic in mathematical statistics.

To illustrate the basic ideas of hypothesis testing, suppose that the parameter δ represents the effect of a treatment and that the statistic d is a sample estimate of δ. Hypothesis testing uses a somewhat convoluted logic starting with a so-called *null hypothesis* that is the opposite (the logical negation) of the conclusion you hope to draw. That desired conclusion is called the *alternative hypothesis* in hypothesis testing. Then the statistical procedure evaluates the evidence of how likely the observed data would be if that null hypothesis were true. If the data observed is unlikely to have occurred if the null hypothesis were true, we reject the null hypothesis and conclude that its opposite (the alternative hypothesis) is true. For example, you might wish to conclude that a treatment is effective (that is, that $\delta \neq 0$). To carry out a test of this hypothesis, we start by constructing a null hypothesis that $\delta = 0$.

We use a test statistic (call if T) to evaluate the likelihood that the data would have occurred if $\delta = 0$. The test statistic T is constructed so that the larger the value of T, the more inconsistent it is with the belief that $\delta = 0$ (for example, it can be an increasing function of the estimate d of δ). We use the sampling distribution of T to situate the observed value of T (call that T_0) in terms of the entire distribution of values of T that could have occurred if $\delta = 0$. If the probability of obtaining a value of T as large (or larger than) T_0 is small, we reject the null hypothesis and conclude that the value of δ is probably not zero. In other words, we accept the alternative hypothesis, which is what we wanted to confirm in the first place.

This description leaves open the question of how small the likelihood that T_0 could have occurred (if $\delta = 0$) needs to be in order to reject the null hypothesis. The threshold for rejecting the null hypothesis is called the *significance level*. From a mathematical perspective, the significance level can be selected to be any value, but a very strong (practically unassailable) convention in education and the social sciences is 5%. If a result could have occurred less than 5% of the time if $\delta = 0$, we should reject the null hypothesis of $\delta = 0$ and accept the alternative hypothesis.

The description above makes it obvious that the conclusion of a hypothesis test can be wrong. In particular, when $\delta = 0$ we get a value of the test statistic T that causes rejection of the null hypothesis 5% of the time. In other words, we make an error of inference 5% of the time when the null hypothesis is true. The important advantage of hypothesis testing is that we know that the chance of making an error of rejecting the null hypothesis when it is true is exactly 5%: We have controlled the rate of such errors.

Hypothesis tests can also make errors in another way: They can fail to reject the null hypothesis when the alternative hypothesis is true. That is, they can fail to detect that a treatment effect is actually nonzero. The probability that a hypothesis test will (correctly) determine that a treatment effect is nonzero is called the *statistical power* of the hypothesis test.

Note that rejection of the null hypothesis is conclusive—it implies that the alternative hypothesis must be accepted. If the null hypothesis is not rejected the results are ambiguous. Failure to reject the null hypothesis *does not* mean that the null hypothesis must be accepted. If the statistical power is low, the test might have a substantial probability of failing to reject the null hypothesis when it is false. For example, if the statistical power of the test was only 20%, then the test has an 80% chance of failing to reject the null hypothesis when it is true, so failure to reject the null hypothesis is far from conclusive evidence that the null hypothesis is true.

A statistical test having high statistical power is essential for hypothesis tests to be conclusive even when they do not reject the null hypothesis. The test statistics we will introduce in this book are generally known to be optimal in the sense of being uniformly most powerful, meaning that no other test (known to us or not yet discovered) can have higher power. This does not mean that these tests have high absolute power, just that they have the highest power possible, given the limitation of the amount of information in the data (and information is actually a well-defined mathematical concept).

The mathematical details of deriving test statistics, their sampling distributions, and computing statistical power can be quite complex and are the domain of mathematical statistics, but it is unnecessary to understand the details in order to use the results that have been developed over the years.

Research design is fundamentally concerned with organizing the data collection for a study so that the results of the investigation will be unambiguous. This book focuses on how to create research designs that will yield unambiguous conclusions about the effects of treatments. We focus on a set of kinds of research designs that are well suited to evaluation studies in educational research and development. These designs are perfectly general and can also be used in other types of research and evaluation.

Effect Size

While hypothesis testing is a fundamental technique in scientific research, good research practice requires that hypothesis tests should be supplemented by descriptive methods such as estimation of effect size. Statements to this effect have been made by both the statistical community (Wasserstein and Lazar, 2016) and the scientific community in education (American Educational Research Association, 2006) and psychology (Wilkinson, 1999). The statistical models for most research designs include a parameter that corresponds to the treatment effect. However, that parameter may not be the best way to represent the treatment effect in a manner that transparently communicates the magnitude of effect. One reason is that different studies may use different measurement instruments so that the same construct (like mathematics achievement) may be measured on different scales in different studies.

While a difference of 5 points on a particular achievement test may be a good way to represent the treatment effect within the context of a single experiment, it may not be easily interpretable to another researchers who is not familiar with that test. For example, is 5 points on a particular version of the Iowa test the same, bigger, or smaller than 5 points on a particular version of the Stanford Achievement test or on a specialized test created by the investigator?

Effect size indices were created to make treatment effects more readily interpretable across studies. The most frequently used effect size in evaluation work is the standardized mean difference, often called Cohen's *d*. In keeping with the notation we discussed earlier, we use the symbol δ to refer to this effect size when it is a population parameter and *d* when we are referring to a sample estimate of δ. In general, we define the effect size δ as

$$\delta = \frac{Treatment\ Effect}{Standard\ Deviation},$$

and *d* as an estimate of δ. Exactly which model parameters correspond to the treatment effect or the standard deviation are somewhat different for different research designs and we define the effect size precisely in the discussion of each particular type of research design.

Effect size has an important connection to hypothesis testing. The power of statistical tests, and therefore the sensitivity of research designs, depends on effect size. Because planning research designs involves assessing whether the research design is sufficiently sensitive to give unambiguous results, understanding how large the effect size of a treatment might be is essential to planning research designs.

Measurement and effect size. The reliability of measurement of the outcome variable has an important relation with effect size. The reason is that measurement error does not influence the treatment effect (on the average) but the noise added by measurement error does increase the standard deviation, decreasing the effect size. Because of that, measurement error decreases (attenuates) effect sizes (see Hedges, 1985). This is analogous to measurement error attenuating the correlation between variables. In fact, the relation between the effect size if the outcome is measured without error (δ_T) and the effect size measured with an unreliable measure (δ_U) having reliability coefficient $\rho_{UU'}$ is

$$\delta_U = \delta_T \sqrt{\rho_{UU'}} \, .$$

Because reliability coefficients are between 0 and 1, inclusive, $\delta_U \leq \delta_T$.

Alignment of measures and effect size. The alignment of the measure of outcome and the construct that the treatment impacts has an influence on effect size. This is important in judging how large the effect size associated with a treatment might be. To understand why this should be true, imagine an outcome measure that has items of two types: One type of item represents material whose learning the treatment was supposed to facilitate (aligned) and one type of item based on content irrelevant to the treatment (nonaligned). A treatment effect might be expected if the test were composed solely on the aligned items, but no treatment effect would be expected if the test were solely on the nonaligned material. From the perspective of evaluating the treatment effect, since no effect is expected on the nonaligned material, that material simply adds noise to the outcome measure. Thus, nonalignment of part of the outcome measure has the same effect as adding measurement error.

This idea has an important application when selecting outcome measures. While a standardized test measuring a broad spectrum of achievement in a domain may have many desirable qualities, it may be poorly aligned to any particular treatment that focuses on a narrower portion of that achievement domain. For example, rates and proportions might be only 10% of the entire grade-specific mathematics achievement domain, but rates and proportions might be the entire focus of a particular treatment. If so, then the treatment effect size measured with a standardized test of the entire mathematics achievement domain could suffer from considerable attenuation compared with the effect size that would be found using an outcome measure that was more highly aligned with the treatment.

The logic given above ignores the fact that a treatment focusing on a specific topic might facilitate learning in other topics not specifically taught. However there is empirical evidence from large scale meta-analyses that effect sizes obtained in research studies using standardized test that focus on broader constructs tend to obtain smaller effect sizes than standardized tests that focus on narrower constructs or experimenter designed tests (that also focus on narrower constructs)—the later two kinds of test presumably selected because they were more aligned to he material the treatment was attempting to teach (Lipsey and Wilson, 1994).

Design Sensitivity

In order for the statistical analysis of data collected to be unambiguous, the design has to be sufficiently sensitive. The sensitivity of the design depends not only on the type of design (which is under the control of the investigator, but also on other factors which are not under the control of the investigator. We call these other factors design parameters because they are properties (literally parameters) of the population from which the data is sampled. There are three different, but complementary, ways to describe design sensitivity, all of which can be useful in planning designs.

One way to describe design sensitivity is statistical power. Statistical power is the chance that if there is a (true or underlying) intervention effect of a given size, the statistical analysis is likely to detect it. Statistical power analysis starts with the size of an effect that is deemed important enough to detect, and a design (e.g., a sample size configuration), and describes the probability that an effect of the given size would result in a statistically significant observed effect. Computation of statistical power can be accomplished by suing tables of power values, but is more easily accomplished using specialized software

Another way to describe design sensitivity is the use of minimal detectable effect size. The minimal detectable effect size is the smallest effect size for which the design has a given statistical power. Using this approach, one starts with design (a sample size configuration) and a desired level of statistical power and describes the smallest effect size for which the design has the specified level of statistical power. Approximate computations of minimum detectable effect size can be done using tables, but exact computations of require specialized software (Bloom, 1995).

A third way to describe design sensitivity is precision. Precision is the standard error of the estimated treatment effect, which determines the width of the confidence interval for the treatment effect. Using precision analysis, one starts with the design (a sample size configuration) and computes the precision that would be expected.

Complex Sampling (Clustering and Autocorrelations)

Statistical Clustering

The most basic sampling mechanism is simple random sampling from a population. In a simple random sample, the sampled elements are independent of one another and every element in the population has the same probability of being selected into the sample. Simple random sampling is the sampling mechanism that is most often discussed in statistics and the most often assumed in deriving the properties of statistical procedures. A very basic fact of simple random sampling is that if the population has mean μ and variance σ^2, then the variance of the mean of a simple random sample of n observations is σ^2/n. Thus, the larger the sample, the smaller the variance of the sample mean. The variance of the sample mean (or its square root, the standard error) is often used to describe the precision or uncertainty of the sample mean (or other statistics). Thus, if we want to estimate the population mean μ, a larger sample is better than a smaller one because it provides a more precise or less uncertain estimate.

While simple random sampling is, well, simple, it is not a good characterization of how many samples are obtained in education research. First, we rarely start with a well-defined population (a point to which we will return). More importantly for this discussion, if students are the population elements of interest, students are typically sampled by first sampling schools, then sampling students within those schools. Such a sampling plan is called two-stage cluster sampling. Two-stage cluster sampling is widely used in survey research for the same reason it is used in education research: It is much more practical than simple random sampling.

To see why, suppose we were interested in doing a study to generalize to a large school district. Because every student would be equally likely to be in the sample, a simple random sample would likely include students in many schools spread throughout the district (e.g., a sample of 50 students might be spread out over 30 schools). It would be very expensive to send data collection personnel to many different schools. The difficulty of conducting a national household survey of 500 people using a simple random sample would be even more daunting. Cluster samples reduce the difficulty in data collection by sampling first aggregate groups (generically called clusters) that are convenient because they are geographically compact (like neighborhoods for a household survey) or administratively useful (like schools for a survey of students).

It should not be surprising that there is a cost for the greater feasibility of cluster samples. The logic is simple: Individuals within the same cluster are more similar to one another than they are to individuals in a different cluster. Thus, the observations within each cluster do not adequately reflect the full variation of the population. Another way of thinking about this is that observations within each cluster are not statistically independent of one another (as required by simple random sampling). We might expect, therefore, that a cluster sample has less information than a simple random sample and would yield less precise estimates.

Return to the population with mean μ and variance σ^2 that we considered before. The variance of a two-stage random sample of n individuals from each of m clusters (a total sample size $N = mn$) is

$$\frac{\sigma^2\left[1+(n-1)\rho\right]}{mn} = \left(\frac{\sigma^2}{N}\right)\left[1+(n-1)\rho\right],$$

where the symbol ρ (the intraclass correlation) is a parameter (ranging from 0 to 1, inclusive) that describes the strength of clustering in the population (see, e.g., Kish, 1965). Note that if $\rho > 0$ and $n > 1$, the variance of the cluster sample is larger than σ^2/N, which is the variance of a simple random sample of the same total size, N.

The quantity $[1 + (n - 1)\,\rho]$ is called the *design effect* in survey research. The design effect reflects how much larger the variance of (how much less precise) the mean from a two-stage cluster sample is than that of a simple random with the same total sample size. Note that the design effect depends on both ρ and the cluster size n. When the clusters are schools and the outcome is an achievement test, values of ρ are typically around 0.15 to 0.25 (Hedges and Hedberg, 2007). Thus, when $n = 20$, design effects are on the order of 3.8 to 5.8. Sometimes survey researchers divide the total sample size by the design effect to yield a quantity they call the *effective sample size*. The effective sample size is the size of a simple random sample that would yield a sample with the same precision as the two-stage cluster sample. When $n = 20$, and ρ is in the range of 0.15 to 0.25, the effective sample size is on forth to one sixth as large as the total sample size.

The variance of the sample mean has implications for research design because treatment effects are defined as the difference between the mean of a treatment group and that of a control group. If the two means are less precise in a study with a two-stage cluster sample, the mean difference (the treatment effect estimate) is also less precise than if the study had used simple random sampling. Because the treatment effect estimate is less precise, the design provides a less sensitive test for treatment effects.

An important example of this principle is the hierarchical or cluster randomized design in which all of the individuals taking part in the study within a school are assigned to the same treatment. An example of this design would be when the treatment schools all used a new curriculum and the control schools used a different curriculum. Such a design can have advantages for efficacy or effectiveness trials. For example, it is sometimes politically problematic, practically difficult, or conceptually impossible to assign different students to different treatments (e.g., for a whole school intervention like whole school positive behavior support). Whatever the advantages cluster randomized designs might have, they involve cluster sampling, they typically require a sample size that is much larger than a design that would assign individual students to treatments (e.g., 4 – 6 times as large in our example). Such very large sample sizes are usually infeasible in an initial field trial of an intervention.

Autocorrelation

In most designs, observations are expected to be independent of one another. One exception is designs using clustered sampling (where observations in the same clusters are correlated). Another important exception is that of designs, such as interrupted time series designs, with autocorrelations. Autocorrelations occur when observations are collected in temporal order and the previous observations are correlated with subsequent observations. The simplest case of autocorrelations occurs when there are observations are collected at only two timepoints, for example in a pretest-posttest design.

In interrupted time series designs, there are observations at several timepoints before and several time points after the treatment (or 'interruption' as the terminology refers to the treatment). In such designs, stronger correlations are expected between adjacent measurements and the further apart the measurements, the weaker correlations are expected to be. The autocorrelation structure is part of the stochastic specification of statistical models for data from interrupted time series designs.

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Chapter 4
Introduction to Randomized Experiments

Randomized experiments are often called the "gold standard" for inference about cause and effect. This is for good reason. Comparative studies to determine whether a treatment or intervention had an effect have been carried out for literally hundreds of years. (e.g., Lind, 173; or Donner and Klar, 2000). In such studies, called observational, correlational, or quasi-experiments, the scientist attempts to make the group receiving treatment as alike as possible *in every important way*, except that one group receives the treatment and the other does not. They do so by either explicit matching or pseudo-matching matching on important variables via statistical adjustment. The problem is that it has proven very difficult to know *what* the important variables are and *how to measure them* adequately enough for effective matching. Random assignment of individuals to treatment assures that, on the average, there is matching on both all variables we know and all variables *we have not thought of*. This profound advantage is why randomized experiments have attained the status of the most powerful tool for causal inference.

Random assignment is not the same as haphazard or unsystematic assignment with an unknown assignment mechanism. There is no way to assure that assignment was random without actually *controlling* the assignment. Statements like, "the assignment was essentially random" are misleading and statistically meaningless. R. A. Fisher, the original proponent of randomized experiments in agriculture noted that

Insert Fisher Quote

How to Randomize

It might seem completely obvious how to randomize, yet our experience in teaching about the design of experiments is that there are confusing subtleties in actually carrying out randomization. Fisher (1935) advocated using a deck of well-shuffled cards. To randomize into two groups, he advised using a deck of cards each card printed with the numbers 1 or the number 2 so that there were equal numbers of each. Then randomization into two groups of n individuals each is carried out by ensuring that the cards are well shuffled and then selecting cards one at a time until to assign individuals to the treatment group named on the cards selected until one group is filled (that is, it has been assigned *n* individuals). At that point, the remaining individuals are assigned to the unfilled group. If there were 2n individuals to begin with, all of the individuals will be assigned at this point. If there were more than 2n individuals, those not assigned are not needed.

When given this explanation of a randomization procedure, many students object when they hear that, when one group is filled, all the remaining individuals are assigned to the other group. This seems to violate their idea of both independence and balance, since "all the people at the end of the process get assigned to the same treatment group." Of course, whether there is a sequence of people assigned to the same treatment group, and how long that sequence might be depends entirely on the (random) assignments that have been made previously. A sequence of individuals can only be assigned to group 2 at the end of the assignment process if a sequence (or sequences totally the same number) has been assigned to group 1 earlier in the assignment. And of course, it is just as likely that the final sequence is assigned to one group as it is to the other.

This way of thinking illustrates common fallacies about how random sequences should look. Research shows that, asked to create sequences of random numbers, people tend to create sequences that are actually much more orderly than they should be, without repeated sequences. In general, people are very poor at being random number generators without technological aides (like cards, random number tables, or computers).

Another way to randomly assign is to use a table of random numbers (included in many, especially older, statistics books). To assign units to either of two treatments, start at a random place in the random number table and assign the first unit to group 1 if the random number is odd and to group 2 if it is even, go to the next random number and use the same procedure to assign the second unit, etc. The same procedure can be used with a random number generator in a computer. However, it is crucial that you start at a different, ideally random place in the random number table (otherwise every 'random' assignment to an experiment will be the same). It is perhaps less obvious that computer random number generators can have the same problem. Computer random number generators start with what is called a *seed value*, which is transformed in a complex way to yield the 'random' number. The random number has no obvious relation to the seed value, but the process always yields the same random number if it starts with the same seed value. Random number generators usually permute the seed value each time they generate a random number so that the next random number they generate is based on a different seed value. But is it crucial that the entire sequence of random numbers that may be generated all depend on the original seed value. If you generate a million random numbers starting with the seed value '1234', you will generate *exactly the same sequence* of random numbers if you start again with the same seed. (Statisticians know this and use it to replicate their simulations exactly in debugging code.) The process is no different than always starting in the same place in the random number table. If you do not know how to set the seed value for a random number generator, then you might be well advised not to use it.

Randomization and Alternation

Although random assignment seems simple, Fisher went to great pains to describe how it should be done because it was not obvious to researchers at the time. The earliest references to some form of random assignment imagined two groups of matched individuals and random assignment of the entire group to treatment (e.g., von Helmont, 1648). Note that this form of assignment invokes chance assignment once for the entire experiment (it is actually a form of what we would now called cluster randomization). Note that random assignment, as described above, assigns each unit independently to treatment or control, invoking the chance assignment once for every unit in the experiment.

During the early part of the 20th century the term random assignment was often used in a loose way that often did not distinguish true randomization (unit by unit) as described above from alternation (assigning every other unit on a list to the treatment group) (see, e.g., Chalmers, 2010). It is important to recognize that alternation is not the same as randomization: It does not provide the same assurance of control over selection bias that randomization does and it does not, in general, assure that the statistical properties of significance tests are the same as those that occur as a consequence of random assignment.

One way to see that alternation does not provide the same assurance of control over selection bias is to imagine a perverse actor organizing the experiment. If that actor was able to arrange the list of units so that every other unit had systematically higher (or lower) scores on a covariate that predicted outcome, they could bias the results of the experiment. This would be easy to do if the list were in order of individuals who were recruited to be part of the experiment.

It could be done even if the list were alphabetical by selectively accepting into the experiment based on the alphabetical status of their names. Alternation would only be formally equivalent to randomization if the list itself were randomly ordered, in which case alternation would be exactly equivalent to individual random assignment.

Statistical Inference and Generalization

Statistical inference plays a crucial role in quantitative research by providing a warrant for the argument that the research results obtained from a particular set of units studied might apply to other units (students or teachers) that have not been studied. However, it is unrealistic to expect statistical inference to carry the entire burden of justifying generalization to all of the targets of generalization that are of practical interest.

One useful description of the situation was offered by Jerome Cornfield and John Tukey in a paper in the *Annals of Mathematical Statistics*. They offered the metaphor of the "islands of inference" (Cornfield and Tukey, 1956). He argued that scientific inference (generalization) is like the problem of trying to cross a river, and statistical inference based on an experimental design does not get us to the other side, but only to an island nearer the other shore. The choice of design can move that island closer or further from the other shore, but it will never get us all the way across. What is always required to get entirely to the other shore is some degree of extra-statistical scientific reasoning. In a program of research, it may be useful to use different designs that place the island successively closer to the other shore over time, reducing the amount of extra-statistical reasoning required to support claims about generalizability.

In developing an educational treatment, laboratory and design experiments appropriately place the island rather far from the other shore (of generalization that are of practical interest) and the statistical inference carries relatively little of the burden of making the generalization to large scale practice (which may be a distant ambition at this point). In efficacy and effectiveness studies, the choice of design moves the island closer to, but still not touching that other shore. In between these two alternatives are designs that might be used in initial field studies that rely on statistical inference to demonstrate that the treatment can have effects on units other than the ones that have been explicitly studied.

The role that statistical inference is called upon to play in generalization is most clearly connected to the role that sampling plays in the design. In all experimental designs, individuals (students or teachers) are sampled. At a minimum, these designs support generalization to other individuals (than those observed) in the population from whom the observed individuals have been sampled. This is the minimal role that statistical inference might play: Supporting the claim that the treatment would probably have effects if another group of individuals had been sampled into the study rather than the ones who were actually observed.

The experimental designs used in efficacy and effectiveness studies introduce another level of sampling in addition to sampling of individuals. They also sample contexts (usually schools or classrooms). Such designs support not only generalization to other individuals in the same contexts as those observed, but to other contexts as well (the population of contexts from which the observed contexts have been sampled). Because this is a more demanding requirement of the statistical inference, larger samples (more contexts and with them more individuals) are required.

The inferential goal of initial field studies is often in the middle space between laboratory or design experiments and efficacy or effectiveness trials. In development evaluations, the goal of the design is permit statistical inference to support inference to other individuals (than those observed) in the same context as that observed in the evaluation study. In such studies it is appropriate to use designs that do not use sampling of contexts in order to use statistical inference to generalize to unobserved contexts. This does not mean that the contexts selected for the study do not matter or that scientific generalizations to other contexts are inappropriate. It does mean that the warrant for those generalizations across contexts must be extra-statistical, that is, conceptual. The logical warrant supporting generalization is something like "the contexts of the target of generalization are similar enough to those in the experiment for the same effects to be found." The credibility of this argument will doubtless depend on the breadth of contexts that are actually observed in the experiment.

Restricted Randomization

There is nothing about the principles of randomized experiments that dictate that every available unit should be randomized. The question of what units to randomized relates not to internal validity but to generalizability. If the object of the experiment is to assess the causal effect of treatment in a larger population, then the principles of representative (e.g., random) sampling should govern the choice of what units to randomize. On the other hand, the object of the investigation may not be to demonstrate that treatment effects generalize to a larger population of units, contexts or times. In initial field trials, the object of the investigation may be to demonstrate that the treatment can work in some units in some context and at some time and would likely work for other units in those same contexts. In a sense, the study may have the objective of showing proof of concept that warrants larger scale evaluations in more generalizable contexts (e.g., and efficacy or effectiveness study).

Demonstration of proof of concept does not necessarily require that the evaluation be carried out so that broader generalization is warranted by the statistical inference. It requires only that the statistical inference demonstrates that the treatment worked in some well-defined situation and the warrant for generalization of results to other situations is conceptual/substantive rather than statistical. Thus, the situations in which evaluations are conducted can be engineered somewhat to improve design sensitivity. For example, an investigator might choose to exclude some students (but only prior to randomization) in order to reduce variation on the pretest, which in turn would be expected to reduce variation on the posttest and increase design sensitivity. Such strategies are discussed in some of the chapters that follow.

Restricted randomization is closely related to rerandomization: Only accepting a restricted set of possible outcomes of the randomization process that yield balance on the covariates (see, e.g., Morgan and Rubin, 2012). Strategies for rerandomization and restricted randomization and the analyses that best exploit them can be rather complex. They have also occasionally been controversial. We discuss only the simplest of these in the chapters that follow, but these are adequate for most evaluation studies addressed by this book.

In efficacy or effectiveness studies, the objective of the research is to demonstrate that the treatment has effects in authentic contexts that are likely to generalize to other populations, contexts, and times. For such studies, it is crucial that the study sample, including the units randomized are like those in the target of generalization. Strategies that modified samples or excluded aggregate units would be much less acceptable in efficacy or effectiveness studies which are intended to pose a greater burden on statistical inference to warrant generalization.

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Chapter 5
Individually Randomized Trials with No Clustering

Introduction

The simplest randomized designs are individually randomized trials with no clustering. In such designs it is possible to assign individuals to treatments independently from one another. This design is not possible if (for practical or political reasons) the individuals in the same classroom or school must be assigned to the same treatment. In that case the design would be a cluster randomized design (one where whole statistical clusters such as school or classrooms are assigned to treatment or some variety of randomized block design (where treatment assignment is carried out withing clusters such as schools).

Statistical Model and Notation

Suppose that the design has n individuals that are randomly assigned to each of the treatment and the control condition. Such a design is called a balanced design because there are equal numbers of individuals assigned to each treatment. Balanced designs are desirable because they are the most sensitive, that is they produce statistical tests that have the highest statistical power and the most precise estimate of treatment effects for a given total sample size. Then, in the conventional notation of the analysis of variance, the model for Y_{ij}, the outcome measurement of the j^{th} individual in the i^{th} treatment, is

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij},\ i = 1, 2; j = 1, \ldots, n, \quad (1)$$

where μ is the grand mean, α_i is the effect of the i^{th} treatment and ε_{ij} is a residual. The residual ε_{ij} is assumed to have a normal distribution with variance σ_ε^2 and the grand mean μ and the treatment effects α_i are taken to be fixed, but unknown constants.

In this design, it is conventional to define the effect size as the standardized mean difference

$$\delta = (\alpha_1 - \alpha_2)/\sigma_\varepsilon,$$

(see, Hedges, 1981).

Let $\overline{Y}_{1\bullet}$ and $\overline{Y}_{1\bullet}$ be the means of the j^{th} treatment and control groups, respectively, and let $\overline{Y}_{\bullet\bullet}$ be the grand mean. This notation with dots in the subscripts may seem confusing, but it has a logic. It takes the values of all four subscripts to identify a particular observation, but one could average over all observations in treatment group 1 to obtain $\overline{Y}_{1\bullet}$ (here that average is denoted by dots in the second subscripts which identify the individual within treatment group to denote averaging over individuals and clusters).

The Statistical Analysis of the Design

In the analysis of this design, the overall treatment effect estimate is

$$\overline{Y}_{1\bullet} - \overline{Y}_{2\bullet} = \frac{1}{n}\sum_{j=1}^{n}\left(Y_{1j} - Y_{2j}\right).$$

The test statistic used for testing treatment effects is

$$F = MST/MSCT,$$

where MST is the so-called treatment mean square defined by

$$MST = SST = n\left(\bar{Y}_{1\bullet} - \bar{Y}_{2\bullet}\right)^2 \Big/ 2,$$

and *MSW* is the so-called within-treatment-groups mean square defined by

$$MSW = SSW/2(n-1) = \sum_{j=1}^{n}\left(Y_{1j} - \bar{Y}_{1\bullet}\right)^2 + \left(Y_{2j} - \bar{Y}_{2\bullet}\right)^2 \Big/ 2(n-1)$$

(see, e.g., Kirk, 1995). When the null hypothesis of no treatment effects is true, the test statistic has the *F*-distribution with 1 degree of freedom in the numerator and $2(n-1)$ degrees of freedom in the denominator. Thus we carry out the significance test by comparing the obtained value of *F* with the appropriate critical value of the *F*-distribution with 1 and $2(n-1)$ degrees of freedom or by using the *p*-value computed by the analysis software.

The standard error (precision) of the treatment effect estimate is

$$SE = \sqrt{\frac{2MSCT}{n}}.$$

Planning Individually Randomized Designs

The object planning a research design is to ensure that if the data is collected according to the design, the analysis of that data will yield unambiguous results. To ensure results will be unambiguous, the design has to be sufficiently sensitive. There are three different, but complementary, ways to describe design sensitivity, all of which can be useful in planning designs.

One way to describe design sensitivity is statistical power. Statistical power is the chance that if there is a (true or underlying) intervention effect of a given size, the statistical analysis is likely to detect it. Statistical power analysis starts with the size of an effect that is deemed important enough to detect, and a design (e.g., a sample size configuration), and describes the probability that an effect of the given size would result in a statistically significant observed effect. Computation of statistical power can be accomplished by suing tables of power values, but is more easily accomplished using specialized software

Another way to describe design sensitivity is the use of minimal detectable effect size. The minimal detectable effect size is the smallest effect size for which the design has a given statistical power. Using this approach, one starts with design (a sample size configuration) and a desired level of statistical power and describes the smallest effect size for which the design has the specified level of statistical power. Approximate computations of minimum detectable effect size can be done using tables, but exact computations of require specialized software.

A third way to describe design sensitivity is precision. Precision is the standard error of the estimated treatment effect, which determines the width of the confidence interval for the treatment effect. Using precision analysis, one starts with the design (a sample size configuration) and computes the precision that would be expected.

General Principles for Planning Individually Randomized Designs

Planning an individually randomized design to evaluate an educational intervention primarily involves choosing a design that will have adequate sensitivity (high statistical power and precision, or small minimal detectable effect size). In principle, design sensitivity depends on the significance level chosen, the effect size expected, and *n*. Although significance level is in principle an arbitrary choice of the investigator, the 0.05 or 5% significance level has been so reified by convention that it is, for all practical purposes, not a choice but the de facto standard. The effect size expected is also not completely a choice but is dictated by the prior experience of the investigator about what size effect is large enough to be important or is the effect size to be expected from the intervention being evaluated. Thus, planning the design amounts to deciding on how many subjects (usually students) to assign to each treatment within each block. That is, planning the design involves choosing n to achieve the desired design sensitivity.

The computation of either statistical power or minimum detectable effect size can be accomplished by using statistical tables that are widely available (see, e.g., Hedges and Rhoads, 2009) or by the use of statistical software that can compute noncentral F-distribution function (see the Appendix). It is probably easiest to use software that is designed to do the computations directly (e.g., **Website reference**). While exact power computations using software are recommended, we offer a few general principles that are helpful in the sections that follow and illustrate them with computations of design sensitivity.

The most sensitive design given a fixed total sample size is a balanced design in which the same number of individuals are assigned to each treatment within each block. Exact design sensitivity depends on n and δ, but in balanced designs. Table 1 shows the statistical power, minimum detectable effect size, and standard error of the estimated treatment effect for total sample sizes from $n = 10$ to $n = 500$ and effect size values of $\delta = 0.40$ and 0.60. The table shows that the minimum detectable effect size and standard error of the treatment effect estimate depend only on n and not the effect size. However, the statistical power does depend on effect size. Or example, and the statistical power of designs with $\delta = 0.6$ is greater than that for designs with $\delta = 0.4$ with the same sample size n. Note also that when sample size becomes large, power converges to 1.0.

Insert Table 1 About here

To provide another perspective on design sensitivity, Table 2 gives statistical power as a function of n and δ. The table shows more clearly increases with δ for any fixed n. It also shows more clearly how power increases with n for any fixed δ. Note that the lower righthand corner of the table is essentially all values of 1.0, which shows that power converges to 1.0 for large δ or large n.

Insert Table 1 About here

The minimum detectable effect size is a convenient way of evaluating design sensitivity for a given candidate design. Figure 1 shows the MDES as a function of n when the target power is 80%. It shows that when $n = 63$, $MDES = 0.5$. Thus, a sample size of at least $n = 63$ would be required to be adequately sensitive (have 80% power). Alternatively, an $n > 150$ (actually, $n = 395$) would be required to obtain an $MDES = 0.2$

Insert Figure 1 About Here

Using Covariates Can Increase Sensitivity and Reduce Sample Size Requirements

Covariates are any variables whose value could not have been influenced by the treatment assignment. The most common covariates are baseline data (such as pretests) collected before the treatment begins. If baseline data on individuals is available, it can be used to increase the sensitivity of the design if the analysis uses statistical adjustment via regression or the analysis of covariance. The use of a covariate (or multiple covariates) essentially has the same impact as increasing the effect size. In particular, if the effect size (with no covariates used) is δ

and correlation (or multiple correlation if there is more than one covariate) between the covariate(s) and outcome is ρ, then the design using covariates is roughly equivalent to increasing the effect size from δ to $\delta / \sqrt{1-\rho^2}$. This is generally an increase because $0 \leq |\rho| \leq 1$, which implies that $\delta / \sqrt{1-\rho^2} \geq \delta$.

Table 3 gives the minimal detectable effect size as a function of group sample size n for balanced designs and values of ρ from 0.1 to 0.9. The table shows that the use of covariates with relatively high correlation with outcome can greatly increase sensitivity and reduce MDES. For example, when n = 50, a covariate with $\rho = 0.7$ (a fairly typical value when both covariate and outcomes are achievement tests of the same domain), the use of a covariate decreases the MDES from 0.57 (with no covariate) to 0.40 (with the covariate).

Insert Table 3 About Here

While covariates add complexity to the analysis, they can substantially increase design sensitivity and reduce sample size requirements. Pretests of measures similar to the outcome are often the most effective covariates (the ones with the largest correlations with the outcome). Note that cognitive tests tend to be rather highly correlated with one another, so it is not essential that the covariate measure exactly the same construct as the outcome to be useful. For example, reading achievement tests may be useful covariates, even in studies using math or science achievement as the outcome. Other variables measured at baseline (such as race or social class measures are typically poorer covariates. Reference values of covariate outcome correlations for academic achievement based on national representative samples of students are given in Hedges and Hedberg (2007).

Unequal Sample Sizes Can Increase Design Sensitivity if it Enables Larger *N*

For a fixed total sample size N, a balanced design (equal allocation of individuals to treatment and control in each block) is the most sensitive (with or without covariates). However, there are situations in which the sample size is limited by the number of individuals who can be treated. For example, there may only be enough field staff to work with a limited number of treatment students or there may be a limited number of computers or other technology necessary to use the intervention. In that case, unequal allocation of individuals to treatment groups (more individuals randomized to control and fewer to treatment) can increase the total sample size and can increase design sensitivity.

Suppose that an unbalanced design assigns n^T individuals to treatment in each block and n^C individuals to control so that the proportion $\pi = n^T/(n^T + n^C)$ of the total number of individuals are assigned to treatment. In a balanced design the total sample size would be $N = n^T/0.5 = 2n^T$, but in an unbalanced design, because $\pi < 0.5$, the total sample size N would be greater than $2n^T$. In fact, the ratio of the total sample size of a design with $\pi = n^T/N$ to that of the balance design would be $1/2\pi$. Thus if $\pi = 0.5$ (the design is balanced) the ratio is unity, as expected, but if $\pi = 0.4$, the total sample size of the design is 25% larger, if $\pi = 0.3$ it is 67% larger, and if $\pi = 0.2$ it is 150% larger than that of a balanced design with the same number of *treated* individuals. This increased total sample size increases design sensitivity.

While imbalance of the design does not literally change the effect size, it has an effect on sensitivity that is equivalent of changing the effect size from δ to $\delta\sqrt{4\pi(1-\pi)}$. Because

$$\delta\sqrt{4\pi(1-\pi)} \leq \delta$$

and only equal to δ if $\pi = 0.5$, imbalance has a tendency to increase sensitivity because of the increased total sample size, but a simultaneous tendency to decrease sensitivity because it does the equivalent of decreasing the effect size. However, unless π is very much smaller than 0.5, the positive impact of imbalance on total sample size is bigger than the negative impact on effect size and the unbalanced design is more sensitive with the same number of treated individuals.

Table 4 shows the minimum detectable effect size as a function of the total number of individuals assigned to treatment for balanced designs ($\pi = 0.5$) and unbalanced designs with various degrees of imbalance ($\pi < 0.5$). For example, if $n^T = 50$ individuals are to be treated, a balanced design would have a minimum detectable effect size of $\delta = 0.57$, but that could be reduced to $\delta = 0.47$ if we assigned $n^C = 50 + 75 = 125$ individuals ($\pi = 0.28$) of the total individuals to treatment. We could lower the minimum detectable effect size by using an even more imbalanced design, but as the table shows that as n^C tends to infinity, the MDES tends to a limiting value that is larger than zero. For example, if $n^T = 50$, it is impossible to obtain a MDES smaller than 0.40, even with an infinite sample size in the control group. This limiting MDES value decreases as n^T increases: It is 0.40 when $n^T = 50$, it is 0.28 when $n^T = 100$, and 0.20 when $n^T = 200$.

Insert Table 4 About Here

Examples

In tis section we offer three examples that illustrate how to use the design principles in planning the design for an evaluation.

Example 1. Consider the evaluation of an informal science curriculum delivered in a booth -like setting in a museum. The curriculum is an interactive exhibit designed to teach physical principles. The particular curriculum of interest emphasizes engagement in simulations controlled by the subjects. It is intended to replace an existing curriculum that is based more on static examples of simulations that are not under the control of the subjects. Each curriculum is designed to engage subjects for about 45 minutes. The outcome is a short test embedded in the curriculum itself, asking subjects to make predictions about physical situations. Because it is a museum setting, there is little prior information about the subjects. The major resource limitation is the number of exhibit booths (10) that can be used. The outcome measure is closely targeted to the concepts being taught and the developers expect a modest effect of about $\delta = 0.4$ as a result of the changes that are more engaging to the subjects.

Consulting Table 2 (or appropriate software) we see that achieving and MDES of 0.4 would require an n of 100 in each group in a balanced design or a total $N = 200$. With 10 booths to administer the treatment or control curriculum, $N = 200$ would imply 20 "turns" of the booths. Given pervious experience of 5 turns for each booth per day, this suggests that data collection could be completed in 4 – 5 days so that a full week of data collection might provide a margin to account for slower accumulation of cases than expected, subjects who did not complete the curriculum or posttest, etc.

We might wonder if there was a better design. Considering the design principles discussed about, there are two principles we might use to increase design sensitivity: Use of covariate or an unbalanced design. Because we have no prior information about the subjects, the use of a covariate to increase design sensitivity is impossible without revising the curriculum. This could be done, but would increase the length of time required to complete the curriculum, possibly discourage completion, and is considered "a last resort."

An unbalanced design could be used, but given that the major resources limitation is the number of booths, an unbalanced design would not enable a larger total sample size. Because unbalanced designs are only more sensitive if they enable a larger total sample size an unbalanced design would confer no advantage in design sensitivity.

Thus, we conclude that a balanced design, anticipating a total sample size of $N = 200$, but collecting data long enough to be sure to obtain at least and $N = 200$ is a good design choice.

Example 2. Consider the evaluation of high school hands on robotics workshop in which students actually build robots. This workshop is resources intensive and enrollment is limited to 20 by the materials necessary and the laboratory space in which the robots are built, but there is a great deal of demand for such a program. This workshop is to be held on Saturdays, drawing students from several local schools. A less resource intensive robotics program involving demonstrations of the design, construction, and use of robots was available and it was decided to use participation in this program as a control condition. The target outcome of this program is knowledge and application of robotic design principles as measured by a group assessment. A medium size effect of $\delta = 0.5$ is expected.

Consulting Table 2 (or using appropriate software), we see that with a balance design and n = 20, the minimum detectable effect size is 0.91, much larger than 0.5. Considering the design principles discussed about, there are two principles we might use to increase design sensitivity: Use of covariate or an unbalanced design. Consulting Table 3 (or using appropriate software), we see that with n = 20 and a balanced design, it would require a covariate with a correlation of over 0.8 to obtain an MDES of 0.5. Correlations between cognitive tests that are that high are not impossible but are very unusual. Thus, it is not clear that it would be possible to obtain a covariate with that high a correlation with the outcome. This leaves he possibility of an unbalanced design with $n^T = 20$ and a larger control group. Consulting Table 4 (or using appropriate software) we see that even an unbalanced design with $n^T = 20$ and an infinite sized control group, the MDES is still 0.63, larger than 0.5.

This suggests that the goal of an adequately sensitive design with $n^T = 20$ is not attainable. An alternative might be to expand the treatment group by running two (consecutive or concurrent) workshops, each of 20 students for a total $n^T = 40$. Consulting Table 2 again, we see that a balanced design with $n = 40$ has an MDES of 0.64, still larger than 0.5. Considering the use of a covariate and Table 3, we see that a covariate with a correlation of greater than 0.6 with the outcome would be required to obtain an MDES less than 0.5. This might be possible, but would require careful construction of the covariate to align with the outcome and it would need to be pilot tested to insure it had as high a correlation as is required.

An alternative is an unbalanced design with a larger enrollment in the control group. Consulting Table 4, we see that an unbalanced design with $n^T = 40$ and $n^C = 40 + 150 = 190$ would have an MDES of 0.49. Thus, such a design would be adequately sensitive. This design has the advantage that it does not involve the development and validation (in terms of predictive power) of a new test specifically for the evaluation—an endeavor that would be time consuming and might not succeed.

Example 3. Consider the evaluation of a treatment that is a redesign of a math teaching application. The redesign covers the same content but is organized differently with different applications and examples. The treatment can be embedded in the application along with the current version and random assignment of students carried out by the application. The application transfers performance information to a served each session, which is saved for the entire school year. The application has imbedded assessments that served as unit evaluations and longer evaluations that will serve as the baseline (pretest) final evaluation. The modifications to the treatment are small but significant and the outcome measure is highly aligned with the treatment so a modest effect size of $\delta = 0.3$ is expected.

Interpolation between values $n = 150$ and $n = 200$ in Table 2 we see that (interpolating between the MDES values for $n = 150$ and $n = 200$), to obtain an MDES of 0.3 in a balanced design would require about $n = 175$ (appropriate software would yield the exact figure of $n = 176$) for a total sample size of $N = 350$. Given the ease of data collection a sample size of this order might be acceptable. Of course, it would be advisable to collect a somewhat larger sample size as precaution against missing or otherwise unusable data.

If this sample size is viewed as too large to be feasible, we might consider two principles to increase design sensitivity: Use of covariate or an unbalanced design. An unbalanced design would seem to have no advantage here because it would not obviously lead to a larger total sample size. If the pretest was reasonably reliable it might be used as a covariate to increase design sensitivity. Consulting Table 3, if it was realistic to think it was correlated at least $\rho = 0.4$, then the needed sample size could be reduced to $n = 150$ (a total of $N = 300$). If it was realistic to think it was correlated at least $\rho = 0.65$, then (interpolating between the MDES values for $\rho = 0.6$ and $\rho = 0.7$) the needed sample size could be reduced to $n = 100$ (a total of $N = 200$). Appropriate software would yield the exact correlation needed as $\rho = 0.66$. In either case increasing the sample size somewhat above he absolute minimum required would be a sensible precaution.

Conclusions

Individually randomized designs can powerful research designs that are scientifically rigorous and are often quite feasible and straightforward to plan and to use in evaluation designs. Designs powerful enough to detect effects of $\delta = 0.50$ can be achieved with sample sizes of n = 70 per group (a total sample size of $N = 140$) with no covariates and $n = 40$ (a total sample size of $N = 80$) using a covariate that is correlated $\rho = 0.7$ with the outcome. Designs that are adequate to detech an effect size of $\delta = 0.20$ require larger sample sizes, but can still be achieved with sample sizes of $n = 200$ (a total sample size of $N = 400$) by using a covariate that is correlated $\rho = 0.7$ with the outcome.

Individually randomized designs are often implemented within single large schools or across several schools if care is taken to assure that the randomization is independent of schools. The independence of schools is not a requirement for carrying out a randomized trial, but randomization that is not independent of schools technically leads to a different experimental design, the randomized blocks design.

While balanced designs provide the greatest sensitivity for a fixed total sample size, unbalanced designs can achieve greater sensitivity in cases where the treatment can only be administered to a limited number of individuals, but a larger number of control individuals can be handled. In this situation, covariates can be combined with an unbalanced design to obtain even greater design sensitivity.

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Appendix

Power computations for this design are based on the sampling distribution of the F-statistic given in the text. When $\delta^2 \neq 0$, and the covariate-outcome correlation is ρ, then F has the noncentral F-distribution with $n^T + n^C - 2$ degrees of freedom and noncentrality parameter

$$\lambda = \frac{n^T n^C n\delta^2}{\left(n^T + n^C\right)\left(1 - \rho^2\right)}$$

(see, e.g., Searle, 1971). Therefore, the statistical power of the level α test of the null hypothesis H_0: $\delta^2 = 0$ is

$$G[c_\alpha | \lambda, n^T + n^C - 2 - q]$$

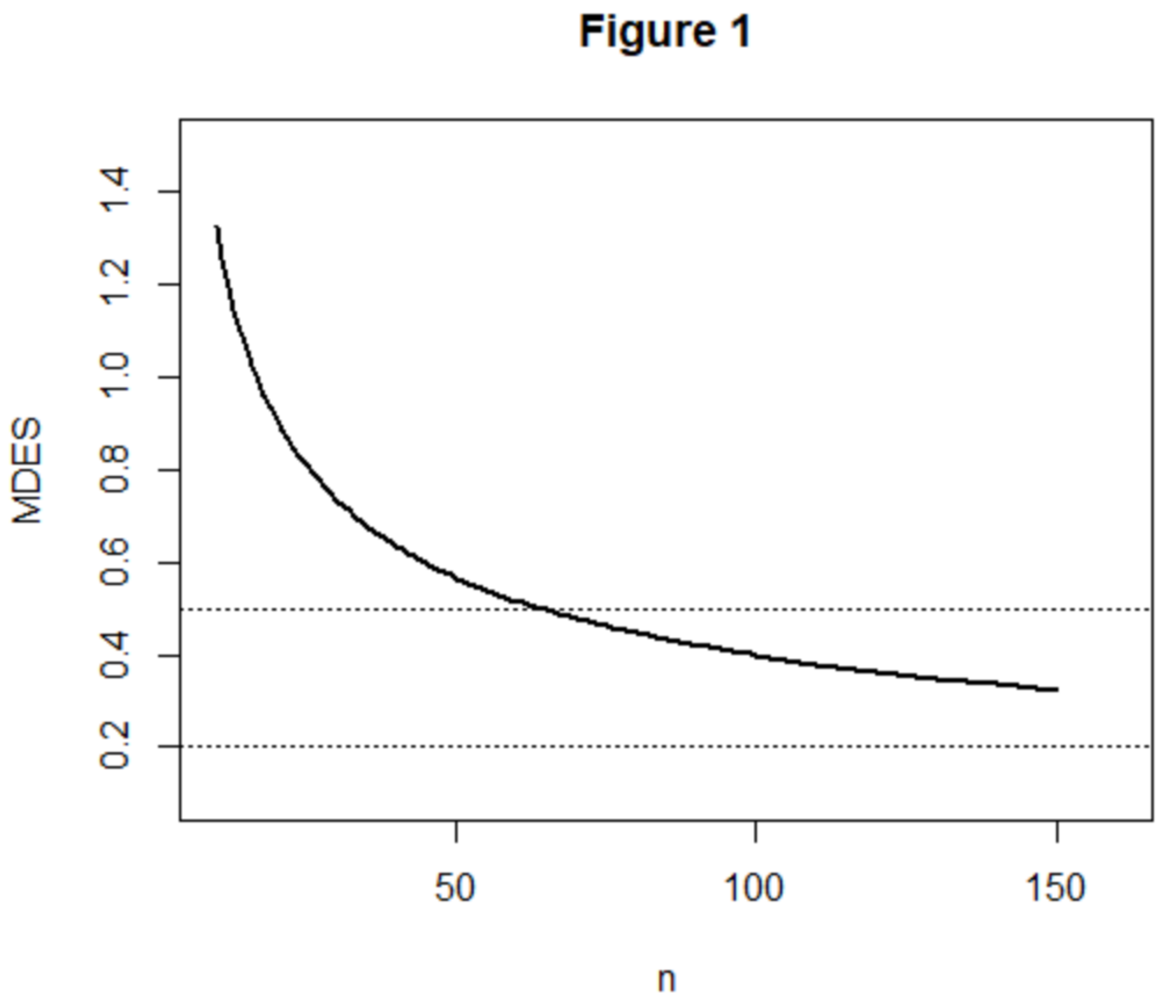
where $q \geq 0$ is the number of covariates (if any), c_α is the $100(1 - \alpha)$ percentile of the central F-distribution with 1 degree of freedom in the numerator and $n^T + n^C - 2 - q$ degrees of freedom in the denominator, and $G[x \mid \lambda, v]$ is the cumulative distribution function of the noncentral F-distribution with 1 degree of freedom in the numerator, v degrees of freedom in the denominator, and noncentrality parameter λ.

The limiting power as n^C tends to infinity with n^T fixed is obtained by noting that, as n^C tends to infinity, λ tends to $\lambda_L = n^T\delta^2/(1 - \rho^2)$. Moreover, as the degrees of freedom of the noncentral F-distribution tends to infinity, the noncentral F-distribution tends to a noncentral chi-squared distribution. Because power is a continuous function of λ, it follows that the limiting power is the power at the limiting λ value λ_L.

The approximate minimal detectable effect size can be obtained using a formula in Bloom (1995) or obtained exactly by iteratively computing power with successively larger values of the effect size until the appropriate power is obtained.

Figure 1

Minimum Detectable effect size as a function of group sample size *n*



Appendix

Power computations for this design are based on the sampling distribution of the F-statistic given in the text. When there is no block treatment interaction and $\delta^2 \neq 0$, then F has the noncentral F-distribution with $n^T + n^C - 2$ degrees of freedom and noncentrality parameter

$$\lambda = \frac{n^T n^C n\delta^2}{\left(n^T + n^C\right)}$$

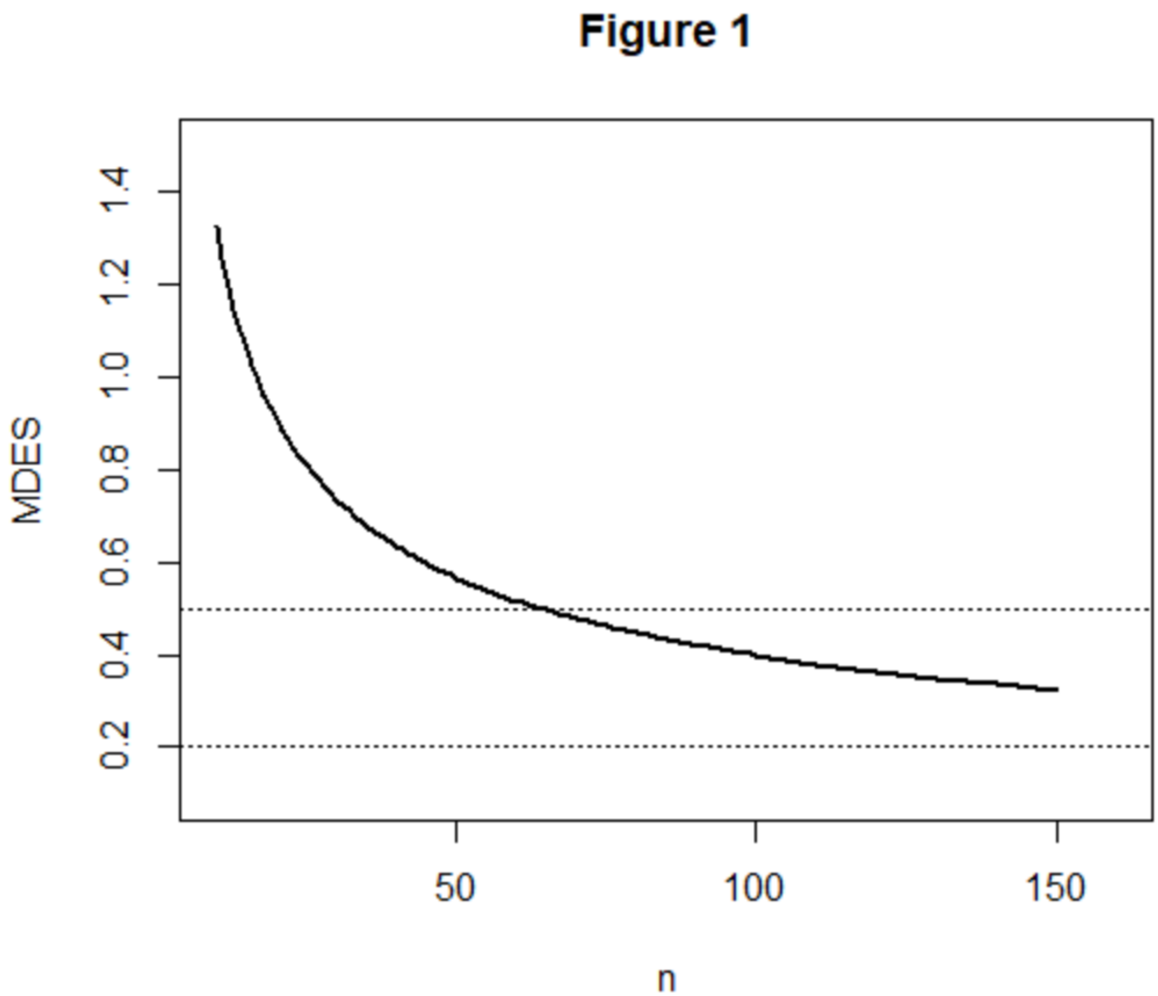
(see, e.g., Searle, 1971). Therefore, the statistical power of the level α test of the null hypothesis H_0: $\delta^2 = 0$ is

$$G[c_\alpha | \lambda, n^T + n^C - 2]$$

where c_α is the $100(1 - \alpha)$ percentile of the central F-distribution with $n^T + n^C - 2$ degrees of freedom and $G[x \mid \lambda, v]$ is the cumulative distribution function of the noncentral F-distribution with v degrees of freedom and noncentrality parameter λ.

The approximate minimal detectable effect size can be obtained using a formulas in Bloom (1995) or by iteratively computing power with successively larger values of the effect size until the appropriate power is obtained.

Figure 1
Minimum Detectable effect size as a function of group sample size *n*



Chapter 6
Randomized Block Designs for Development Research

The randomized block design, which is also called the multisite design arises when assignment to treatments is made within statistical clusters or blocks such as schools or classrooms (see, e.g., Kirk, 1995). It also arises when individuals are matched into groups by covariates such as social class or prior achievement and assignment to treatments is made within these matched groups. This design can be considerably more sensitive than the individually randomized design if the individuals within blocks are mor like one another (in terms of outcome scores) than they are to individuals in other blocks.

This design involves what might be conceived as a two-stage cluster sample, but it is handled in a special way. In the randomized block design, assignments to treatment are made within clusters, so that if clusters are schools, every school includes some individuals assigned to treatment and some to the control condition. The most obvious implementation of this design would be one in which individual students are assigned to treatments within schools, but it could also be implemented by assigning some students in each classroom to each treatment in each school.

Layout of the Design

The term "block" (sometimes called plot) in experimental design comes from the origin of experimental designs in agriculture, where field experiments were literally conduced in agricultural fields and blocks or plots were literally plots of ground within those fields. The layout of the design is a diagram of how the field might be divided into plots within those fields. Figure 1 illustrates the layout of a randomized block design, with shaded areas corresponding to treated portion of the blocks and unshaded portions of blocks corresponding to control portions of blocks. Note that the spaces between the blocks are intended to convey that the blocks (plots of land) might be physically distant from one another to ensure independence.

Insert Figure 1 About Here

In agriculture, there are variations in outcomes due to micro-variations in characteristics such as soil fertility, drainage, and temperature. These characteristics are related physical proximity and the physical proximity of the points where observations are taken within blocks assures that observations within blocks are likely to be more similar than they are to observations from different blocks. Thus observations within blocks are matched to one another.

In an educational field study, the ideas of layout and blocks are metaphorical rather than physical, but the same concept of matching applies. Individuals within blocks (e.g., schools) are likely to be more similar than individuals in different blocks, thus assignment to treatments within blocks is a kind of assignment among individuals who are matched to one another by virtue of being in the same blocks.

Statistical Model and Notation

Suppose that the design has m blocks, and the n individuals in each block are randomly assigned to each of the treatment and the control condition. Such a design is called a balanced design because there are equal numbers of individuals assigned to each treatment within each block. Balanced designs are desirable because they are the most sensitive, that is they produce statistical tests that have the highest statistical power and the most precise estimate of treatment effects for a given sample size. Then, in the conventional notation of the analysis of variance, the model for Y_{ijk}, the outcome measurement of the k^{th} individual in the j^{th} block and the i^{th} treatment, is

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \varepsilon_{ijk},\ i = 1, 2; j = 1, \ldots, m; k = 1, \ldots, n, \tag{1}$$

where μ is the grand mean, α_i is the effect of the i^{th} treatment, β_j is the effect of the j^{th} block, $\alpha\beta_{ij}$ is the interaction effect of the i^{th} treatment and the j^{th} block, and ε_{ijk} is a residual. The residual ε_{ijk} is assumed to have a normal distribution with variance σ_ε^2 and the grand mean μ and the treatment effects α_i are taken to be fixed, but unknown constants.

Let $\overline{Y}_{1j\bullet}$($j = 1, \ldots, m$) and $\overline{Y}_{2j\bullet}$ ($j = 1, \ldots, m$) be the means at the j^{ih} block in the treatment and control groups, respectively, let $\overline{Y}_{1\bullet\bullet}$ and $\overline{Y}_{2\bullet\bullet}$ be the overall (group) means in the treatment and control groups, respectively, and let $\overline{Y}_{\bullet\bullet\bullet}$ be the grand mean. This notation with dots in the subscripts may seem confusing, but it has a logic. It takes the values of all three subscripts to identify a particular observation, but one could average over all observations in treatment group 1 to obtain $\overline{Y}_{1\bullet\bullet}$ (here that average is denoted by dots in the second and third subscripts which identify the block and individual within block within treatment group to denote averaging over individuals and blocks). Alternatively, one could average over only the observations in treatment group 2 within block j to obtain $\overline{Y}_{2j\bullet}$ (here the dot appears only in the third subscript to denote averaging only over individuals).

How the Randomized Block Design Increases Precision

Precision of treatment effect estimates is best understood in terms of what counts as "error" or *imprecision*. Imprecision is all the variation within treatment groups. This is the "noise" that statistical tests compare to the signal of treatment effects to see if the treatment effects are big enough to be reliable (i.e., statistically significant). If we just assign individuals to treatments ignoring the blocks, variation across blocks is part of the imprecision. The randomized block design makes treatment-control comparisons between individuals in the same block (and then averages these block-specific treatment effects across blocks). Therefore, variation across blocks is not part of the imprecision in the randomized block design. Because there are typically differences between average scores across blocks, randomized block designs have smaller amounts of imprecision and yield more precise estimates of treatment effects than do designs that do not match within blocks.

Fixed Versus Random Block Effects

An important distinction in experimental design is whether to conceive effects as being fixed or random. This distinction has a variety of general ramifications but one that matters for the randomized block design is how *statistical* (as opposed to substantive) inference is used in the logic of inference from the study. Two different possibilities conform to the blocks fixed and blocks random distinction.

Blocks fixed. One possibility (the blocks fixed model) is that the statistical inference is to the treatment effect in the blocks that are actually in the experiment. In this case, we might say that the statistical inference is to the average treatment effect in these blocks. Inference to other blocks not in the experiment is an extra-statistical matter that depends on substantive issues, such as whether the target of generalization sufficiently similar to the experiment that generalization is reasonable. Statistical inference can assess the uncertainty in inference about whether there was a treatment effect in the blocks observed in the experiment but not the uncertainty associated with the similarity (or not) of application of treatment effects to blocks not in the experiment.

Blocks random. A more complex possibility (the blocks random model) is that the blocks in the experiment are conceived as a random sample from a population of blocks and that inference is intended to be to that population of blocks (most of which are unobserved). In this case, the statistical inference is about the treatment effect in that universe of blocks (e.g., to the average treatment effect in that population of blocks). This model is appealing because it invokes the sampling theory of generalization to explain why results of this experiment generalize to other blocks. However, the blocks in field experiments are seldom chosen by random sampling. Thus, researchers making inferences are faced with the question of "from what population of blocks could those in the sample have possibly been a random sample?" This is an extra-statistical mater that depends on substantive considerations much like those in the blocks fixed model.

Choosing between the blocks fixed and blocks random models. The blocks fixed model is simpler both conceptually and statistically. In many initial field trials undertaken as part of intervention development, the primary question is whether the intervention can produce an effect under favorable circumstances. This is in contrast with the efficacy and effectiveness questions of whether the intervention can produce effects in typical circumstances or whether it can be made to work at scale. Thus, the blocks fixed model often appropriate for initial field trials. For that reason, we focus the bulk of our attention on the blocks fixed model in this paper.

The blocks random model stretches statistical inference to accomplish a more demanding goal of assessing generalizability to a larger population of blocks rather than just inferring to the observed blocks. Consequently, designs with more blocks and larger sample sizes are typically required to achieve adequate sensitivity in the blocks random model. The blocks random model is often more appropriate when the inference goal is to understand what the intervention effect might be when it is scaled up to a much larger implementation (but attention to the sampling of blocks is important to support such inferences).

The Statistical Analysis of Randomized Block Designs

In any analysis of the randomized block design, it is important to recognize that it is possible to estimate the treatment effect within each block. For example, the treatment effect estimate for the j^{th} block is

$$\overline{Y}_{1j\bullet} - \overline{Y}_{2j\bullet}, j = 1, \ldots, m.$$

Regardless of whether blocks are fixed or random, the overall treatment effect estimate is the average of the block-specific treatment effect estimates

$$\frac{1}{m}\sum_{j=1}^{m}\left(\overline{Y}_{1j\bullet} - \overline{Y}_{2j\bullet}\right)$$

and if (as in our model) the design is balanced

$$\frac{1}{m}\sum_{j=1}^{m}\overline{Y}_{1j\bullet} - \overline{Y}_{2j\bullet} = \overline{Y}_{1\bullet\bullet} - \overline{Y}_{2\bullet\bullet}.$$

The statistical tests for treatment effects differ somewhat depending on whether blocks are fixed or random.

Blocks fixed. If blocks are fixed, then the test statistic used for testing treatment effects is $F = MSA/MSW$,
where MSA is the so-called treatment mean square defined by

$$MSA = SSA = mn\left(\bar{Y}_{1\bullet\bullet} - \bar{Y}_{2\bullet\bullet}\right)^2 \Big/ 2,$$

and MSW is the so-called within cells mean square defined by

$$MSW = SSW/(2mn-2m) = \sum_{i=1}^{2}\sum_{j=1}^{m}\sum_{k=1}^{n}\left(Y_{ijk} - \bar{Y}_{\bullet\bullet\bullet}\right)^2 \Big/ 2m(n-1)$$

(see, e.g., Kirk, 1995). When the null hypothesis of no treatment effects is true, the test statistic has the F-distribution with 1 degree of freedom in the numerator and $N - 2m$ degrees of freedom in the denominator. Thus we carry out the significance test by comparing the obtained value of F with the appropriate critical value of the F-distribution with 1 and $N - 2m$ degrees of freedom or by using the p-value computed by the analysis software.

The standard error (precision) of the treatment effect estimate is

$$SE_F = \sqrt{\frac{2MSE}{mn}}.$$

Blocks random. If blocks are random, then the test statistic used for testing treatment effects is

$F = MSA/MSAB$,

where MSA is the so-called treatment mean square defined above and $MSAB$ is the so-called block-treatment interaction mean square defined by

$$MSAB = SSAB/(m-1) = n\sum_{j=1}^{m}\left[\left(\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}\right) - \left(\bar{Y}_{1\bullet\bullet} - \bar{Y}_{2\bullet\bullet}\right)\right]^2 \Big/ 2(m-1)$$

(see, e.g., Kirk, 1995). When the null hypothesis of no treatment effects is true, the test statistic has the F-distribution with 1 degree of freedom in the numerator and $m - 1$ degrees of freedom in the denominator. Thus we carry out the significance test by comparing the obtained value of F with the appropriate critical value of the F-distribution with 1 and $m - 1$ degrees of freedom or by using the p-value computed by the analysis software.
The standard error (precision) of the treatment effect estimate in the blocks random model is

$$SE_R = \sqrt{\frac{2MSAB}{mn}}.$$

Comparing blocks fixed and blocks random analyses. The number of degrees of freedom in the denominator is smaller in the blocks random model than in the blocks fixed model [($m - 1$) versus ($N - 2m$)]. Because the critical values are an increasing function of denominator degrees of freedom, this means that the critical values are larger for the blocks random test than the blocks fixed test. Also, there is a factor on n in $MSAB$ that tends to make it larger than MSE, so that $MSAB$ is typically larger than MSW. Thus the blocks random test statistic tends to be smaller (because it has a larger denominator) but requires a larger value to be declared statistically significant (because the critical value is lager), so the blocks random test tends to be less sensitive than the blocks fixed test.

Note that the blocks random test statistic uses the same numerator (MSA) but a different denominator ($MSAB$ versus MSW) than the blocks fixed test statistic. Examining the formula for $MSAB$, notice that $MSAB$ is actually the $n/2$ times the sum of squared differences of the block specific treatment effects, the $\left(\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}\right)$ values from the average treatment effect $\left(\bar{Y}_{1\bullet\bullet} - \bar{Y}_{2\bullet\bullet}\right)$, divided by $m - 1$. That is, *MSAB* is $n/2$ times the variance of the $\left(\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}\right)$ values. Another way of writing the *F*-statistic is that it is the square of the one-sample *t*-statistic applied to the block-specific treatment effect estimates

$$t = \frac{\sqrt{m}\left(\frac{1}{m}\sum_{j=1}^{m}\left(\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}\right)\right)}{S_{\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}}},$$

where $S_{\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}}$ is the standard deviation of the block-specific treatment effect estimates.

Because SE_F and SE_R differ in that SE_F is computed using *MSW* and SE_R is computed using *MSAB,* and because *MSAB* tends to be larger than *MSW*, it follows that SE_R is typically larger than SE_F.

The fact that the blocks random model produces less sensitive tests and less precise estimates is not a paradox. These differences stem from the differences in the statistical inferences each model is trying to support. In the case of the blocks fixed model, inferences are about the specific blocks that are observed. This is an "easier" inference that the inferences to a larger population of blocks that the blocks random model is trying to accomplish.

Unbalanced designs. The statistical analysis methods we described above apply to balanced designs. The extension of these methods to unbalanced designs is rather straightforward in the blocks fixed model. Let n_{ij} refer to the sample size in the *i*th treatment group in the *j*th block. In that case, the test statistic is

$F = MST/MSE,$

where

$$MST = \frac{\tilde{n}}{m}\left(\sum_{j=1}^{m}\bar{Y}_{1j\bullet} - \bar{Y}_{2j\bullet}\right)^2,$$

$$MSE = \frac{\tilde{n}}{m^2}MSE,$$

and

$$\tilde{n} = \sum_{j=1}^{m}\frac{\left(n_{1j} + n_{2j}\right)}{n_{1j}n_{2j}}.$$

Here the test statistic has 1 degree of freedom in the numerator and $N - 2m$ degrees of freedom in the denominator. Thus we carry out the significance test by comparing the obtained value of *F* with the appropriate critical value of the *F*-distribution with 1 and $N - 2m$ degrees of freedom or by using the *p*-value computed by the analysis software.

The precision of the treatment effect estimate in the unbalanced blocks fixed analysis is

$$SE = \sqrt{\frac{\tilde{n}MSE}{m^2}}.$$

The analysis of unbalanced design if the blocks random case is more complex. The simplest and most flexible analyses are probably accomplished using hierarchical linear models (see, e.g., Raudenbush and Bryk, 2002).

Planning Randomized Block Designs with Fixed Block Effects

The object planning a research design is to ensure that if the data is collected according to the design, the analysis of that data will yield unambiguous results. To ensure results will be unambiguous, the design has to be sufficiently sensitive. There are three different, but complementary, ways to describe design sensitivity, all of which can be useful in planning designs.

One way to describe design sensitivity is statistical power. Statistical power is the chance that if there is a (true or underlying) intervention effect of a given size, the statistical analysis is likely to detect it. Statistical power analysis starts with the size of an effect that is deemed important enough to detect, and a design (e.g., a sample size configuration), and describes the probability that an effect of the given size would result in a statistically significant observed effect. Computation of statistical power can be accomplished by suing tables of power values, but is more easily accomplished using specialized software

Another way to describe design sensitivity is the use of minimal detectable effect size. The minimal detectable effect size is the smallest effect size for which the design has a given statistical power. Using this approach, one starts with design (a sample size configuration) and a desired level of statistical power and describes the smallest effect size for which the design has the specified level of statistical power. Approximate computations of minimum detectable effect size can be done using tables, but exact computations of require specialized software.

A third way to describe design sensitivity is precision. Precision is the standard error of the estimated treatment effect, which determines the width of the confidence interval for the treatment effect. Using precision analysis, one starts with the design (a sample size configuration) and computes the precision that would be expected.

General Principles for Planning Randomized Block Designs with Fixed Block Effects

Planning a randomized block design with fixed block effects to evaluate an educational intervention primarily involves choosing a design that will have adequate sensitivity (high statistical power and precision, or small minimal detectable effect size). In principle, design sensitivity depends on the significance level chosen, the effect size expected, m, and n. Although significance level is in principle an arbitrary choice of the investigator, the 0.05 or 5% significance level has been so reified by convention that it is, for all practical purposes, not a choice but the de facto standard. The effect size expected is also not completely a choice but is dictated by the prior experience of the investigator about what size effect is large enough to be important or is the effect size to be expected from the intervention being evaluated. Thus, planning the design amounts to deciding on how many blocks or sites (which are typically schools) and how many subjects (usually students) to assign to each treatment within each block. That is, planning the design involves choosing m and n to achieve the desired design sensitivity.

The computation of either statistical power or minimum detectable effect size can be accomplished by using statistical tables that are widely available (see, e.g., Hedges and Rhoads, 2009) or by the use of statistical software that can compute noncentral F-distribution function (see the Appendix). It is probably easiest to use software that is designed to do the computations directly (e.g., **Website reference**). While exact power computations using software are recommended, we offer a few general principles that are helpful in the sections that follow and illustrate them with computations of design sensitivity.

Sensitivity Depends Mostly on δ, and $N = 2mn$, Very Little on m or n or Fixed N

The most sensitive design given a fixed total sample size N is a balanced design in which the same number of individuals are assigned to each treatment within each block. Exact design sensitivity depends on m, n, and δ, but in balanced designs, sensitivity essentially depends only on the total sample size N in balanced designs. Table 1 shows the statistical power, minimum detectable effect size, and standard error of the estimated treatment effect for a fixed total sample size of $N = 400$ and effect size values of $\delta = 0.20$, 0.25, and 0.30. The table shows that the minimum detectable effect size and standard error of the treatment effect estimate are the same (to two decimal places) whether $m = 2$ and $n =100$ or $m = 100$ and $n = 2$ (or any values in between where $2mn = 400$). For a given effect size δ, values of statistical power differ only in the third decimal place.

Insert Table 1 About here

There is one other pattern in the table that is worth noting. Both the minimum detectable effect size and the standard error of the estimated treatment effect are the same for all three values of δ. This is because neither depends on δ. For example, the minimum detectable effect size describes the smallest effect the design could detect (with, e.g., 80% power) and does not presuppose any specific effect size. Similarly, the calculation

The reason sensitivity depends primarily on N and not m or n individually, is that statistical power depends on the so-called noncentrality parameter

$$\lambda = \delta\sqrt{\frac{N}{4}}$$

and, to a lesser extent, on the degrees of freedom of the test $f = N - 2m$ (see the Appendix). Because the noncentrality parameter is constant for fixed N, only the degrees of freedom change with m and n. When N is large enough, the degrees of freedom remain relatively large regardless of the size of m and have relatively little effect on power.

The fact that power depends mainly on total sample size means that the researcher is free to use any design that has sufficient sample size to obtain the power that they require. There is a convention in social sciences that power of 80% or better is desirable. Thus, the researcher may choose a design that produces 80% power by finding the total sample size N that leads to 80% power for the effect size they wish to detect.

Equivalently, they could find a total sample size N that has a minimal detectable effect size (at 80% power) as small or smaller than the effect size they wish to detect. Figure 1 gives the minimal detectable effect size as a function of total sample size N (for $m = 10$ blocks, but results would be virtually identical for other values of m). One important thing this figure reveals is that adequate power can be achieved with a balanced design of 5 to 10 blocks (schools) so long as there is an adequate total sample size. Specifically to obtain 80% power to detect effect sizes of $\delta = 0.2$, a total sample size of 760 is needed, to detect an effect size of $\delta = 0.25$, a total sample size of $N = 500$ is needed, and to detect an effect size of $\delta = 0.30$ a total sample size of $N = 340$ is needed.

Insert Figure 1 About Here

Once the total sample size is determined, there is the choice of how many blocks and how many individuals to assign to each treatment within each block (in other words, the choice of m and n so that $2mn = N$). Because this choice does not affect design sensitivity, it must be made on other considerations. An obvious consideration is practical: It is usually more difficult to carry out a study in a larger number of schools (if schools are blocks) than a smaller number. There is however another consideration: Generalizibilty (sometimes called external validity). Because this is an extra-statistical consideration when block effects are fixed, it cannot be addressed technically, but it is none the less important. The larger the number of schools, the greater the presumed generalizability of the results. The choice of a design must reconcile these two opposing considerations.

Using Covariates Can Increase Sensitivity and reduce Sample Size Requirements

Covariates are any variables whose value could not have been influenced by the treatment assignment. The most common covariates are baseline data (such as pretests) collected before the treatment begins. If baseline data on individuals is available, it can be used to increase the sensitivity of the design if the analysis uses statistical adjustment via regression or the analysis of covariance. The use of a covariate (or multiple covariates) essentially has the same impact as increasing the effect size. In particular, if the effect size (with no covariates used) is δ and correlation (or multiple correlation if there is more than one covariate) between the covariate(s) and outcome is ρ, then the design using covariates is roughly equivalent to increasing the effect size from δ to $\delta\big/\sqrt{1-\rho^2}$. This is generally an increase because $0 \leq |\rho| \leq 1$, which implies that $\delta\big/\sqrt{1-\rho^2} \geq \delta$.

Figure 2 gives the minimal detectable effect size as a function of total sample size N (for m = 10 blocks, but results would be virtually identical for other values of m) for four values of ρ. The solid line reflects no covariates (ρ = 0), the dashed line reflects ρ = 0.4, the dotted line reflects ρ = 0.6, and the final line (which is dotted and dashed) reflects ρ = 0.7. The dashed line is somewhat lower than the solid line reflecting that the minimum detected effect size is somewhat smaller when using a covariate with a correlation of ρ = 0.4 with the outcome. For example, when ρ = 0.4, a sample size of only N = 320 is needed to detect and effect of δ = 0.30, but with no covariates a sample size of N = 340 is needed. The impact of a covariate with a correlation of ρ = 0.6 is somewhat larger. For example, a sample size of only N = 400 inC <- s needed to detect an effect of δ = 0.2 using a covariate of ρ = 0.6, but with no covariates, a sample size of N = 500 is needed.

Insert Figure 2 About Here

While covariates add complexity to the analysis, they can substantially increase design sensitivity and reduce sample size requirements. Pretests of measures similar to the outcome are often the most effective covariates (the ones with the largest correlations with the outcome). Note that cognitive tests tend to be rather highly correlated with one another, so it is not essential that the covariate measure exactly the same construct as the outcome to be useful. For example, reading achievement tests may be useful covariates, even in studies using math or science achievement as the outcome, outcome. Other variables measured at baseline (such as race or social class measures are typically poorer covariates (see Hedberg and Hedges, 2007).

Unequal Allocation to Treatments Can Have Advantages if it Enables Larger *N*

For a fixed total sample size N, a balanced design (equal allocation of individuals to treatment and control in each block) is the most sensitive (with or without covariates). However, there are situations in which the sample size is limited by the number of individuals who can be treated. For example, there may only be enough field staff to work with a limited number of treatment teachers or there may be a limited number of computers or other technology necessary to use the intervention. In that case, unequal allocation of individuals to treatment groups (more individuals randomized to control and fewer to treatment) can increase the total sample size and can increase design sensitivity.

Suppose that an unbalanced design assigns n^T individuals to treatment in each block and n^C individuals to control in each block so that the proportion $\pi = n^T/(n^T + n^C)$ of individuals in each block are assigned to treatment and a total of $N^T = N\pi$ individuals are assigned to treatment overall. In a balanced design the total sample size would be $N = n^T/0.5 = 2n^T$, but in an unbalanced design, because $\pi < 0.5$, the total sample size N would be greater than $2n^T$. In fact, the ratio of the total sample size of a design with $\pi = N^T/N$ to that of the balance design would be $1/2\pi$. Thus if $\pi = 0.5$ (the design is balanced) the ratio is unity, as expected ,but if $\pi = 0.4$, the total sample size of the design is 25% larger, if $\pi = 0.3$ it is 67% larger, and if $\pi = 0.2$ it is 150% larger than that of a balanced design with the same number of *treated* individuals. This increased total sample size increases design sensitivity.

While imbalance of the design does not literally change the effect size, it has an effect on sensitivity that is equivalent of changing the effect size from δ to $\delta\sqrt{4\pi\left(1-\pi\right)}$. Because

$$\delta\sqrt{4\pi\left(1-\pi\right)} \le \delta$$

and only equal to δ if $\pi = 0.5$, imbalance has a tendency to increase sensitivity because of the increased total sample size, but a simultaneous tendency to decrease sensitivity because it does the equivalent of decreasing the effect size. However, unless π is very much smaller than 0.5, the positive impact of imbalance on total sample size is bigger than the negative impact on effect size and the unbalanced design is more sensitive with the same number of treated individuals.

Figure 3 shows the minimum detectable effect size as a function of the total number of individuals assigned to treatment for balanced designs ($\pi = 0.5$) and unbalanced designs with various degrees of imbalance ($\pi < 0.5$). For example, if $N^T = 250$ individuals are to be treated, a balanced design would have a minimum detectable effect size of $\delta = 0.25$, but that could be reduced to $\delta = 0.21$ if we assigned $\pi = 0.3$ (30%) of the total individuals to treatment. This would mean assigning $N^T = 250$ individuals to treatment and $N^C = 583$ for a total sample size of $N = 833$. We could lower the minimum detectable effect size to $\delta = 0.20$ by using an even more imbalanced design and assigning $\pi = 0.2$ (20%) of the total individuals to treatment. In this case $N^T = 250$ individuals would still be assigned to treatment, but $N^C = 1{,}000$ individuals would have to be assigned to the control group, for a total sample size of $N = 1{,}250$.

Insert Figure 3 About Here

While greater sensitivity (for a fixed number of treated individuals) can be achieved with highly imbalanced designs, such designs necessarily involve larger sample sizes than balanced designs achieving the same sensitivity. For example, a balanced design with no covariates and a total sample size of $N = 760$ could achieve the same sensitivity ($MDES = 0.21$) as the unbalanced design with a total sample size of $N = 1{,}250$. The balanced design would need to treat 110 *more* individuals (for a total of $N^T = 380$), but would need 620 *fewer* individuals in the control group ($N^C = 380$ instead of 1,000). If the capacity to treat individuals is limited by cost or other considerations and the capacity of the control group is not, the unbalanced design might be a good design choice.

There is not always unlimited capacity to include individuals in the control group. If, for example, instructional or other conditions in the control group are monitored as part of the experiment, or if the cost of assessment of control individuals is high (e.g., if individually administered assessments are used), there can be significant costs involved with the control group. In such cases the financial and practical cost of obtaining the required sample size by adding individuals to the control group will need to be balanced by the costs of adding individuals to the treatment group will have to be balanced against one another.

While small degrees of imbalance are unlikely to be apparent, there may be political considerations in using highly unbalanced designs. For example, if the blocks are schools, it may be difficult for school personnel to justify participating in an experiment in which very few students in the school actually receive the treatment. Depending on the details of the treatment, there may also be practical difficulties in implementing a treatment if too few individuals are assigned to receive it at each site.

On the other hand, there may be situations that are well suited to a highly imbalanced design. For example, if the treatment seems highly desirable and can be individually administered (e.g., an expensive stand-alone technology intervention), a design offering a lottery for a very few lucky students to receive the treatment (with an understanding that most would not be able receive it) might be a very attractive design if the outcome was easy to measure (e.g., from data already collected by the schools).

Planning Randomized Block Designs with Random Block Effects

It is much more complex to plan randomized block designs with random block effects than to plan those with fixed block effects. Sensitivity in these designs is not primarily a function of total sample size and effect size, but depends importantly on both the number of blocks and the number of individuals within the blocks. Sensitivity also depends on the degree of treatment effect heterogeneity across blocks, something that is difficult to know in advance. Because of these difficulties, a complete treatment of planning designs with random block effects is beyond the scope of this paper. However, some generalizations are possible.

In general, randomized block designs with random block effects will require larger sample sizes than designs with fixed block effects to obtain the same sensitivity to detect a given effect size δ. Designs with random block effects will usually require a minimum number of blocks (e.g., more than 5 – 10 blocks) to achieve adequate sensitivity to detect all but very large effects. If a great deal of treatment effect heterogeneity is expected, designs with random block effects will require large numbers of blocks to obtain high sensitivity.

This brief discussion is not intended to be an argument that designs with random block effects have few virtues. Recall the discussion of the logic of generalization from designs with random block effects. These designs are intended to incorporate into their statistical inference, the uncertainty associated with generalization to a population of other blocks (most of which are not observed). The penalty for that more expansive role for statistical inference is greater complexity and more extensive sample size requirements.

Examples

Conclusions

Randomized block designs with fixed block effects can powerful research designs that are scientifically rigorous, but are often quite feasible and straightforward to plan and to use in evaluation designs. They are often implemented with schools taking the role of blocks, and designs with as few as 5 – 10 blocks can be sensitive enough to detect modest effects of treatments. Sensitivity can be increased by the use of covariates like pretests, but this modestly increases the complexity of the statistical analyses. While balanced designs provide the greatest sensitivity for a fixed total sample size, unbalanced designs can achieve greater sensitivity in cases where the treatment can only be administered to a limited number of individuals, but a larger number of control individuals can be handled.

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Appendix

Power computations for this design are based on the sampling distribution of the *F*-statistic given in the text. When there is no block treatment interaction and $\delta^2 \neq 0$, then F has the noncentral *F*-distribution with $2m(n-1)$ degrees of freedom and noncentrality parameter

$$\lambda = \frac{mn\delta^2}{2}$$

(see, e.g., Searle, 1971). Therefore, the statistical power of the level α test of the null hypothesis H_0: $\delta^2 = 0$ is

$$G[c_\alpha | \lambda, 2m(n-1)]$$

where c_α is the 100$(1-\alpha)$ percentile of the central *F*-distribution with $2m(n-1)$ degrees of freedom and $G[x \mid \lambda, \nu]$ is the cumulative distribution function of the noncentral *F*-distribution with ν degrees of freedom and noncentrality parameter λ.

The approximate minimal detectable effect size can be obtained using a formulas in Bloom (1995) or by iteratively computing power with successively larger values of the effect size until the appropriate power is obtained.

Figure 1
Minimum detectable effect size as a function of total sample size *N* (with *m* = 10)

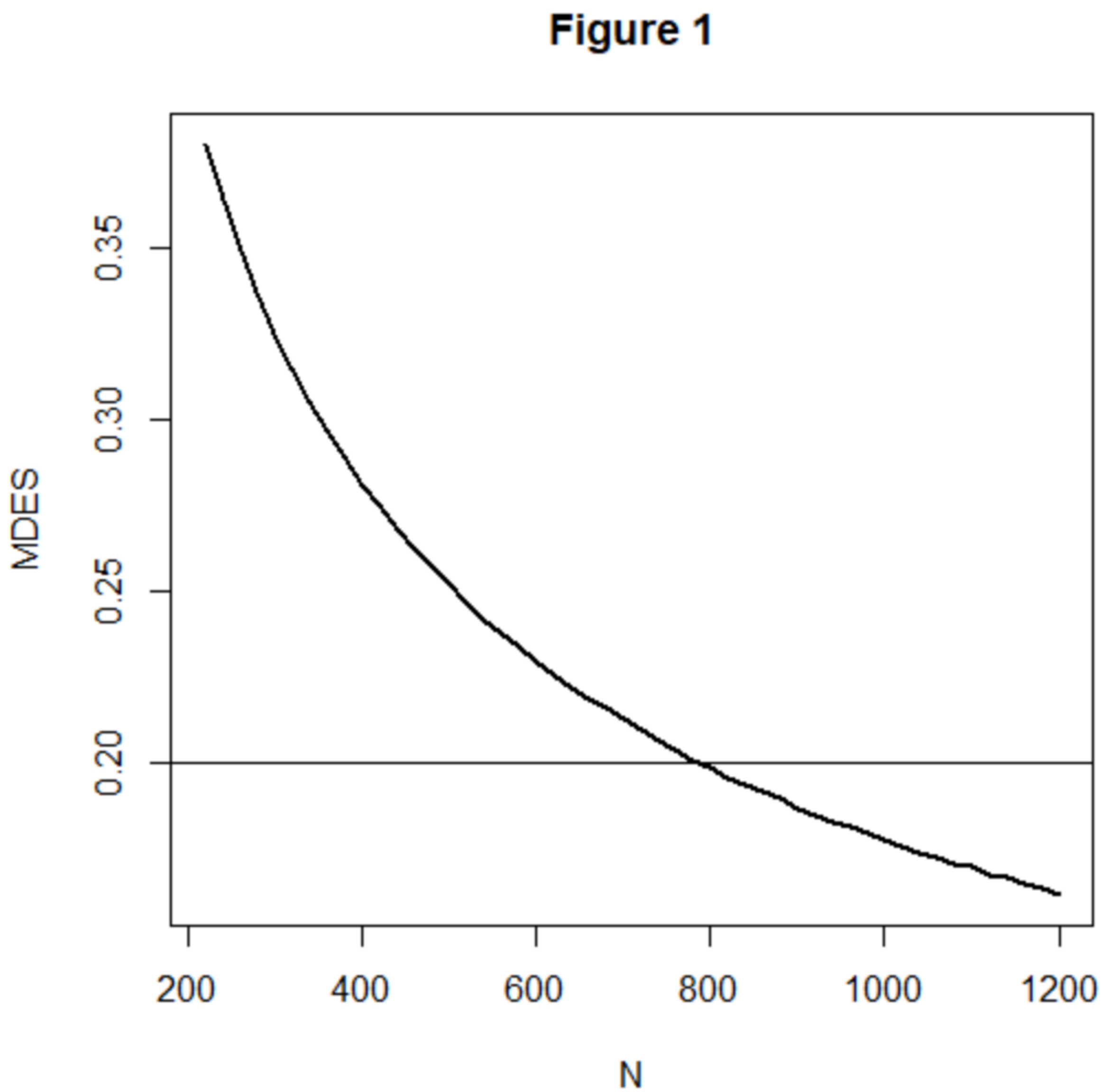


Figure 2
Minimum detectable effect size as a function of total sample size N (with m = 10), for ρ = 0 (no covariates, solid line), ρ = 0.4 (dashed line), ρ = 0.6 (dotted line), and ρ = 0.7 (dashed and dotted line).

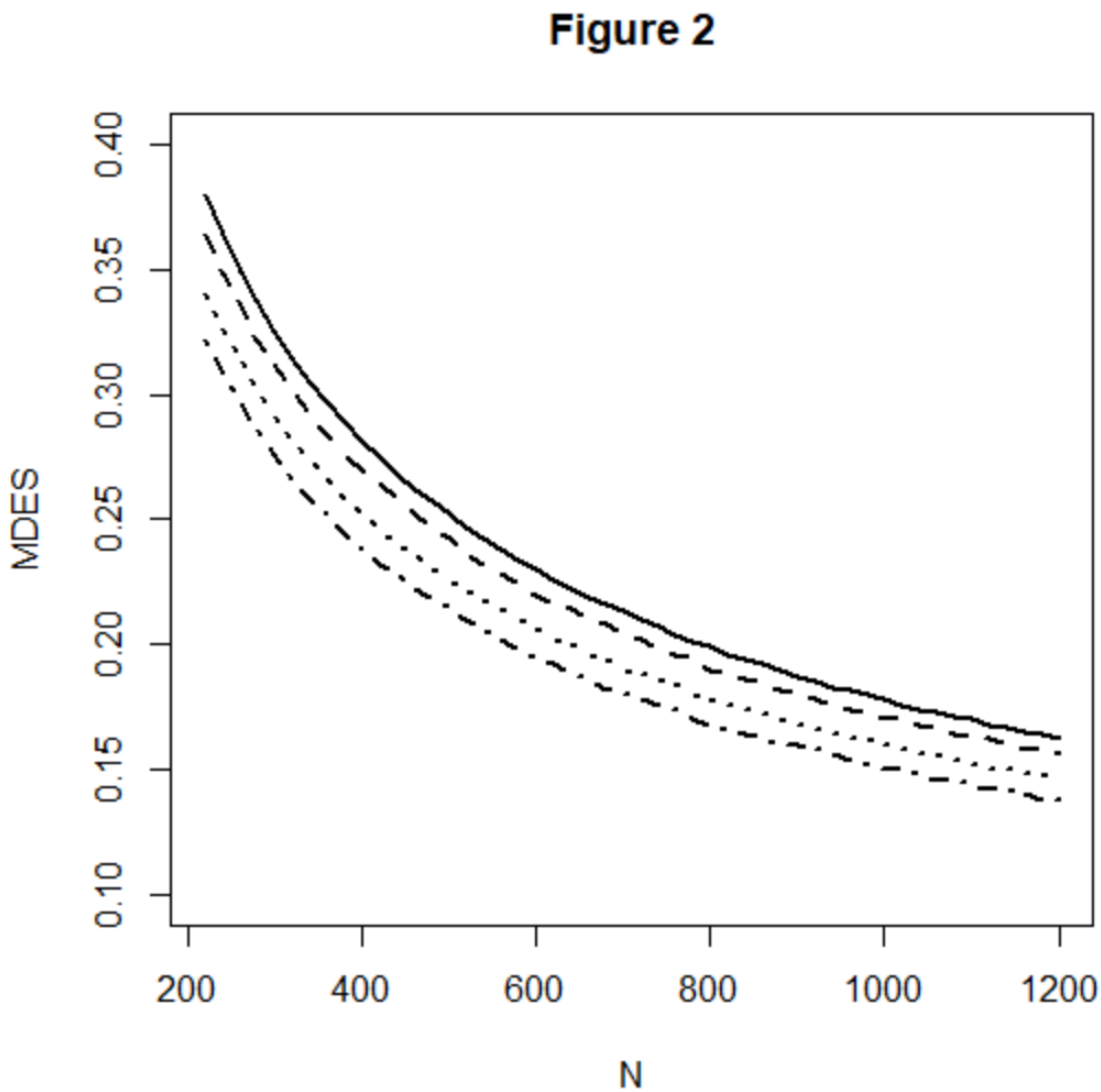
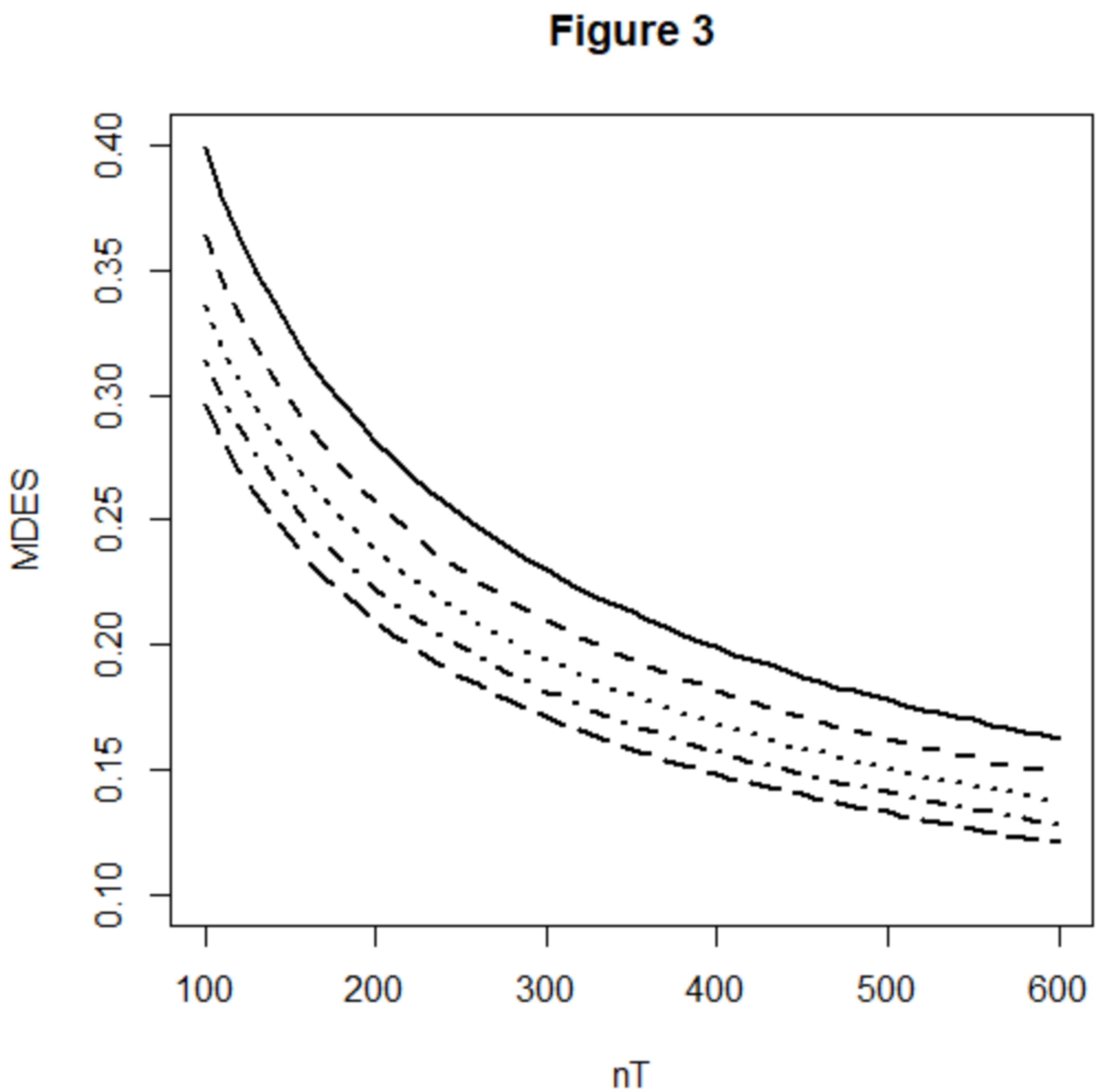


Figure 3
Minimum detectable effect size as a function of treatment group sample size n^T, when the treatment group is a proportion π =0.5 (solid line), $\pi = 0.4$ (dashed line), $\pi = 0.3$ (dotted line), $\pi = 0.2$ (dashed and dotted line), and $\pi = 0.1$ (long dashed line) of the total sample.



Chapter 7
Randomized Block Designs Assigning Clusters to Treatment

The Randomized Block Design Assigning Clusters to Treatments

The hierarchical or cluster randomized design (which assigns all individuals within a block or cluster to the same treatment) and the randomized block design (which assigns some individuals to each treatment within each block or cluster) are two basic experimental designs from which other designs can be constructed. One important elaboration of these two basic designs adds a nested factor with each block by treatment combination of the randomized block design. We will call the levels of the nested factor “clusters” for simplicity (an alternative would be to call them “subblocks”). Thus, the same treatment is assigned to all of the individuals within each cluster and each block contains clusters that are assigned to each treatment. In the terminology of experimental design, blocks are crossed with treatments and clusters are nested within block by treatment combinations (see, e.g., Kirk, 1995). This design is sometimes called a multi-site cluster randomized trials, but this terminology is somewhat ambiguous.

This design arises, for example, when schools are the blocks, classrooms within schools are the clusters, and entire classrooms (all the individuals within a classroom) are assigned to the same treatment. One can conceive the sampling of individuals as either a three-stage cluster sample (if blocks are considered to have random effects) or a stratified two-stage cluster sample (if the blocks are considered to have fixed effects). If blocks are considered to have random effects, then the sampling of blocks (e.g., schools) is considered to be the first stage of sampling, but if blocks are considered to have fixed effects, the blocks represent strata of a population made up entirely of the blocks in the experiment.

In either case, clusters (e.g., classrooms) are considered to have random effects. This idea might seem counterintuitive, particularly if all of the classrooms within a school are actually assigned to one of the treatments. The logic that classes have random effects in this design is similar to that for arguing that individuals have random effects, even though we may include all of the individuals in the school in the experiment. We wish to do statistical inference about the school, not just the *particular students* that happen to be in the school on this occasion. In making this inference, we imagine an inference across many samples of students that might have been sampled into the school. Similarly, in making an inference about the school, we imagine not just this particular set of classes (which are of course made up of students), but many classes that might have been made up (sampled) to collectively make up the school.

The question of whether blocks (schools) have fixed or random effects depends on whether the statistical inference is about this particular set of schools or a larger population of schools from which the schools in the experiment are a (random) sample. Considering the schools to have fixed effects seems particularly wee suited to the inference goal of determining whether a treatment *can* produce an effect, which is often the goal of initial field trials in development research. Considering the schools to have random effects seems particularly well suited to the goal of determining whether the treatment can produce and effect in a larger universe of schools than where it was explicitly tested, which is often the goal of efficacy or effectiveness studies.

Layout of the Design

The term "block" (sometimes called plot) in experimental design comes from the origin of experimental designs in agriculture, where field experiments were literally conduced in agricultural fields and blocks or plots were literally plots of ground within those fields. The layout of the design is a diagram of how the field might be divided into plots within those fields. Figure 1 illustrates the layout of a randomized block design, with shaded areas corresponding to treated portion of the blocks and unshaded portions of blocks corresponding to control portions of blocks. Note that the spaces between the blocks are intended to convey that the blocks (plots of land) might be physically distant from one another to ensure independence.

Insert Figure 1 About Here

In agriculture, there are variations in outcomes due to micro-variations in characteristics such as soil fertility, drainage, and temperature. These characteristics are related physical proximity and the physical proximity of the points where observations are taken within blocks assures that observations within blocks are likely to be more similar than they are to observations from different blocks. Thus, observations within blocks are matched to one another and the observations within clusters are matched to one another at an even finer level.

In an educational field study, the ideas of layout and blocks are metaphorical rather than physical, but the same concept of matching applies. Individuals within blocks (e.g., schools) are likely to be more similar than individuals in different blocks and individuals within the same classes are likely to be even more like one another than individuals who are just in the same school (but different classes). Thus assignment of clusters to treatments within blocks is a kind of assignment among individuals who are matched to one another by virtue of being in the same school, but assignment of clusters (classes) to treatment means that individuals receiving different treatments in the same school are also different from one another because they are in different classrooms.

Statistical Model and Notation

Suppose that the design has m blocks, and that p clusters with n individuals each from each block are randomly assigned to each of the treatment and the control condition. Such a design is called a balanced design because there are equal numbers of clusters and individuals assigned to each treatment within each block. Balanced designs are desirable because they are the most sensitive, that is they produce statistical tests that have the highest statistical power and the most precise estimate of treatment effects for a given total sample size. Then, in the conventional notation of the analysis of variance, the model for Y_{ijkl}, the outcome measurement of the l^{th} individual in the k^{th} cluster in the j^{th} block and the i^{th} treatment, is

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + \alpha\beta_{ij(k)} + \alpha\gamma_{ik} + \varepsilon_{ijkl}, \quad i = 1, 2; j = 1, \ldots, m;$$
$$k = 1, \ldots, p; l = 1, \ldots, n, \tag{1}$$

where μ is the grand mean, α_i is the effect of the i^{th} treatment, β_j is the effect of the j^{th} cluster, $\alpha\beta_{ij(k)}$ is the interaction effect of the i^{th} treatment and the j^{th} cluster within the k^{th} block, $\alpha\gamma_{ik}$ is the interaction effect of the i^{th} treatment with the k^{th} block, and ε_{ijkl} is a residual. The residual ε_{ijkl} is assumed to have a normal distribution with variance σ_ε^2 and the grand mean μ and the treatment effects α_i are taken to be fixed, but unknown constants.

In this design, $\alpha\beta_{ij(k)}$ is the interaction effect of the i^{th} treatment and the j^{th} cluster within the k^{th} block, represents the variance of cluster means within treatments within blocks. It therefore has a random effect, which has variance σ_C^2. It is conventional to define the effect size as

$$\delta = (\alpha_1 - \alpha_2) \Big/ \sqrt{\sigma_\varepsilon^2 + \sigma_C^2} \,,$$

which corresponds to the standardized mean difference effect size used in cluster randomized trials (see, Hedges, 2007).

Let $\overline{Y}_{1jk\bullet}$($j$ = 1, …, p; k = 1, …, p) and $\overline{Y}_{1jk\bullet}$($j$ = 1, …, p; k = 1, …, p) be the means of the j^{th} treatment and control group clusters in the k^{th} block, respectively, $\overline{Y}_{1j\bullet\bullet}$($j$ = 1, …, m) and $\overline{Y}_{2j\bullet\bullet}$ (j = 1, …, m) be the means of the treatment and control groups in at the j^{jh} block, respectively, let $\overline{Y}_{1\bullet\bullet\bullet}$ and $\overline{Y}_{2\bullet\bullet\bullet}$ be the overall (group) means in the treatment and control groups, respectively, and let $\overline{Y}_{\bullet\bullet\bullet\bullet}$ be the grand mean. This notation with dots in the subscripts may seem confusing, but it has a logic. It takes the values of all four subscripts to identify a particular observation, but one could average over all observations in treatment group 1 to obtain $\overline{Y}_{1\bullet\bullet\bullet}$ (here that average is denoted by dots in the second, third, and fourth subscripts which identify the block, cluster, and and individual within block and treatment combination within treatment group to denote averaging over individuals and blocks).

How the Randomized Block Design Increases Precision

Precision of treatment effect estimates is best understood in terms of what counts as “error” or *imprecision*. Imprecision is all the variation within treatment groups. This is the “noise” that statistical tests compare to the signal of treatment effects to see if the treatment effects are big enough to be reliable (i.e., statistically significant). If we just assign individuals to treatments ignoring the blocks, variation across blocks is part of the imprecision. This design makes treatment-control comparisons between individuals in the same block (and then averages these block-specific treatment effects across blocks). Therefore, variation across blocks is not part of the imprecision in the randomized block design. Because there are typically differences between average scores across blocks, this designs have smaller amounts of imprecision and yield more precise estimates of treatment effects than do designs (such as cluster randomized designs assigning entire blocks to different treatments) that do not match within blocks.

On the other hand, this design does not make treatment-control comparisons within *clusters*, therefore differences between clusters (within treatment-block combinations) do contribute to the imprecision of treatment effect estimates. Another way of putting this is that the overall treatment effect estimate is the average of the treatment effect estimates computed within blocks. But the treatment effect estimate computed within a block is the difference between the means of treated and untreated individuals, respectively, within that block. Each of these means is the average of a clustered sample of p clusters of n individuals each.

Fixed Versus Random Block Effects

An important distinction in experimental design is whether to conceive effects as being fixed or random. This distinction has a variety of general ramifications but one that matters for the randomized block design is how *statistical* (as opposed to substantive) inference is used in the logic of inference from the study. Two different possibilities conform to the blocks fixed and blocks random distinction.

Blocks fixed. One possibility (the blocks fixed model) is that the statistical inference is to the treatment effect in the blocks that are actually in the experiment. In this case, we might say that the statistical inference is to the average treatment effect in these blocks. Inference to other blocks not in the experiment is an extra-statistical matter that depends on substantive issues, such as whether the target of generalization sufficiently similar to the experiment that generalization is reasonable. Statistical inference can assess the uncertainty in inference about whether there was a treatment effect in the blocks observed in the experiment but not the uncertainty associated with the similarity (or not) of application of treatment effects to blocks not in the experiment.

Blocks random. A more complex possibility (the blocks random model) is that the blocks in the experiment are conceived as a random sample from a population of blocks and that inference is intended to be to that population of blocks (most of which are unobserved). In this case, the statistical inference is about the treatment effect in that universe of blocks (e.g., to the average treatment effect in that population of blocks). This model is appealing because it invokes the sampling theory of generalization to explain why results of this experiment generalize to other blocks. However, the blocks in field experiments are seldom chosen by random sampling. Thus, researchers making inferences are faced with the question of "from what population of blocks could those in the sample have possibly been a random sample?" This is an extra-statistical mater that depends on substantive considerations much like those in the blocks fixed model.

Choosing between the blocks fixed and blocks random models. The blocks fixed model is simpler both conceptually and statistically. In many initial field trials undertaken as part of intervention development, the primary question is whether the intervention can produce an effect under favorable circumstances. This is in contrast with the efficacy and effectiveness questions of whether the intervention can produce effects in typical circumstances or whether it can be made to work at scale. Thus, the blocks fixed model often appropriate for initial field trials. For that reason, we focus the bulk of our attention on the blocks fixed model in this paper.

The blocks random model stretches statistical inference to accomplish a more demanding goal of assessing generalizability to a larger population of blocks rather than just inferring to the observed blocks. Consequently, designs with more blocks and larger sample sizes are typically required to achieve adequate sensitivity in the blocks random model. The blocks random model is often more appropriate when the inference goal is to understand what the intervention effect might be when it is scaled up to a much larger implementation (but attention to the sampling of blocks is important to support such inferences).

The Statistical Analysis of the Design

In the analysis of this design, it is important to recognize that it is possible to estimate the treatment effect within each block. For example, the treatment effect estimate for the j^{th} block is

$$\overline{Y}_{j\bullet\bullet} - \overline{Y}_{2j\bullet\bullet}, j = 1, \ldots, m.$$

Regardless of whether blocks are fixed or random, the overall treatment effect estimate is the average of the block-specific treatment effect estimates

$$\frac{1}{m}\sum_{j=1}^{m}\left(\overline{Y}_{1j\bullet\bullet} - \overline{Y}_{2j\bullet\bullet}\right)$$

and if (as in our model) the design is balanced

$$\frac{1}{m}\sum_{j=1}^{m}\overline{Y}_{1j\bullet\bullet} - \overline{Y}_{2j\bullet\bullet} = \overline{Y}_{1\bullet\bullet\bullet} - \overline{Y}_{2\bullet\bullet\bullet}.$$

The statistical tests for treatment effects differ somewhat depending on whether blocks are fixed or random.

Blocks fixed. If blocks are fixed, then the test statistic used for testing treatment effects is
F = *MSA/MCT*,
where *MSA* is the so-called treatment mean square defined by

$$MSA = SSA = mpn\left(\overline{Y}_{1\bullet\bullet} - \overline{Y}_{2\bullet\bullet}\right)^2 \Big/ 2,$$

and *MSCT* is the so-called cluster-treatment within blocks interaction mean square defined by

$$MSCT = SSCT \big/ 2m\left(p-1\right) = n\sum_{j=1}^{m}\sum_{k=1}^{p}\left(\overline{Y}_{1jk\bullet} - \overline{Y}_{1j\bullet\bullet}\right)^2 + \left(\overline{Y}_{2jk\bullet} - \overline{Y}_{2j\bullet\bullet}\right)^2 \Big/ 2m\left(p-1\right)$$

(see, e.g., Kirk, 1995). When the null hypothesis of no treatment effects is true, the test statistic has the *F*-distribution with 1 degree of freedom in the numerator and $2m(p-1)$ degrees of freedom in the denominator. Thus we carry out the significance test by comparing the obtained value of *F* with the appropriate critical value of the *F*-distribution with 1 and $2m(p-1)$ degrees of freedom or by using the *p*-value computed by the analysis software.

The standard error (precision) of the treatment effect estimate is

$$SE_F = \sqrt{\frac{2MSCT}{mn}}.$$

A technical note is that the *F*-test above and the standard error of the estimated treatment effect are exactly valid when there are no block-treatment interactions, that is when the treatment effect does not vary across blocks. When there are block treatment interactions, the test and standard error are slightly conservative. In other words, true *p*-values of this test are smaller than nominal and the true standard error of the estimated treatment effect is smaller than that computed from the formula above. Unless there are very large differences in average treatment effects across blocks, the impact of these differences ae rather small.

Blocks random. If blocks have random effects, then there is no natural test statistic in classical analysis of variance for testing treatment effects across blocks. It is clear that the test that is valid for fixed block effects is not valid (it would reject too often and have a higher actual significance level than the nominal significance level). However, as we have emphasized earlier, for initial field trials in development research, the model with fixed block effects is probably more appropriate anyway.

Planning Designs with Fixed Block Effects

The object planning a research design is to ensure that if the data is collected according to the design, the analysis of that data will yield unambiguous results. To ensure results will be unambiguous, the design must be sufficiently sensitive. There are three different, but complementary, ways to describe design sensitivity, all of which can be useful in planning designs.

One way to describe design sensitivity is statistical power. Statistical power is the chance that if there is a (true or underlying) intervention effect of a given size, the statistical analysis is likely to detect it. Statistical power analysis starts with the size of an effect that is deemed important enough to detect, and a design (e.g., a sample size configuration), and describes the probability that an effect of the given size would result in a statistically significant observed effect. Computation of statistical power can be accomplished by suing tables of power values, but is more easily accomplished using specialized software

Another way to describe design sensitivity is the use of minimal detectable effect size. The minimal detectable effect size is the smallest effect size for which the design has a given statistical power. Using this approach, one starts with design (a sample size configuration) and a desired level of statistical power and describes the smallest effect size for which the design has the specified level of statistical power. Approximate computations of minimum detectable effect size can be done using tables, but exact computations of require specialized software.

A third way to describe design sensitivity is precision. Precision is the standard error of the estimated treatment effect, which determines the width of the confidence interval for the treatment effect. Using precision analysis, one starts with the design (a sample size configuration) and computes the precision that would be expected.

Design Sensitivity as a Function of *m, p, n*, and *δ*

Design sensitivity minimum detectable effect size and precision depend on the significance level chosen, the effect size expected, the classroom-within-school intraclass correlation ρ, m, p, and n. Statistical power depends, in addition, on the effect size δ. Design sensitivity values for $p = 2$, $\rho = 0.05$ (typical values) a few m, n, and δ values are given in Table 1. The table has 6 horizontal panels, each panel has the same pattern of m values but a different value of n. The top 3 panels have $\delta = 0.2$ while the bottom 3 panels have $\delta = 0.4$. Examining values within each panel, see that power is a strongly increasing function of m and MDES and *SE* are decreasing functions of m. Comparing values within either the top 3 or the bottom 3 panels, see that power is an increasing function of n and MDES and *SE* are decreasing functions of n. The values in this table suggest that the effect of n on design sensitivity diminishes as n becomes larger. This is true: The impact of n on design sensitivity reaches a point of diminishing returns and design sensitivity tends to an asymptote as n becomes large. The details of these general patterns may be more apparent in Appendix tables A1, A2, and A3.

General Principles for Planning Randomized Block Designs with Fixed Block Effects

Planning a randomized block design with fixed block effects to evaluate an educational intervention primarily involves choosing a design that will have adequate sensitivity (high statistical power and precision, or small minimal detectable effect size). In principle, design sensitivity depends on the significance level chosen, the effect size expected, the classroom-within-school intraclass correlation ρ, m, p, and n. Although significance level is in principle an arbitrary choice of the investigator, the 0.05 or 5% significance level has been so reified by convention that it is, for all practical purposes, not a choice but the de facto standard. The effect size expected is also not completely a choice but is dictated by the prior experience of the investigator about what size effect is large enough to be important or is the effect size to be expected from the intervention being evaluated. The classroom-within-school intraclass correlation can be influenced by the investigator. Thus, planning the design amounts to minimize the intraclass correlation ρ and deciding on how many blocks or sites (which are typically schools), how many clusters p (which are typically classrooms) to assign to each treatment within each block, and how many subjects (usually students) n to choose to compose each cluster. That is, planning the design involves minimizing ρ and choosing m, p, and n to achieve the desired design sensitivity.

The computation of either statistical power or minimum detectable effect size can be accomplished by using statistical tables that are widely available (see, e.g., Hedges and Rhoads, 2009) or by the use of statistical software that can compute noncentral F-distribution function (see the Appendix). It is probably easiest to use software that is designed to do the computations directly (e.g., **Website reference**). However, caution is necessary in using software, particularly for complex designs like this one, because notation and terminology differ among software packages and the same terms (e.g., "effect size" do not mean the same things for all software packages (see, e.g., Spybrook, Hedges, and Borenstein, 2014). While exact power computations using software are recommended, we offer a few general principles that are helpful in the sections that follow and illustrate them with computations of design sensitivity.

The Classroom-Within-School Intraclass Correlation

Design sensitivity depends strongly on classroom-within-school intraclass correlation: The smaller the intraclass correlation, he higher the design sensitivity. The classroom-within-school intraclass correlation ρ is defined as the variance of the classroom means σ_C^2 divided by the total variation of outcome observations (Y_{ijkl} scores) within blocks (schools), that is

$$\rho = \sigma_C^2/(\sigma_C^2 + \sigma_\varepsilon^2).$$

If the classrooms within schools are simply sampled, then ρ is determined by the natural distribution of class means within schools. This distribution will be determined partly by the distribution of scores and by policies that determine how students are allocated into classrooms. For example, schools that track classrooms by academic achievement would be expected to have high intraclass correlations. Empirical evidence suggest that typical classroom level intraclass correlations across a range of schools are on the order of 0.1 to 0.2 and tend to be smaller in high socio-economic status schools than in lower SES schools (see, e.g., Nye, Konstantopoulos, and Hedges, 2004).

It is possible to influence ρ when designing an experiment by matching classes before random assignment to treatments. For example, if a pretest X is available, it may be possible to select classroom that have very similar average pretest scores to be the experimental sample. Alternatively, it is possible to compose the experimental sample by selecting a subset of individuals from each classroom whose mean pretest scores match closely across classrooms. Note that the matching is carried out within-schools, so that classrooms need to match only other classrooms in the same school. They need not match those in different schools.

It is generally not possible to make $\rho = 0$ because matching is carried out on *pretest* scores, but the intraclass correlation of interest is defined in terms of the between- and within-classroom variances of the *outcome* (posttest) scores. If the pretest-posttest correlation is $\rho_{Y\backslash X}$ then the variance of classroom mean scores on the outcome Y within clusters (schools) for a given pretest value X is

$$(1 - \rho_{Y\backslash X}^2)\sigma_\varepsilon^2/n.$$

Therefore, if all the class means matched exactly on the pretest X, the intraclass correlation would be

$$\rho = \frac{\left(1-\rho_{Y|X}^2\right)\sigma_\varepsilon^2/n}{\left(1-\rho_{Y|X}^2\right)\sigma_\varepsilon^2/n+\sigma_\varepsilon^2} = \frac{1-\rho_{Y|X}^2}{\left(1-\rho_{Y|X}^2\right)+n}.$$

For example, if $\rho_{Y\backslash X}^2 = 0.5$ (so that $\rho_{Y\backslash X} \approx 0.71$) and $n = 20$, then $\rho \approx 0.02$. It is important that the ρ value obtained after matching depends on n, so that is n is smaller, ρ becomes larger. For example, if $\rho_{Y\backslash X}^2 = 0.5$ but $n = 10$, then $\rho = 0.05$, and if $n = 5$, $\rho = 0.09$. Thus, the impact of the within cluster sample size n on ρ will need to be considered in selecting the appropriate n for the design.

If the outcome variable and pretest are cognitive tests like achievement tests, a pretest-posttest correlation of $\rho_{Y\backslash X} \approx 0.7$ is quite sensible, and hence it should be possible to obtain an intraclass correlation of $\rho = 0.05$. Empirical evidence suggests that background variables like race, gender, and crude social class indicators (like free or reduced-price lunch eligibility) will have correlations with outcomes and thus even perfect matching on such covariates will lead larger intraclass correlations (see Hedges and Hedberg, 2007). In other words, matching on a covariate that is a pretest is strongly recommended.

The Cluster Size *n* and Number of Classrooms per School *p* Have Natural Limits

Although p and n are under the control of the investigator in principle, they have natural limits in the setting where blocks are schools and classrooms are clusters. Classroom larger than 25 or 30 are unusual, not every student may be permitted to participate in a field experiment (and some students may be omitted from the experimental sample in order to match classrooms), so it would probably be unwise to plan an experiment with more than $n = 20$ students per classroom. Because matching between classrooms is most effective when n is as large as possible, it is unwise to plan an experiment with $n < 10$ students per classroom (because $n = 10$ corresponds to an intraclass correlation of $\rho = 0.05$, which is about the largest value which is wise).

Similarly, schools do not have an unlimited number of classrooms in each grade, so that schools with more than 6 – 8 classrooms per grade are unusual. Not every classroom will be suitable for (e.g., some may be tracked or be special purposes classrooms) or will be willing to participate in a field trial. Thus, it would probably be unwise to plan an experiment with more than $p = 3$ classes per school. Because p must be greater than 1 to ensure there are nonzero degrees of freedom for error (in other words to estimate an error term for the test statistic), this mean that $p = 2$ or $p = 3$ are the only practical choices for a balanced design, except in unusual circumstances of vary large schools. We will consider in a subsequent section the possibility of using an unbalanced design with a total of 3 classes per school (1 class allocated to one treatment group and 2 classes allocated to the other treatment group), so that p is effectively 1.5 (but the details of design sensitivity computations are slightly different in unbalanced designs).

The minimum detectable effect size is a decreasing function of both m and n, but depends more strongly on m than n. Figure 2 gives the minimum detectable effect size (MDES) when $\rho = 0.05$ and $p = 2$ as a function on m for $n = 10$, $n = 12$, $n = 15$, and $n = 20$. The figure shows that the MDES when $m = 5$ decreases from 0.53 (when $n = 10$) to 0.0.43 (when $n = 20$). It also shows that when $n = 15$, the MDES decreases from 0.47 (when $m = 5$) to 0.31 (when $m = 10$). The figure can be used to plan a design when $p = 2$ by identifying the combinations of m and n that give an MDES as small as required, which will assure that the design with that *m, n*, and p, Appendix tables A4 and A5 give some more detailed values of MDES as a function of m and n for $p = 2$ and $p = 3$.

Insert Figure 2 About Here

Figure 3 gives the minimum detectable effect size (MDES) when $\rho = 0.05$ and $p = 3$ as a function on m for $n = 10$, $n = 12$, $n = 15$, and $n = 20$. The figure shows that the MDES when $m = 5$ decreases from 0.53 (when $n = 10$) to 0.0.43 (when $n = 20$). It also shows that when $n = 15$, the

MDES decreases from 0.47 (when $m = 5$) to 0.31 (when $m = 10$). The figure can be used to plan a design when $p = 3$ by identifying the combinations of m and n that give an MDES as small as required, which will assure that the design with that m, n, and p,

Insert Figure 3 About Here

Optimal Sample Size Allocation

There is not a monotonic relationship between total sample size and indices of design sensitivity in this design as there is in simpler designs. Instead, larger total sample sizes do not always yield designs that are more sensitive. For example, a design with $p = 2$, $m = 2$, and $n = 20$ (a total sample size of 160) has an MDES of 0.83 while a design with $p = 2$, $m = 3$, and $n = 10$ (a smaller total sample size of 120) has a smaller MDES: 0.74.

There is a complex relationship between m, p, and n with design sensitivity. Moreover, examination of figures 1 and 2 suggests that several different combinations of m and n can yield the same MDES. For example, a minimum detectable effect size of 0.40 can be achieved with $p = 2$, $m = 4$, and $n = 20$ (a total sample size of 320) or $p = 2$, $m = 5$, and $n = 12$ (a total sample size of 240). Similarly, a minimum detectable effect size of 0.40 can also be achieved with $p = 3$, $m = 4$, and $n = 16$ (a total sample size of 384) or $p = 3$, $m = 5$, and $n = 11$ (a total sample size of 330). In general, a larger p will result in a smaller minimal detectable effect size, but the investigator is typically severely limited in choice of p because there must be at least $2p$ classrooms participating in the experiment from each school. Sometimes (when schools are large enough) $p \geq 3$ will be possible, but often $p = 2$ will be the only option. This still leaves the question of what combination of m and n should be chosen.

One way to choose a sample size configuration is to choose it based on the "cost" of adding clusters or blocks versus the cost of adding individuals. In other words, we should choose the allocation of students to classrooms and therefore the number of classrooms and schools in a way that yields the most sensitive design for a fixed cost (see, e.g., Raudenbush, 1997 or Hedges and Borenstein, 2014).

The most straightforward concept of concept of cost is the marginal monetary cost of adding a classroom or a student to the study. For example, each classroom added to a study might incur costs for teacher incentives, training of the teacher, teacher materials, staff traveling to classroom to set up equipment, collect tests or student work, etc. Similarly, each student added might incur costs for tests or scoring of tests, evaluation of student work, etc. There is a vaguer concept of cost that amounts to a subjective sense of the added practical difficulty of adding another classroom or student. With either concept of cost, the assessment of costs is only approximate, and any design implications based on them should also be understood as approximate. Generally, the addition of a new classroom is associated with a higher marginal cost than adding another student to a classroom that is already in the study.

Understanding the limitations of this kind of analysis, a conventional model for optimal design assumes that the total cost C is

$$C = 2mnpc_1 + 2mpc_2,$$

where c_1 is the cost of adding another student to the study and c_2 is the marginal cost of adding another classroom to the study. Under this model, the within-classroom sample size n_O that maximizes design sensitivity (precision of the estimated treatment effect) for a fixed cost (the optimal n) for a given cost is

$$n_O = \sqrt{\left(\frac{c_2}{c_1}\right)\left(\frac{1-\rho}{\rho}\right)},$$

where ρ is the classroom within school intraclass correlation. This expression implies that n_O is larger when the cost of adding a new classroom increases but n_O is smaller when the intraclass correlation ρ is smaller.

We argued earlier that matching to produce a small intraclass correlation is small provides substantial advantages for increasing net design sensitivity and that the cost of adding classrooms is likely to be higher (typically several to dozens of times higher) than that of adding individuals to the experiment. Thus, intraclass correlation values of $\rho = 0.02$ to 0.05 and cost ratios from 5 to 50 are often reasonable. Such intraclass correlation and cost ratios imply optimal allocations will be in the range of 10 to 50 students per classroom, with the highest numbers often not feasible in most school settings. A tabulation of optimal allocations for cost rations c_2/c_1 and intraclass correlations $\rho = 0.02$ to 0.05 is given in Table 4.

Insert Table 2 About Here

Unbalanced Designs May Be a Feasible Way to Increase Design Sensitivity

For a fixed total sample size, a balanced design (equal allocation of individuals to treatment and control in each block) is the most sensitive. However, there are situations in which the sample size is limited by the number of classes that are available within schools or by limitations in how many classes can be treated (e.g., because of limitations of personnel needed for training or implementation or of equipment). In that case, unequal allocation of classes to treatment groups (more classes randomized to treatment and fewer to control or vice versa) can increase the total sample size and can increase design sensitivity. For example, if only 3 classes per school can be expected to participate in the experiment, allocation of $p = 2$ classes to both treatment and control per school is impossible, but allocating only $p = 1$ classroom per treatment leaves no degrees of freedom for the error term, which would make the analysis impossible. Allocation of $p^T = 2$ and $p^C = 1$ classes per cluster utilizes all available classrooms per school and makes an analysis possible. Although it might seem counterintuitive, the design sensitivity is the same regardless of whether there are more treatment or more control classes (that is the design sensitivity is the same for $p^T = 2$ and $p^C = 1$ as it is for $p^T = 1$ and $p^C = 2$.

Suppose that an unbalanced design assigns p^T clusters (classes) to treatment in each block and p^C clusters (classes) to control in each block so that the proportion $\pi = p^T/(p^T + p^C)$ of clusters in each block are assigned to treatment and a total of $N^T = N\pi$ individuals are assigned to treatment overall. In a balanced design the total sample size would be $N = mnp^T/0.5 = 2mn^T$, but in an unbalanced design, because $\pi < 0.5$, the total sample size N would be greater than $2mn^T$. In fact, the ratio of the total sample size of a design with $\pi = N^T/N$ to that of the balance design would be $1/2\pi$. Thus if $\pi = 0.5$ (the design is balanced) the ratio is unity, as expected ,but if π = 1/3, the total sample size of the design is 50% larger, if $\pi = 1/4$ it is 100% larger than that of a balanced design with the same number of *treated* individuals. This increased total sample size increases design sensitivity.

While imbalance of the design does not literally change the effect size, it has an effect on sensitivity that is equivalent of changing the effect size from δ to $\delta\sqrt{4\pi(1-\pi)}$. Because

$$\delta\sqrt{4\pi(1-\pi)} \leq \delta$$

and is only equal to δ if $\pi = 0.5$, imbalance has a tendency to increase sensitivity because of the increased total sample size, but a simultaneous tendency to decrease sensitivity because it does the equivalent of decreasing the effect size. However, unless π is very much smaller than 0.5, the positive impact of imbalance on total sample size is bigger than the negative impact on effect size and the unbalanced design is more sensitive with the same number of *treated* individuals.

Table 3 shows the minimum detectable effect size as a function of the total number of the number of blocks m and the number of individuals per cluster n when there are $p^T = 2$ and $p^C = 1$ clusters per block (or vice versa) and $\rho = 0.05$. For example, if $\rho = 0.05$, $m = 5$, $n = 20$, an unbalanced design with $p^T = 2$ and $p^C = 1$ (two classes are assigned to treatment and one to control in each school, or vice versa) for a total sample size of $N = 300$, the minimum detectable effect size would be $\delta = 0.57$. For comparison, Table A2 shows that a design with $\rho = 0.05$, $m = 5$, $n = 15$, a balanced design with $p^T = p^C = 2$ (also total sample size of $N = 300$) has a minimum detectable effect size of $\delta = 0.47$. As expected, the unbalanced design is less sensitive than a balanced design with the same total sample size. Table A6 shows the minimum detectable effect size as a function of m and n when $\rho = 0.05$, for unbalanced designs with $p^T = 3$ and $p^C = 1$.

Insert Table 3 About Here

Statistical Adjustment for Covariates Can Substitute for Matching of Classes

Covariates are any variables whose value could not have been influenced by the treatment assignment. The most common covariates are baseline data (such as pretests) collected before the treatment begins. If baseline data on individuals is available, it can be used to increase the sensitivity of the design if the analysis uses statistical adjustment via regression or the analysis of covariance. In a design such as this, covariates primarily increase design sensitivity by reducing between-cluster (between-class) variation in outcome scores. The use of a covariate (or multiple covariates) essentially has the same impact as matching and there is no additional benefit to using covariates in the analysis if the classes have been matched on the covariate prior to assignment to treatment. However statistical adjustment with a pretest or other covariates could be considered as an alternative to matching.

Examples

Conclusions

References

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Appendix

The following tables provide additional information about minimal detectable effect sizes for planning designs.

Insert Appendix Tables A1 – A6 About Here

Power Computations

Power computations for this design are based on the sampling distribution of the *F*-statistic given in (). When there is no block treatment interaction and $\delta^2 \neq 0$, then *F* has the noncentral *F*-distribution with $2m(p-1)$ degrees of freedom and noncentrality parameter

$$\lambda = \frac{mp^T p^C n\delta^2}{\left(p^T + p^C\right)\left[1 + \left(n-1\right)\rho\right]}$$

(see, e.g., Searle, 1971). Therefore, the statistical power of the level α test of the null hypothesis H_0: $\delta^2 = 0$ is

$$G[c_\alpha | \lambda, 2m(p-1)]$$

where c_α is the $100(1-\alpha)$ percentile of the central *F*-distribution with $2m(p-1)$ degrees of freedom and $G[x \mid \lambda, v]$ is the cumulative distribution function of the noncentral *F*-distribution with v degrees of freedom and noncentrality parameter λ.

The approximate minimal detectable effect size can be obtained using a formulas in Bloom (1995) or by iteratively computing power with successively larger values of the effect size until the appropriate power is obtained.

Figure 2
Minimum detectable effect size as a function of m for $p = 2$, $\rho = 0.05$, and $n = 10$ (dashed line), $n = 12$ (dotted line), $n = 15$ (dashed and dotted line), and $n = 20$ (long dashed line)

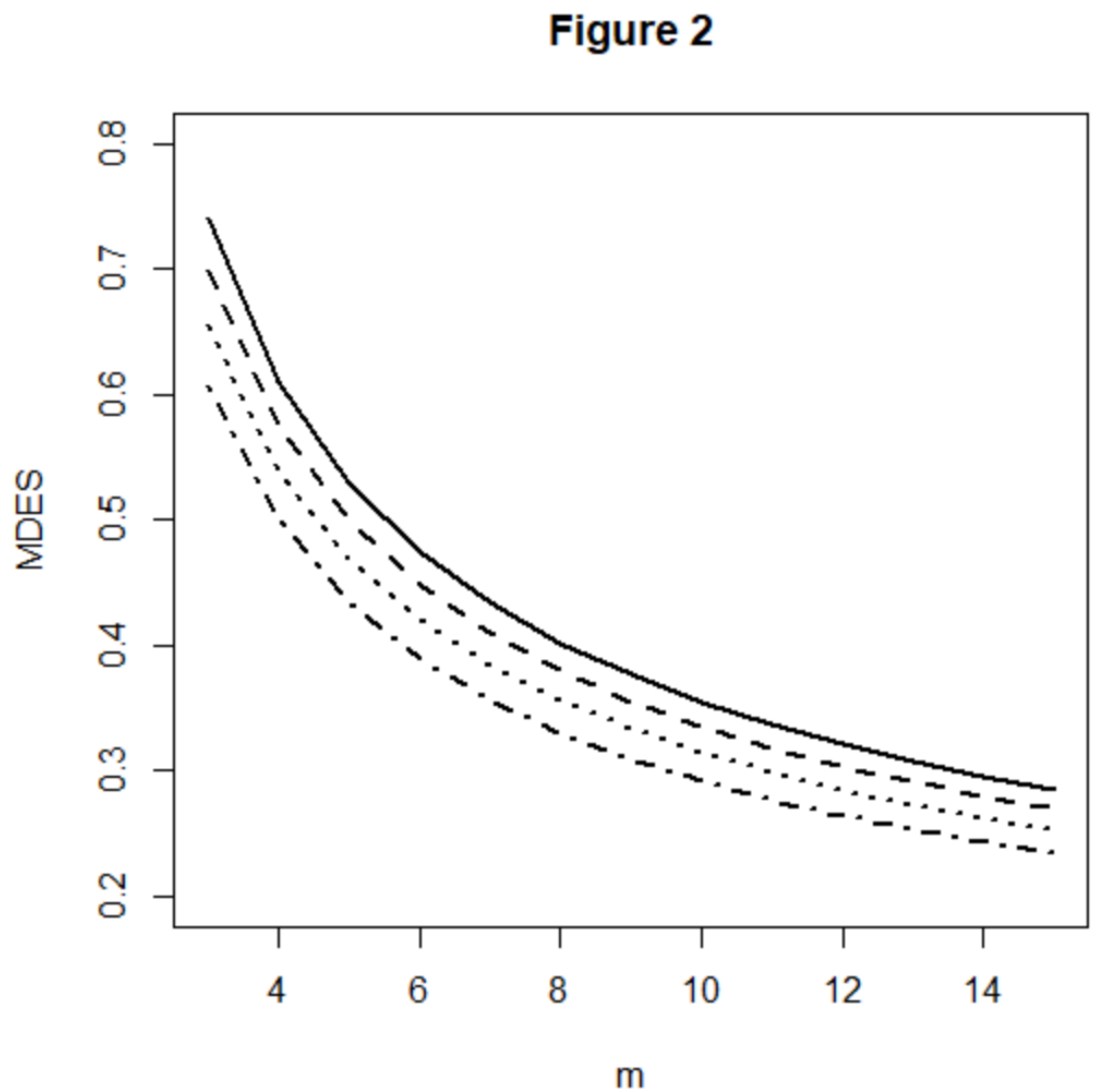
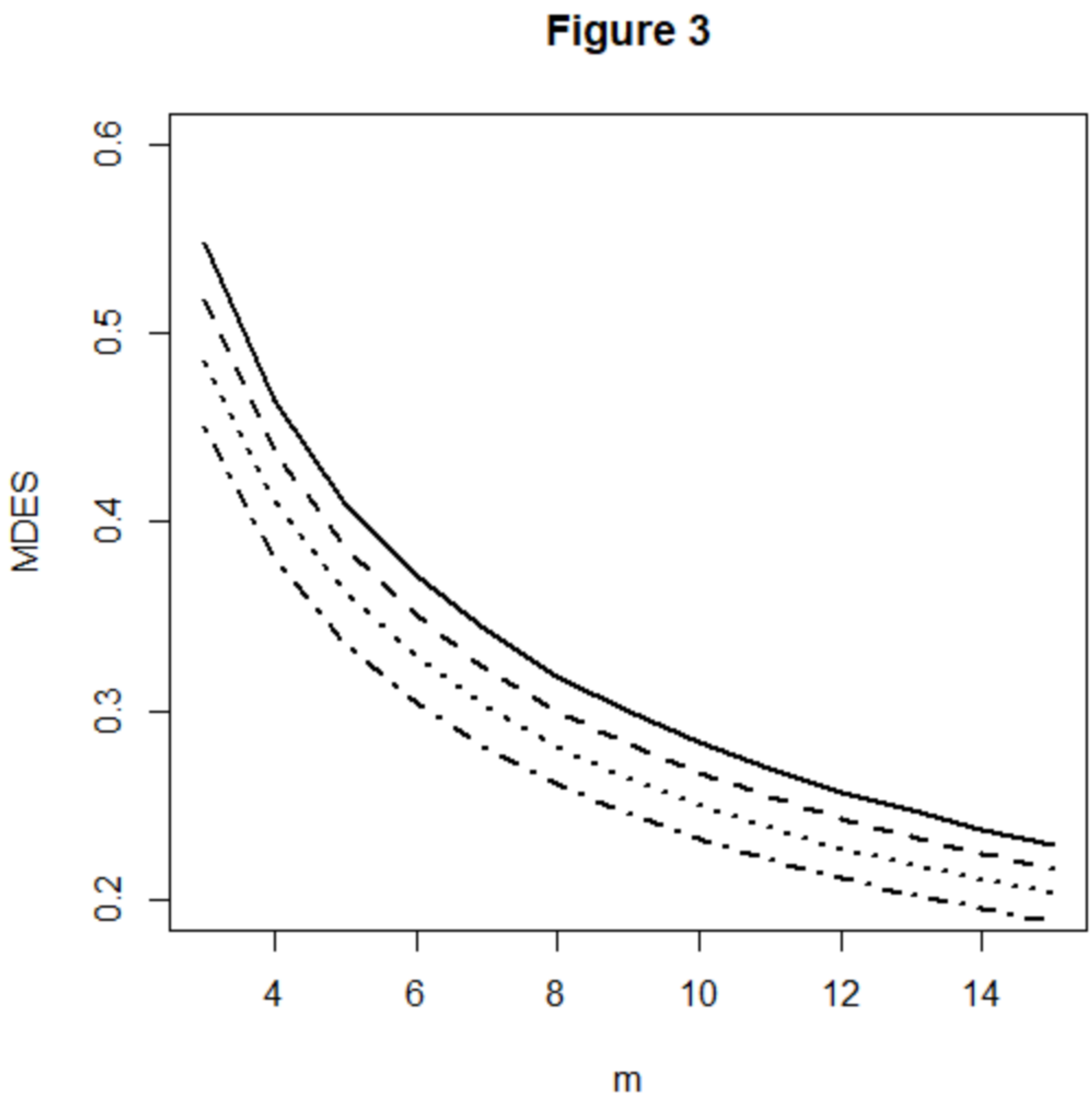


Figure 3
Minimum detectable effect size as a function of *m* for $p = 3$, $\rho = 0.05$, and $n = 10$ (dashed line), $n = 12$ (dotted line), $n = 15$ (dashed and dotted line), and $n = 20$ (long dashed line)



Chapter 8
Cluster Randomized Trials

The Cluster Randomized Design

The hierarchical or cluster randomized design (which assigns all individuals within a block or cluster to the same treatment). In the terminology of experimental design, blocks or clusters are nested within treatments (see, e.g., Kirk, 1995).

This design arises, for example, when schools are blocks and entire classrooms (all the individuals within a classroom) are assigned to the same treatment. One can conceive the sampling of individuals as a two-stage cluster sample. Clusters (e.g., schools) are considered to have random effects. This idea might seem counterintuitive, particularly if all of the schools within a district are actually part of the experiment. The logic is similar to that of considering students to have random effects, even though we sample all of the students in the school, namely that we wish to make a statistical inference about the school, not just the *particular students* that happen to be in the school on this occasion. In making this inference, we imagine an inference across many samples of students that might have been sampled into the school.

The fact that schools must be considered to have random effects in the cluster randomized design is a weakness in development studies where the inference goal is to determine whether the treatment can have an effect under favorable circumstances. Other designs (e.g., randomized block designs) can capitalize on their ability to consider schools as having fixed effects to obtain greater design sensitivity for the same sample size. Yet there are sometimes practical reasons that a cluster randomized design is the only alternative. For example, when the intervention cannot be administered to a subset of the school for theoretical reasons (e.g., whole school interventions), political reasons (e.g., a highly desirable intervention), or practical reasons.

Layout of the Design

The term “block” (sometimes called plot, rather than a cluster) in experimental design comes from the origin of experimental designs in agriculture, where field experiments were literally conduced in agricultural fields and blocks or plots were literally plots of ground within those fields. The layout of the design is a diagram of how the field might be divided into plots within those fields. Figure 1 illustrates the layout of a cluster randomized design, with shaded areas corresponding to treated blocks and unshaded areas corresponding to control blocks. Note that the spaces between the blocks are intended to convey that the blocks (plots of land) might be physically distant from one another to ensure independence.

Insert Figure 1 About Here

In agriculture, there are variations in outcomes due to micro-variations in characteristics such as soil fertility, drainage, and temperature. These characteristics are related physical proximity and the physical proximity of the points where observations are taken within blocks assures that observations within blocks are likely to be more similar than they are to observations from different blocks. Thus, observations within blocks are matched to one another.

In an educational field study, the ideas of layout and blocks are metaphorical rather than physical, but the same concept of matching applies. Individuals within blocks (e.g., schools) are likely to be more similar than individuals in different blocks. Thus assignment of clusters to treatments within blocks is a kind of assignment among individuals who are matched to one another by virtue of being in the same school.

Statistical Model and Notation

Suppose that the design has $2m$ blocks with n individuals in each block, and m blocks are randomly assigned to each of the treatment and the control condition. Such a design is called a balanced design because there are equal numbers of clusters and individuals assigned to each treatment. Balanced designs are desirable because they are the most sensitive, that is they produce statistical tests that have the highest statistical power and the most precise estimate of treatment effects for a given total sample size. Then, in the conventional notation of the analysis of variance, the model for Y_{ijk}, the outcome measurement of the k^{th} individual in the j^{th} cluster in the i^{th} treatment, is

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{ijk},\ i = 1, 2; j = 1, \ldots, m; k = 1, \ldots, n; l = 1, \ldots, n, \qquad (1)$$

where μ is the grand mean, α_i is the effect of the i^{th} treatment, $\beta_{j(i)}$ is the effect of the j^{th} cluster within the i^{th} treatment, and ε_{ijk} is a residual. The residual ε_{ijk} is assumed to have a normal distribution with variance σ_ε^2 and the grand mean μ and the treatment effects α_i are taken to be fixed, but unknown constants.

In this design, $\beta_{j(i)}$ is the effect of the j^{th} cluster within the i^{th} treatment. It therefore has a random effect, which has variance σ_C^2. It is conventional to define the effect size as the standardized mean difference

$$\delta = (\alpha_1 - \alpha_2)\Big/\sqrt{\sigma_\varepsilon^2 + \sigma_C^2},$$

(see, Hedges, 2007).

Let $\overline{Y}_{1j\bullet}$ ($j = 1, \ldots, m$) and $\overline{Y}_{1j\bullet}$ ($j = 1, \ldots, m$) be the means of the j^{th} treatment and control group clusters, respectively, let $\overline{Y}_{1\bullet\bullet}$ and $\overline{Y}_{2\bullet\bullet}$ be the overall (group) means in the treatment and control groups, respectively, and let $\overline{Y}_{\bullet\bullet\bullet}$ be the grand mean. This notation with dots in the subscripts may seem confusing, but it has a logic. It takes the values of all four subscripts to identify a particular observation, but one could average over all observations in treatment group 1 to obtain $\overline{Y}_{1\bullet\bullet}$ (here that average is denoted by dots in the second and third subscripts which identify the cluster and individual within block and treatment combination within treatment group to denote averaging over individuals and clusters).

What Contributes to Uncertainty of the Treatment Effect in the Cluster Randomized Design

Precision of treatment effect estimates is best understood in terms of what counts as "error" or *imprecision*. Imprecision is all the variation within treatment groups. This is the "noise" that statistical tests compare to the signal of treatment effects to see if the treatment effects are big enough to be reliable (i.e., statistically significant). Because we assign individuals to treatments within clusters, variation across clusters is part of the imprecision of the treatment effect. Although it may be somewhat counterintuitive, in a balanced design, the precision of the estimated treatment effect computed from the $2mn$ individuals is exactly the same as the precision of the treatment effect computed from the $2m$ cluster means (much smaller number of "observations").

The Statistical Analysis of the Design

In the analysis of this design, the overall treatment effect estimate is

$$\frac{1}{m}\sum_{j=1}^{m}\left(\overline{Y}_{1j\bullet}-\overline{Y}_{2j\bullet}\right)=\frac{1}{mn}\sum_{j=1}^{m}\sum_{k=1}^{n}\left(Y_{1jk}-Y_{2jk}\right)$$

and if (as in our model) the design is balanced

$$\frac{1}{m}\sum_{j=1}^{m}\overline{Y}_{1j\bullet}-\overline{Y}_{2j\bullet}=\overline{Y}_{1\bullet\bullet}-\overline{Y}_{2\bullet\bullet}.$$

The test statistic used for testing treatment effects is

$F = MST/MSCT$,

where MST is the so-called treatment mean square defined by

$$MST = SST = mn\left(\overline{Y}_{1\bullet\bullet}-\overline{Y}_{2\bullet\bullet}\right)^2 \Big/ 2,$$

and $MSCT$ is the so-called clusters-within-treatment mean square defined by

$$MSCT = SSCT/2\left(m-1\right)=n\sum_{j=1}^{m}\left(\overline{Y}_{1j\bullet}-\overline{Y}_{1j\bullet}\right)^2+\left(\overline{Y}_{2j\bullet}-\overline{Y}_{2j\bullet}\right)^2 \Big/ 2\left(m-1\right)$$

(see, e.g., Kirk, 1995). When the null hypothesis of no treatment effects is true, the test statistic has the F-distribution with 1 degree of freedom in the numerator and $2(m – 1)$ degrees of freedom in the denominator. Thus we carry out the significance test by comparing the obtained value of F with the appropriate critical value of the F-distribution with 1 and $2(m – 1)$ degrees of freedom or by using the p-value computed by the analysis software.

The standard error (precision) of the treatment effect estimate is

$$SE=\sqrt{\frac{2MSCT}{mn}}.$$

Planning Cluster Randomized Designs

The object planning a research design is to ensure that if the data is collected according to the design, the analysis of that data will yield unambiguous results. To ensure results will be unambiguous, the design must be sufficiently sensitive. There are three different, but complementary, ways to describe design sensitivity, all of which can be useful in planning designs.

One way to describe design sensitivity is statistical power. Statistical power is the chance that if there is a (true or underlying) intervention effect of a given size, the statistical analysis is likely to detect it. Statistical power analysis starts with the size of an effect that is deemed important enough to detect, and a design (e.g., a sample size configuration), and describes the probability that an effect of the given size would result in a statistically significant observed effect. Computation of statistical power can be accomplished by suing tables of power values, but is more easily accomplished using specialized software

Another way to describe design sensitivity is the use of minimal detectable effect size. The minimal detectable effect size is the smallest effect size for which the design has a given statistical power. Using this approach, one starts with design (a sample size configuration) and a desired level of statistical power and describes the smallest effect size for which the design has the specified level of statistical power. Approximate computations of minimum detectable effect size can be done using tables, but exact computations of require specialized software.

A third way to describe design sensitivity is precision. Precision is the standard error of the estimated treatment effect, which determines the width of the confidence interval for the treatment effect. Using precision analysis, one starts with the design (a sample size configuration) and computes the precision that would be expected.

Design Sensitivity is a Function of *m, n*, and *δ*

Design sensitivity minimum detectable effect size and precision depend on the significance level chosen, the effect size expected, the cluster-within-treatment intraclass correlation ρ, m, and n. Statistical power depends, in addition, on the effect size δ. Design sensitivity values for $\rho = 0.05$ (typical values) a few m, n, and δ values are given in Table 1. Table 1 shows some general patterns of design sensitivity for various combinations of m and n that yield a fixed total sample size of $N = 400$ and effect size values of $\delta = 0.20$, 0.35, and 0.50. There are three horizontal panels in the table, each panel has the same effect size δ and shows the statistical power, minimum detectable effect size, and standard error of the estimated treatment effect for different configurations of m and n. Looking within each panel, note that each measure of design sensitivity depends strongly on m (for a fixed N), with power increasing and *MDES* decreasing as m increases. Comparing the corresponding rows of the three panels, note that statistical power increases as effect size increases, but MDES and precision (*SE*) depend only on m and n but are unchanged. That is because they are not predicated by any particular δ and thus do not depend on δ.

Insert Table 1 About Here

Design sensitivity also depends strongly on the intraclass correlation ρ. Appendix Table A1 presents design sensitivity values for the same combinations of m, n, and δ, but when $\rho = 0.10$. Comparing the corresponding values of the two tables, note that designs are less sensitive (power is smaller, MDES is larger and estimates of treatment effect are less precise, that is *SE* is larger, when $\rho = 0.10$ than when $\rho = 0.05$).

General Principles for Planning Cluster Randomized Designs

Planning a randomized block design with fixed block effects to evaluate an educational intervention primarily involves choosing a design that will have adequate sensitivity (high statistical power and precision, or small minimal detectable effect size). In principle, design sensitivity depends on the significance level chosen, the effect size expected, the classroom-within-school intraclass correlation ρ, m, and n. We call these three parameters design parameters because they essentially determine the design.

Although significance level is in principle an arbitrary choice of the investigator, the 0.05 or 5% significance level has been so reified by convention that it is, for all practical purposes, not a choice but the de facto standard. The effect size expected is also not completely a choice but is dictated by the prior experience of the investigator about what size effect is large enough to be important or is the effect size to be expected from the intervention being evaluated. The intraclass correlation can be influenced by the investigator. Thus, planning the design amounts to minimize the intraclass correlation ρ and deciding on how many blocks or sites (which are typically schools), to assign to each treatment within each block, and how many subjects (usually students) n to choose to compose each cluster. That is, planning the design involves minimizing ρ and choosing m and n to achieve the desired design sensitivity.

The computation of either statistical power or minimum detectable effect size can be accomplished by using statistical tables that are widely available (see, e.g., Hedges and Rhoads, 2009) or by the use of statistical software that can compute noncentral F-distribution function (see the Appendix). It is probably easiest to use software that is designed to do the computations directly (e.g., **Website reference**). However, caution is necessary in using software, particularly for complex designs like this one, because notation and terminology differ among software packages and the same terms (e.g., "effect size" do not mean the same things for all software packages, see, Spybrook, Hedges, and Borenstein, 2014). While exact power computations using software are recommended, the relations between m, n, and ρ with design sensitivity are rather complex. In the sections that follow, we offer a few general principles that are helpful in planning designs and illustrate them with computations of design sensitivity.

The Classroom-Within-School Intraclass Correlation

Design sensitivity depends strongly on the intraclass correlation: The smaller the intraclass correlation, the higher the design sensitivity. The classroom-within-school intraclass correlation ρ is defined as the variance of the classroom means σ_C^2 divided by the total variation of outcome observations (Y_{ijk} scores) across clusters (schools), that is

$$\rho = \sigma_C^2/(\sigma_C^2 + \sigma_\varepsilon^2).$$

If the schools are simply sampled, then ρ is determined by the natural distribution of school means. This distribution will be determined partly by the distribution of scores and by policies that determine how students are allocated into schools. For example, if schools were a diverse mix of some with higher and other with lower SES student bodies and therefore academic achievement would be expected to have high intraclass correlations.

Empirical evidence suggest that typical school level intraclass correlations across a range of schools are on the order of 0.2 nationally, and tend to be smaller (on the order of 0.1) in high socio-economic status schools than in lower SES schools (see, e.g., Nye, Konstantopoulos, and Hedges, 2004). There is also empirical evidence that the intraclass correlations within school districts is often smaller. For example, Hedberg and Hedges (2014) found that the mean intraclass correlation within medium sized elementary school districts was often less than 0.05.

It is possible to influence ρ when designing an experiment by matching the group of schools to be assigned before random assignment to treatments. If a pretest X is available, it may be possible to select schools that have very similar average pretest scores to be the experimental sample. Such pretest data at the school aggregate level is often available from administrative records or on state websites. If individual level data is available, it may be possible to compose the experimental sample by selecting a subset of individuals from each school whose mean pretest scores match closely across schools.

It is generally not possible to make $\rho = 0$ because matching is carried out on *pretest* scores, but the intraclass correlation of interest is defined in terms of the between- and within-school variances of the *outcome* (posttest) scores. If the pretest-posttest correlation is $\rho_{Y\backslash X}$ then the variance of school mean scores on the outcome Y for a given pretest value X is

$$(1 - \rho_{Y\backslash X}^2)\sigma_\varepsilon^2/n.$$

Therefore, if all the school means matched exactly on the pretest X, the intraclass correlation would be

$$\rho = \frac{\left(1-\rho_{Y|X}^2\right)\sigma_\varepsilon^2/n}{\left(1-\rho_{Y|X}^2\right)\sigma_\varepsilon^2/n+\sigma_\varepsilon^2} = \frac{1-\rho_{Y|X}^2}{\left(1-\rho_{Y|X}^2\right)+n}.$$

For example, if $\rho_{Y\backslash X}{}^2 = 0.5$ (so that $\rho_{Y\backslash X} \approx 0.71$) and $n = 20$, then $\rho \approx 0.02$. It is important that the ρ value obtained after matching depends on n, so that is n is smaller, ρ becomes larger. For example, if $\rho_{Y\backslash X}{}^2 = 0.5$ but $n = 50$, then $\rho = 0.01$, and if $n = 100$, $\rho = 0.005$. Thus, the impact of the within cluster sample size n on ρ will need to be considered in selecting the appropriate n for the design.

If the outcome variable and pretest are cognitive tests like achievement tests, a pretest-posttest correlation of $\rho_{Y\backslash X} \approx 0.7$ is quite sensible, and hence it should be possible to obtain an intraclass correlation of $\rho = 0.01$ to 0.05. Empirical evidence suggests that background variables like race, gender, and crude social class indicators (like free or reduced-price lunch eligibility) will have correlations with outcomes and thus even perfect matching on such covariates will lead larger intraclass correlations (see Hedges and Hedberg, 2007). In other words, matching on a covariate that is a pretest is strongly recommended.

The Number *m* of Schools Will be Limited in Development Studies

Perhaps the most stringent limit in initial field trials in development studies is that it is usually infeasible to involve large numbers of schools. Thus, the number of schools is usually limited to 5 to 10, an occasionally as many as 20. Because the total number of schools is $2m$, this means that $m = 5 – 10$ is about the maximum that is likely to be feasible. Thus, even though sensitivity depends strongly on m, and it is important to obtain as large a number of schools as feasible, there are usually practical limits on how much m can be used to increase design sensitivity.

Increasing Cluster Size *n* Can Increase Design Sensitivity if *ρ* is Relatively Small

While there are typically sharp limits on the number of schools, it is often feasible to obtain a lager number of students in each school (for example, several classrooms). The relation between n and design sensitivity is complex. Design sensitivity increases with n, but begins to level off so that it tends to an asymptote. The asymptote depends on the intraclass correlation ρ, being smaller when ρ is larger. Thus, it can be impossible to obtain adequate sensitivity for some m and ρ regardless of how large n (or the total sample size) becomes. Table 2 presents the smallest possible *MDES* as a function of m and ρ. Two general patterns are evident. As m increases, the smallest possible MDES decreases and as ρ increases the minimum MDES increases. For example, if $\rho = 0.05$, the minimum MDES decreases from 1.27 when $m = 2$ to 0.30 when $m = 10$. Similarly, for $m = 5$, the minimum MDES increases from 0.20 when $\rho = 0.01$ to 0.64 when $\rho = 0.10$. This table makes clear that for values of m that are often feasible in development studies ($m \leq 10$), it is important to ensure that ρ is as small as possible in order to have adequate design sensitivity to detect the effect sizes that are plausible (e.g., $\delta \leq 0.5$).

Insert Table 2 About Here

The values in table 2 illustrate what design sensitivities are possible with large enough n, but do not indicate what specific values of n would be required to have any particular design sensitivity. Table 3 presents minimum detectable effect size as a function of m and n when ρ = 0.05. For example, the table shows that when $m = 5$, the MDES < 0.50 for $n \geq 100$ and when $m = 8$, MDES $\leq$ 0.35 when $n \geq 200$. Note that MDES decreases rapidly as n increases when n is small (e.g., $n = 10 – 20$), but MDES decreases very little as n increases when n is large (e.g., 100 – 200). Appendix table A2 gives the minimum detectable effect size as a function of m and n when $\rho = 0.02$. That table shows that the MDES is considerably smaller if ρ can be made as small as 0.02. In this case when $m = 5$, the MDES $\leq$ 0.35 for $n \geq 100$ and when $m = 8$, MDES $\leq$ 0.25 when $n \geq 200$. It is important to note that very stringent matching before randomization is required to obtain an intraclass correlation of $\rho = 0.02$. Appendix table A3 gives the minimum detectable effect size as a function of m and n when $\rho = 0.075$. That table shows that the MDES is considerably when ρ becomes larger, illustrating the importance of careful matching to reduce the intraclass correlation.

Insert Table 3 About Here

Unbalanced Designs May Be a Feasible Way to Increase Design Sensitivity

For a fixed total sample size, a balanced design (equal allocation of individuals to treatment and control in each block) is the most sensitive. However, there are situations in which the sample size is limited by the number of schools that can be treated (e.g., because of limitations of personnel needed for training or implementation or of equipment). In that case, unequal allocation of schools to treatment groups (more classes randomized to treatment and fewer to control or vice versa) can increase the total sample size of schools and can increase design sensitivity. For example, if only one third of the schools willing to participate in the experiment can be treated, allocating only one-third of the schools to treatment and two-thirds of the schools to control makes it possible to obtain an overall school sample size that is 50% larger. Although it might seem counterintuitive, the design sensitivity is the same regardless of whether there are more treatment or more control schools.

Suppose that an unbalanced design assigns m^T clusters (classes) to treatment and m^C clusters (classes) to control in each block so that the proportion $\pi = m^T/(m^T + m^C)$ of clusters are assigned to treatment and a total of $N^T = N\pi$ individuals are assigned to treatment overall. In a balanced design the total sample size would be $N = nm^T/0.5 = 2nm^T$, but in an unbalanced design, because $\pi < 0.5$, the total sample size N would be greater than $2nm^T$. In fact, the ratio of the total sample size of a design with $\pi = N^T/N$ to that of the balance design would be $1/2\pi$. Thus if $\pi = 0.5$ (the design is balanced) the ratio is unity, as expected, but if $\pi = 1/3$, the total sample size of the design is 50% larger, if $\pi = 1/4$ it is 100% larger than that of a balanced design with the same number of *treated* individuals. This increased total sample size increases design sensitivity.

While imbalance of the design does not literally change the effect size, it has an effect on sensitivity that is equivalent of changing the effect size from δ to $\delta\sqrt{4\pi\left(1-\pi\right)}$. Because

$$\delta\sqrt{4\pi\left(1-\pi\right)} \leq \delta$$

and is only equal to δ if $\pi = 0.5$, imbalance has a tendency to increase sensitivity because of the increased total sample size, but a simultaneous tendency to decrease sensitivity because it does the equivalent of decreasing the effect size. However, unless π is very much smaller than 0.5, the positive impact of imbalance on total sample size is bigger than the negative impact on effect size and the unbalanced design is more sensitive with the same number of *treated* individuals.

Table 4 shows the minimum detectable effect size as a function of the number of treatment clusters m^T and the number of the number of control clusters m^C with $n = 50$ individuals per cluster when $\rho = 0.05$. The table shows $m^C \geq m^T$, but the MDES is identical for the corresponding pairs of m^T and m^C if $m^T \geq m^C$. For example, the table shows that an unbalanced design with $m^T = 4$ and $m^C = 4 + 4 = 8$ (and a total sample size of $N = 600$) would have a minimum detectable effect size of $\delta = 0.50$, but table 4 shows that a balanced design with the same number of treated clusters $\rho = 0.05$, $m^T = m^C = 4$, and $n = 75$, (with the same total sample size of $N = 600$) would have an MDES of $\delta = 0.60$. In other words, the unbalanced design achieves greater design sensitivity with the same number of treatment clusters and the same total sample size but it requires a larger number of control clusters. Appendix tables A4 and A5 show the minimum detectable effect size as a function of m^T and m^C when $\rho = 0.05$ and $n = 75$ or $n = 100$, respectively.

Insert Table 4 About Here

Statistical Adjustment for Covariates Can Substitute for Matching of Classes

Covariates are any variables whose value could not have been influenced by the treatment assignment. The most common covariates are baseline data (such as pretests) collected before the treatment begins. If baseline data on schools is available, it can be used to increase the sensitivity of the design if the analysis uses statistical adjustment via regression or the analysis of covariance. In a design such as this, covariates primarily increase design sensitivity by reducing between-cluster (between-school) variation in outcome scores. The use of a covariate (or multiple covariates) essentially has the same impact as matching and there is no additional benefit to using covariates in the analysis if the classes have been matched on the covariate prior to assignment to treatment. Moreover, using a covariate at the cluster level reduces the degrees of freedom available in the error term, which has a tendency to reduce statistical power. Therefore, although statistical adjustment with a pretest or other covariates could be considered as an alternative to matching, it may not be as helpful as matching when the total number of clusters is very small (less than 15).

Examples

Conclusions

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Appendix

The following tables provide additional information about minimal detectable effect sizes for planning designs.

Insert Appendix Tables A1 – A5 About Here

Power Computations

Power computations for this design are based on the sampling distribution of the F-statistic given in the text. When there is no block treatment interaction and $\delta^2 \neq 0$, then F has the noncentral F-distribution with $m^T + m^C - 2$ degrees of freedom and noncentrality parameter

$$\lambda = \frac{m^T m^C n \delta^2}{\left(m^T + m^C\right)\left[1 + \left(n-1\right)\rho\right]}$$

(see, e.g., Searle, 1971). Therefore, the statistical power of the level α test of the null hypothesis H_0: $\delta^2 = 0$ is

$$G[c_\alpha | \lambda, m^T + m^C - 2]$$

where c_α is the $100(1 - \alpha)$ percentile of the central F-distribution with $m^T + m^C - 2$ degrees of freedom and $G[x \mid \lambda, v]$ is the cumulative distribution function of the noncentral F-distribution with v degrees of freedom and noncentrality parameter λ.

The limiting power as n tends to infinity (for fixed m^T and m^C) is obtained by noting that λ tends to

$$\lambda_{Max} = \frac{m^T m^C \delta^2}{\left(m^T + m^C\right)\rho}.$$

Because power is a continuous function of λ, the limiting power is the power when $\lambda = \lambda_{max}$.

The approximate minimal detectable effect size can be obtained using a formulas in Bloom (1995) or by iteratively computing power with successively larger values of the effect size until the appropriate power is obtained.

Figure 2
Minimum detectable effect size as a function of m for $p = 2$, $\rho = 0.05$, and $n = 10$ (dashed line), $n = 12$ (dotted line), $n = 15$ (dashed and dotted line), and $n = 20$ (long dashed line)

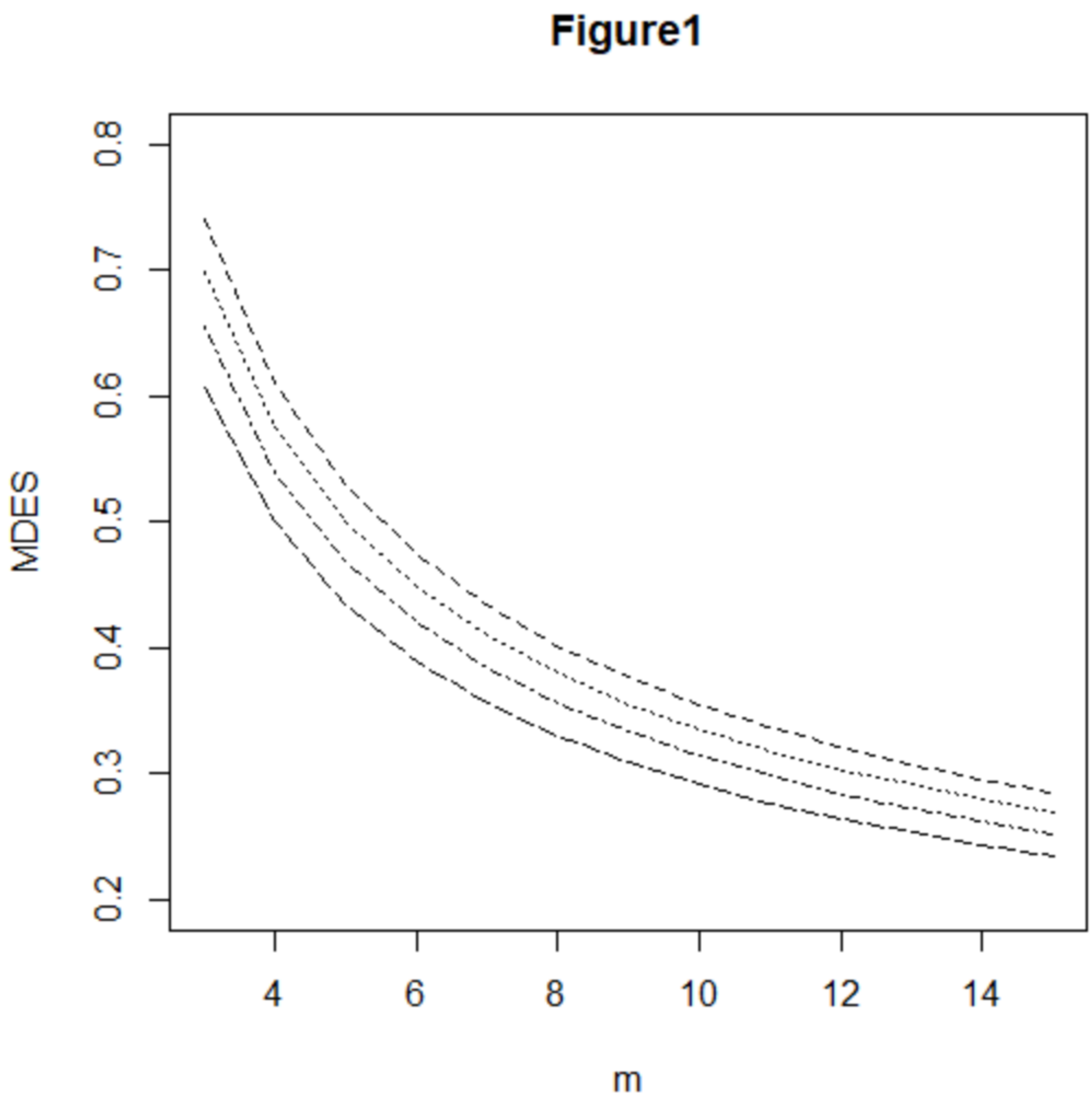
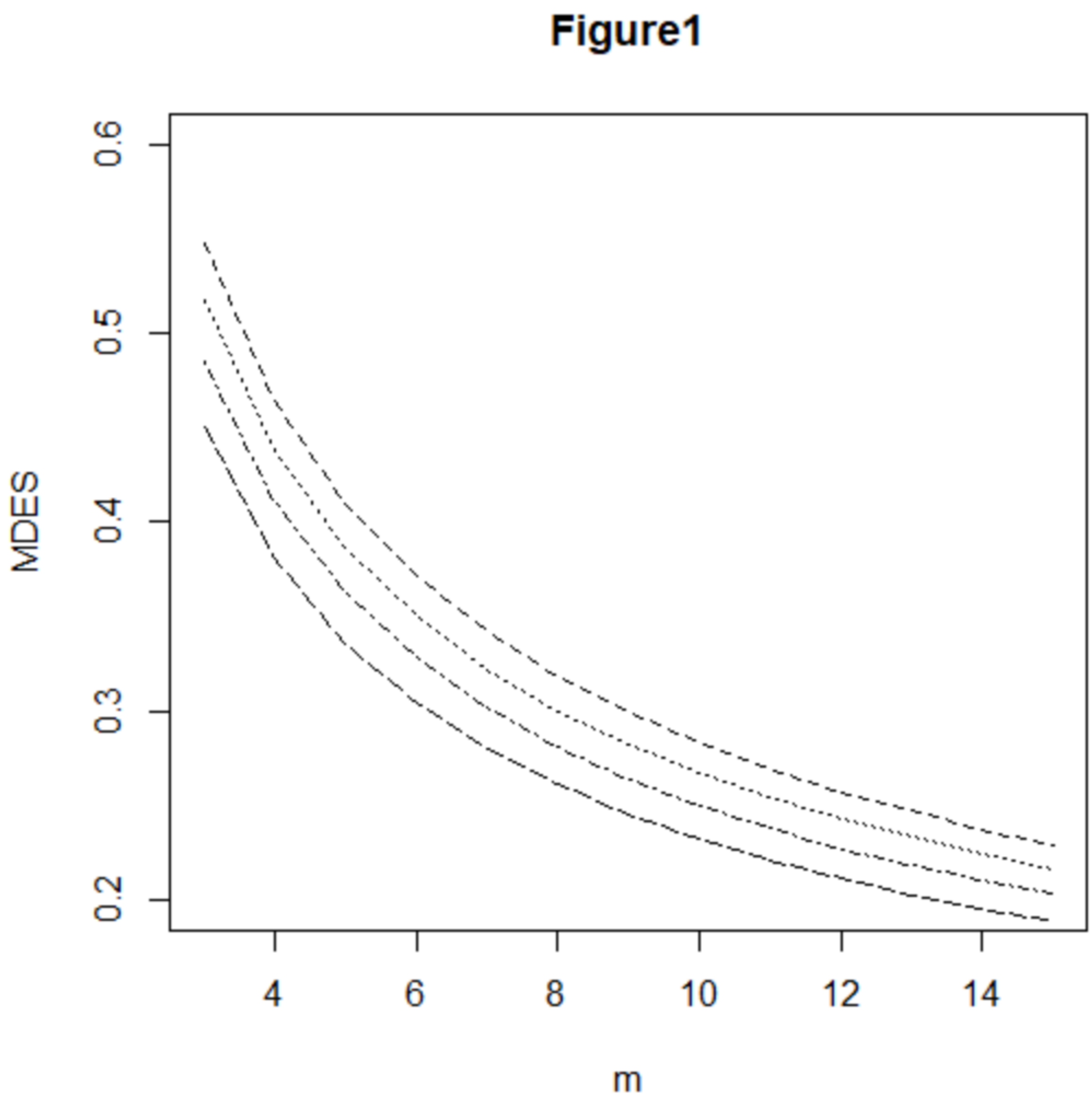


Figure 3
Minimum detectable effect size as a function of m for $p = 3$, $\rho = 0.05$, and $n = 10$ (dashed line), $n = 12$ (dotted line), $n = 15$ (dashed and dotted line), and $n = 20$ (long dashed line)



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Chapter 9
Introduction to Quasi-experiments

While randomized experiments have important advantages in assuring internal validity, they are not always possible. When experiments are not possible, other research designs must be used that do not involve manipulation (assignment of units to treatments that is not random). Such designs are often called *quasi-experimental designs* to reflect the fact that they mimic certain characteristics of experimental designs. As we discussed in chapter 2, it is often useful to imagine a hypothetical experiment when evaluating the validity of a study without randomization, a perspective sed in this section.

There are many different types of quasi-experimental designs, but this book focuses on a few of them that are most likely to be useful in educational evaluations associated with development work. The designs included are difference in differences designs, interrupted time series designs, or nonequivalent control group designs. We do not include, for example designs involving assignment based on a covariate (such as regression discontinuity designs) because they are unlikely to be used in initial field trials.

Some researchers believe that the quasi-experimental designs discussed here are easier to implement than the experimental designs discussed in the previous section. We believe that quasi-experimental designs ae in fact easier to implement poorly, but much more difficult to implement well. All quasi-experimental designs involve assumptions that must be true of these designs are to have high internal validity. These assumptions are specific to each design and range from difficult to impossible to verify empirically.

This does not mean that quasi-experimental designs are without value. It merely means that a great deal of effort usually is required to build a strong argument that the designs are internally valid. Usually this process (and the data that must be collected to support it) make the implementation of quasi-experimental designs more difficult to implement well than experimental alternatives. The statistical analyses necessary for the evaluation of quasi-experiments are almost always more demanding than those necessary for the analysis of similar experiments. The reason is that the statistical analysis (rather than the design) carries a greater burden in assuring the validity of the conclusions of quasi-experiment than it does in assuring the validity of conclusions from randomized experiments.

In both randomized experiments and quasi-experiments, statistical inference provides a summary of the data (e.g., and estimate oof the treatment effect), a characterization of the uncertainty of that treatment effect, and a decision theoretic procedure for deciding if the treatment effect is nonzero. In uncompromised randomized experiments, the *design* assures that the difference between the treatment group mean and the control group mean is an unbiased estimate of the causal effect of the treatment. Note that this statement makes no reference to any data analysis method (other than referring to the mean difference) or no other assumptions necessary. More elaborate analyses may increase precision but are unnecessary to derive an estimate of the causal effect.

But in quasi-experiments, additional assumptions are always necessary in order for the design to yield an estimate of the causal effect of the treatment. For some quasi-experimental designs these are assumptions about the mechanism generating the data (such as the common shocks and parallel trends assumptions for difference in differences analyses). In other quasi-experimental designs these assumptions also include assumptions about the functional form of relations between covariates and outcomes (such as assumptions about adequacy of matching variables and linear regressions when using covariate adjustments in nonequivalent control group designs).

Much of the evidence that supports the warrant for causal inference in quasi-experiments involve analyses to check the validity of the required assumptions. Some assumptions are essentially uncheckable empirically, but their plausibility can be supported by analytic work. For example, consider the assumption that the covariates used for matching are adequate to control for the validity threat of selection. This assumption cannot be proved definitively by empirical means. It can be made more plausible by a scientific argument about which factors could have influenced selection into treatment groups and that are also likely to be correlated with the outcome. If empirical evidence can be assembled that these factors do not differ between treatment and comparison group (at least after statistical adjustment for the covariates used in matching) or they are uncorrelated with the outcome, then the validity of the causal inference from the quasi-experiment is more plausible.

Note that assembling evidence to support the validity of quasi-experiments typically requires additional data collection and can be analytically complex. Moreover, it is not entirely definitive, it merely increases the plausibility of internal validity, it does not prove it. None of the additional work is necessary when using experimental designs (although some degree of checking that the experiment has not been compromised by attrition or outside influences is good practice, as it is in quasi-experiments too).

Chapter 10
The Difference in Differences Design

The difference in differences design is a generalization of the pretest-posttest design. As the name implies, the pretest posttest design evaluates an intervention effect by measuring the outcome first before and then after the treatment and estimates the treatment effect as the difference between the posttest and pretest means. This design may be adequate in situations where there are no other intervening causes that might change he outcome this is seldom plausible in educational settings where other learning (unrelated to treatment), maturation, and historical effects cannot be ruled out. Consequently, the pretest posttest design was deemed a pre-experimental design by Campbell and Stanley (1963), meaning it had less internal validity than experimental designs or the many quasi-experimental designs that they discussed.

The difference in difference design can be more valid than the pretest-posttest design because the comparison group can provide a control for the other intervening causes (other than the treatment) that might produce changes in he outcome between pretest and posttest. Clearly, the comparison group can only control for other intervening causes if it experiences the same intervening causes experienced by the treatment group. Describing the intervening causes as "shocks," this idea is sometimes codified as the assumption of "common shocks"—that the treatment and comparison groups experience he same shocks between pretest and posttest (see, e.g., Angrist and Pischke, 2009).

Another assumption necessary to assure the validity of the difference in differences design is that the two groups are changing at the same are over time, that is that the slopes of their growth trajectories over time are the same. Because identical slopes would mean that the growth trajectories were parallel (the trajectories different, perhaps, because their pretest values were different), this assumption is sometimes codified as the assumption of "parallel trends" (see, e.g., Angrist and Pischke, 2009).

Recall that the average causal effect of a treatment is the difference between (in this case) the mean pretest to posttest change in the treatment group (when that group experiences the treatment) and the mean change that the treatment group would have experienced if (counterfactually) if it had not experienced the treatment. Both the assumption of common shocks and that of parallel trends are necessary to render change in the comparison group the same as the change that would have been observed in the treatment group if there had been no intervention.

Note that neither of these two assumptions is easy to verify. It is difficult to imagine an empirical strategy to verify the common shocks assumption. It is possible to measure a variety of variables that might constitute shocks in both groups and determine whether the mean values of those variables are similar in the two groups (the logic is similar to matching). However, it is difficult to know that *all possible* variables that could constitute shocks have been measured. Similarly, it is difficult to empirically verify that the growth trajectories of both groups are the same. Often the outcome is assessed via an expensive or time-consuming assessment such as an achievement test. Determining the growth trajectories of each groups would require assessments of the outcome variable at many (or at least several) points in time. It may be impossible or infeasible to administer more than two such tests (he pretest and the posttest) because of the burden testing places on students and teachers and because there may not be more than two or three different versions of the test that are strongly equated to one another to support their use in estimating growth trajectories.

Statistical Model and Notation

Suppose that the study design is a two groups, each of which has a pretest and a posttest and let Y_i^T (i = 1, …, n^T) and Y_i^C (i = 1, …, n^C) be the a^{th} posttest observation in the treatment and control groups and let X_a^T (a = 1, …, n^T) and X_i^C (i = 1, …, n^C) be the i^{th} pretest observation in the in the treatment and control groups, respectively. The total number of (X, Y) observations is $N = n^T + n^C$.

Suppose that the both the pretest and the posttest observations are normally distributed observations within treatment groups, so that

$$Y_i^T \sim \mathrm{N}(\mu_Y^T, \sigma^2) \text{ and } Y_i^C \sim \mathrm{N}(\mu_Y^C, \sigma^2)$$

and

$$X_i^T \sim \mathrm{N}(\mu_X^T, \sigma^2) \text{ and } X_i^C \sim \mathrm{N}(\mu_X^C, \sigma^2).$$

Note that the within-group variance σ^2 is the same in both groups for both pretest and posttest. The assumptions about Y are the set of assumptions used to justify the use of the t-test or F-test in analysis of variance. Suppose further that the (within-treatment group) correlation between X and Y was ρ.

Let $\overline{Y}_{\bullet}^{T}$ and $\overline{Y}_{\bullet}^{C}$ be the posttest means in the treatment and control groups, and let $\overline{X}_{\bullet}^{T}$ and $\overline{X}_{\bullet}^{C}$ be the pretest means in the treatment and control groups, respectively. Define the (pooled) within-treatment group variance of Y, S_Y^2 via

$$S_Y^2 = \frac{\sum_{a=1}^{n^T}\left(Y_a^T - \overline{Y}_{\bullet}^{T}\right)^2 + \sum_{a=1}^{n^C}\left(Y_a^C - \overline{Y}_{\bullet}^{C}\right)^2}{N-2}, \qquad (1)$$

and define the (pooled) within-treatment group variance of X, S_X^2 via

$$S_X^2 = \frac{\sum_{a=1}^{n^T}\left(X_a^T - \overline{X}_{\bullet}^{T}\right)^2 + \sum_{a=1}^{n^C}\left(X_a^C - \overline{X}_{\bullet}^{C}\right)^2}{N-2}. \qquad (2)$$

The Statistical Analysis of Difference in Difference Designs

Several different analyses are possible. The simplest analysis would be to conduct an analysis using gain scores (also called a difference scores). Such an analysis essentially uses the gain scores

$$G_i^T = Y_i^T - X_i^T \text{ and } G_i^C = Y_i^C - X_i^C \qquad (3)$$

as the outcome variable. Denote the mean gain in the treatment and control groups via $\overline{G}_{\bullet}^{T}$ and $\overline{G}_{\bullet}^{C}$ respectively and the pooled within groups variance of the gains by S_G^2. To test the null hypothesis if no treatment effect, that is

$$\mathrm{H}_0\text{: } (\mu_Y^T - \mu_X^T) - (\mu_Y^C - \mu_X^C),$$

the usual test statistic for testing hypotheses about the treatment effect is

$$F = \frac{n^T n^C}{n^T + n^C}\left(\frac{\overline{G}_{\bullet}^{T} - \overline{G}_{\bullet}^{C}}{S_G}\right)^2, \qquad (4)$$

which has a noncentral F-distribution with one degree of freedom in the numerator and $N-2$ degrees of freedom in the denominator and noncentrality parameter

$$\lambda = \left(\frac{n^T n^C}{n^T + n^C}\right)\frac{\left[\left(\mu_Y^T - \mu_X^T\right) - \left(\mu_Y^C - \mu_X^C\right)\right]^2}{2(1-\rho)\sigma^2}. \quad (5)$$

In an uncompromised randomized experiment, $\mu_X{}^T = \mu_X{}^C$ and therefore,

$$\left(\mu_Y^T - \mu_X^T\right) - \left(\mu_Y^C - \mu_X^C\right) = \mu_Y^T - \mu_Y^C,$$

but if the experiment is compromised, or if the design is not randomized, then the difference in mean gains might be regarded as a more valid treatment effect estimand the posttest mean difference.

An alternate, but equivalent, analysis uses multiple regression analysis. To explain this analysis, a slightly different notation for the observations is needed. In this notation, the $N = 2n^T + 2n^C$ observations of X and Y are denoted $Z_1, \ldots, Z_N$, and the regression model

$$Z_i = \beta_0 + \beta_1 T_i + \beta_2 POST_i + \beta_3 (T_i * POST_i) + \varepsilon_i, \quad (6)$$

is estimated, where T_i is a dummy variable taking the value 1 if Z_i is a treatment group X or Y and zero otherwise, $POST_i$ is a dummy variable taking the value 1 if Z_i is a Y (posttest score) and zero if it is an X (pretest score). Here β_3 is the treatment effect and it is identical to

$$(\mu_Y{}^T - \mu_X{}^T) - (\mu_Y{}^C - \mu_X{}^C)$$

and the regression estimate of β_3 is identical to $\overline{G}_{\bullet}^T - \overline{G}_{\bullet}^C$. Consequently, the statistical significance test of the hypothesis that $\beta_3 = 0$ is identical to the statistical significance test based on the gain scores explicated above.

Effect Size

Because the quantity

$$\delta = \frac{\left(\mu_Y^T - \mu_X^T\right) - \left(\mu_Y^C - \mu_X^C\right)}{\sigma}$$

occurs as part of the noncentrality parameter that determines the power of the test for treatment effect, δ is the natural effect size measure to use in planning experiments. If the treatment and comparison groups had the same mean pretest score, that is if $\mu_X{}^T = \mu_X{}^C$, then the difference in gain scores would just be the difference in posttest scores so that

$$\left(\mu_Y^T - \mu_X^T\right) - \left(\mu_Y^C - \mu_X^C\right) = \mu_Y^T - \mu_Y^C$$

and the effect size δ would become

$$\frac{\mu_Y^T - \mu_Y^C}{\sigma},$$

which is the effect size used in planning randomized experiments (often called Cohen's **d**).

General Principles for Planning of Difference in Difference Designs

Planning a difference in difference design to evaluate an educational intervention primarily involves choosing a design that will have adequate sensitivity (high statistical power and precision, or small minimal detectable effect size). In principle, design sensitivity depends on the significance level chosen, the effect size δ, n^T, n^C, and ρ. The conventional significance level will generally be 5% and the conventional level of statistical power desired will be 80%.

The computation of either statistical power or minimum detectable effect size can be accomplished by using statistical tables that are widely available (see, e.g., Hedges and Rhoads, 2009) or by the use of statistical software that can compute noncentral *F*-distribution function (see the Appendix). It is probably easiest to use software that is designed to do the computations directly (e.g., **Website reference**). While exact power computations using software are recommended, we offer a few general principles that are helpful in the sections that follow and illustrate them with computations of design sensitivity.

The most sensitive design given a fixed total sample size N is a balanced design in which the same number of individuals in the treatment and comparison group, that is $n^T = n^C = n$. Exact design sensitivity depends on, n, and δ, but in balanced designs, sensitivity essentially depends only on the total sample size N in balanced designs. Table 1 shows the statistical power, minimum detectable effect size, and standard error of the estimated treatment effect for sample sizes of $N = 20$ to 400, effect size values of $\delta = 0.20$ and 0.50, and pretest-posttest correlations of $\rho = 0$, 0.5, and 0.7. The table shows that Table 1 shows that the design sensitivity increases as N increases (as expected). That is, the power increases and the minimum detectable effect size and standard error of the treatment effect estimate decrease as N increases. It also shows that design sensitivity decreases as ρ increases.

Insert Table 1 About here

Unequal Allocation to Treatments Can Have Advantages if it Enables Larger *N*

For a fixed total sample size N, a balanced design (equal allocation of individuals to treatment and control) is the most sensitive. However, there are situations in which the sample size is limited by the number of individuals who can be treated. For example, there may only be enough field staff to work with a limited number of treatment teachers or there may be a limited number of computers or other technology necessary to use the intervention. In that case, unequal allocation of individuals to treatment groups (more individuals in the comparisons group and fewer in the treatment group) can increase the total sample size and can increase design sensitivity.

Suppose that an unbalanced design has n^T individuals in the treatment group and n^C individuals on the control group so that the proportion $\pi = n^T/(n^T + n^C)$ of the total number of individuals are in the treatment group. In a balanced design the total sample size would be $N = n^T/0.5 = 2n^T$, but in an unbalanced design, because $\pi < 0.5$, the total sample size N would be greater than $2n^T$. In fact, the ratio of the total sample size of a design with $\pi = n^T/N$ to that of the balance design would be $1/2\pi$. Thus if $\pi = 0.5$ (the design is balanced) the ratio is unity, as expected ,but if $\pi = 0.4$, the total sample size of the design is 25% larger, if $\pi = 0.3$ it is 67% larger, and if $\pi = 0.2$ it is 150% larger than that of a balanced design with the same number of *treated* individuals. This increased total sample size increases design sensitivity.

Table 2 shows the minimum detectable effect size as a function of the number of individuals assigned to treatment for balanced designs ($\pi = 0.5$) and unbalanced designs with various degrees of imbalance ($\pi < 0.5$). For example, if $n^T = 50$ individuals are to be treated, a balanced design would have a minimum detectable effect size of $\delta = 0.81$, but that could be reduced to $\delta = 0.68$ if we assigned $\pi = 0.3$ (30%) of the total individuals to treatment. This would mean assigning $n^T = 250$ individuals to treatment and $n^C = 117$ for a total sample size of $N = 167$. We could lower the minimum detectable effect size to $\delta = 0.63$ by using an even more imbalanced design and assigning $\pi = 0.2$ (20%) of the total individuals to treatment. In this case $n^T = 50$ individuals would still be assigned to treatment, but $N^C = 200$ individuals would have to be assigned to the control group, for a total sample size of $N = 250$.

Insert Table 2 About Here

Checking the Assumptions of the Designs (beware of matching)

Because pretests and posttests are imperfectly correlated. Individuals that are selected because they are above (or below) the mean can be expected to decrease (or increase) their scores on the posttest, even if there are no outside influences (e.g., if the treatment has no effect). Matching is used to create comparison groups by selecting individuals from a reservoir of potential comparison group members. If the pretest mean of the comparison group is higher than that of the treatment group, selecting individuals for the comparison group that match those in the treatment group will require selecting individuals whose pretest scores are above their group's (the reservoir from which comparison group members are selected) mean. Thus, we can predict that, if there were no outside influences, their posttest scores will be closer to the mean than were their pretest scores. In other words, in the absence of a treatment effect, the gain in the comparison group will be negative, making the difference in gains (the estimated treatment effect) look smaller.

If the mean of the reservoir from which the comparison group is selected has a lower mean than that of the treatment group, the gain in the matched comparison group will tend to be larger than zero, making the difference in the gains the bias is in the opposite direction, making the treatment effect (the estimated treatment effect) look larger. Such regression effects in quasi-experiments have been known for a long time (see, e.g., Campbell and Erlebacher, 1970).

As we noted in chapter 2, if μ_X and μ_Y are the means of X and Y, the variances of X and Y are the same, and ρ is the correlation between X and Y in the reservoir from which the comparison group is composed, then the regression equation for predicting Y from X is

$$\hat{Y} = \mu_Y + \rho\left(X - \mu_X\right) .$$

This equation shows that if $\rho = 0$ there is no regression effect, but the stronger the correlation between X and Y, the stronger the regression effect. If the mean of the reservoir from which the comparison group is selected has a lower mean than that of the treatment group, the bias is in the opposite direction, making the treatment effect look larger

Note that regression effects in matching occur when two things are both true. The first is that the pretest is substantially correlated with the posttest. If the pretest is not correlated with the posttest matching cannot improve precision. The second is that the reservoir from which the comparison group is selected has a different mean than that of the treatment group. This condition is more complex. When groups are not created by random assignment, the underlying pretest means of the treatment group and the comparison reservoir are unlikely to be identical, thus regression is likely to introduce some bias. If the mean difference is sufficiently small, the bias created by regression will be small, but the safest strategy is not to use matching in difference in difference designs (Daw and Hatfield, 2018).

Using Covariates Can Increase Sensitivity and Reduce Sample Size Requirements

As with other designs we have discussed, the use of covariates can increase sensitivity. However, in difference in differences designs, unlike randomized experiments, the use of covariates *can reduce* sensitivity if the covariates are related to the size of the treatment effect. This happens, for example, if the treatment is more effective for students with some characteristics (high or low values on the covariate) than for others.

The intuition of how this can happen hinges on the idea that statistical adjustment is a form of linear regression. Adding a covariate to the regression model reduces residual variation and thus has a tendency to increase the precision of the estimate of the coefficient that is the treatment effect. But, if there is collinearity between the covariate and the treatment indicator, adding a covariate as a predictor has a tendency to decrease the precision of the treatment effect estimate. In randomized experiments, randomization guarantees that the covariate is always uncorrelated with the treatment effect indicator, so there is no collinearity and covariates essentially always increase precision. In quasi-experiments, there is no assurance that covariates will not be correlated with treatment assignment, and thus using covariate can either increase or decrease design sensitivity.

When there is only one covariate, call it U, the use of that covariate will increase design sensitivity if

$$\frac{1-\rho_{UY}^2}{1-\rho_{UT}^2} < 1,$$

where ρ_{UY} is the correlation between the covariate U and the outcome score Y, and ρ_{UT} is the correlation between the covariate U and the treatment indicator T. Otherwise, addition of the covariate U will decrease design sensitivity. Although it may not seem natural to think of the correlation between a continuous variable like U and a binary variable like a treatment effect indicator, it can be rewritten as

$$\rho_{UT} = \frac{\mu_U^T - \mu_U^C}{4\sigma_U},$$

where $\mu_U{}^T$ and $\mu_U{}^C$ are the treatment and control group means on U and σ_U is the standard deviation of U. The numerator in the ratio reflects how much the covariate decreases the variance of the treatment effect estimate due to decreased error variation while the reciprocal of the denominator reflects how much the collinearity of the covariate and treatment indicator increases the variance of the treatment effect estimate.

It may be helpful to interpret ρ_{UT} as $\rho_{UT} = \delta_U/4$, where δ_U is the standardized mean difference between treatment and control on the covariate U, a Cohen's d for the covariate. If we think of 0.5 as a medium sized effect (as Cohen suggests) then the ρ_{UT} value associated with a medium sized effect is only $\rho_{UT} = 0.12$ and the ρ_{UT} value associated with a medium sized effect is only $\rho_{UT} = 0.12$ and ρ_{UT} value associated with an effect of $\delta_U = 1.0$ is only $\rho_{UT} = 0.25$, so that even covariates with only modest correlations with the outcome Y (say $\rho_{UT} > 0.5$) generally have the potential to increase design sensitivity.

Examples

Conclusion

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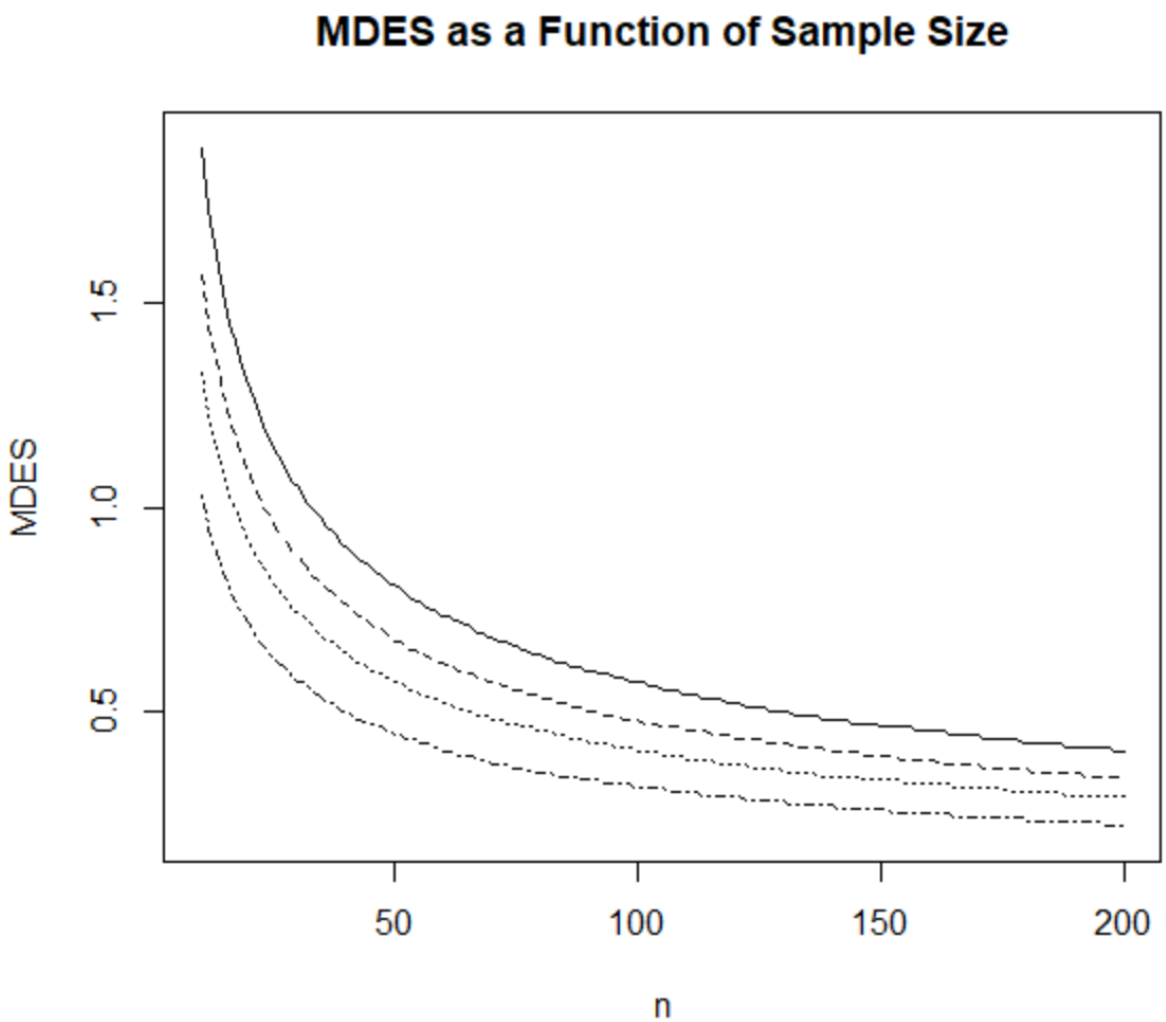
Figure 1
MDES as a function of group sample size n when $\rho = 0$ (solid line), $\rho = 0.3$ (dotted line), $\rho = 0.5$, dashed line), and $\rho = 0.7$ (dashed and dotted line)



Chapter 11
Interrupted Time Series Designs

Interrupted time series designs can be viewed as a generalization of the pretest posttest design in which there is more than one measurement at different timepoints before and more than one measurement at different timepoints after the treatment (or interruption as it is called in these designs). Like difference in difference designs, the units measured may be aggregate units involving different individuals at each timepoint. In some cases, (as in so-called single case designs in psychology or medicine) the individuals may be the same over time. Regardless, the logic of the design is that if there is a treatment effect, we expect to see that the pattern of the measurements after the interruption is different (higher or lower) that that before the interruption and the difference is the treatment (interruption) effect.

Like the difference in difference design, the interrupted time series design requires the assumption that the measurements at different timepoints on the same units are correlated. This is not a mathematical requirement, but an empirical one, necessitated by the fact that, empirically repeated measurements on the same units *are* correlated. This phenomenon is called *autocorrelation*. The pattern of these correlations can be very complex and difficult to estimate from the observed data, particularly from a short series of observations. The dominant consideration is that adjacent measurements are typically more highly correlated than measurements that are further apart in time.

The analysis of interrupted time series can be exceedingly complex mathematically. We present only the simplest such analyses. However, these are likely to be adequate in situations where there is only a relatively short series of observations over time (e.g., no more than 10 – 20), which limits the complexity of models that can be used. However, even in the simplest of these models it is difficult to express the statistical models without the use of matrix algebra.

Autocorrelation

In most research designs, different observations are independent (uncorrelated). In some designs such as the different in difference design, pretest and posttest observations made on te same individuals are understood to be correlated. In the interrupted time series design, all of he observations are understood to be corelated. Thus, some model for those correlations is required. Many such models are possible, but a simple and frequently adequate model is that of normally distributed observations with first order autocorrelation or NAR(0, σ, ρ)(See, e.g., Amemiya, 1985). NAR(0, σ, ρ) implies that the errors follow what is known as a normal autoregressive model of order 1. This implies that the covariance matrix of the error vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_{a+b})'$ is

$$\boldsymbol{\Sigma} = \frac{\sigma^2}{1-\rho^2}\begin{pmatrix} 1 & \rho & \cdots & \rho^{a+b-1} \\ \rho & 1 & \cdots & \rho^{a+b-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho^{a+b-1} & \rho^{a+b-2} & \cdots & 1 \end{pmatrix}.$$

This first order autocorrelation model captures the fact that the the correlation between observations one timepoint apart is proportional to ρ, while the correlation between observations two timepoints apart is proportional to ρ^2, etc., and the correlation diminishes proportionately as the distance between timepoints increases. Because ρ^x tends to zero as x becomes large, observations that are sufficiently far apart are essentially uncorrelated.

Effects of Autocorrelation

Autocorrelation introduces some phenomena that are possibly unexpected. There is less information in the sample than when observations are independent. Thus, the variance of the mean of n observations is not the variance of an individual observation divided by n. Similarly, the degrees of freedom of the sum of squares about the mean is less than $n - 1$.
That is, the usual variance $S^2 = \sum (y_i - \bar{y})^2 \big/ (n-1)$ does not estimate the variance because of the autocorrelations. In fact,

$$\sum_{i=1}^{a} (y_i - \bar{y})^2 \Bigg/ \left(a - \frac{1}{a} \sum_{j=1}^{a} \sum_{i=1}^{a} \rho^{|i-j|} \right)$$

is an unbiased estimator of the variance of a observations. Note that, if $\rho = 0$, the denominator reduces to $a - 1$ and the variance estimator educes to the usual S^2, so that the expression

$$a - \frac{1}{a} \sum_{j=1}^{a} \sum_{i=1}^{a} \rho^{|i-j|}$$

is analogous to the $(n - 1)$ in the usual variance estimator. It does not follow that these are the effective degrees of freedom.

A statistical concept that characterizes the amount of information in the sample is its *effective degrees of freedom*, which is closely related to the idea of effective sample size that was discussed in connection with two-stage cluster samples. The effective degrees of freedom h_a in a sample with first order autocorrelation ρ are given by

$$h_a = \frac{\left(a - \frac{1}{a} \sum_{j=1}^{a} \sum_{i=1}^{a} \rho^{|i-j|} \right)^2}{\sum_{i=1}^{a} \sum_{j=1}^{a} \rho^{2|i-j|} - \frac{2}{a} \sum_{l=1}^{a} \sum_{m=1}^{a} \sum_{i=1}^{a} \rho^{|i-l|} \rho^{|i-m|} + \frac{1}{a^2} \left(\sum_{j=1}^{a} \sum_{i=1}^{a} \rho^{|i-j|} \right)^2}.$$

Note that when $\rho = 0$, $h_a = a - 1$ as expected and when ρ tends to 1, h_a tends to zero (which makes sense since all the observations tend to the same value and the sum of squares about the mean tends to zero).

These formulas are not very enlightening, but Table 1 shows the effective degrees of freedom in a sample of n observations as a function of the autocorrelation. The table shows that as the autocorrelation increases, the effective degrees of freedom in the sample decrease. For example, a sample of $n = 10$ observations has 9 degrees of freedom when $\rho = 0$, 6.23 degrees of freedom when $\rho = 0.5$, but only 3.00 degrees of freedom when $\rho = 0.9$. Thus, as is the case with clustered samples, the amount of information in an autocorrelated sample can be much less than the sample size would suggest.

Insert Table 1 About Here

Model and Notation

Let us assume that there are a total of $a + b$ observations (a before interruption, b after). Suppose that the data vector is $\mathbf{y} = (y_1, y_2, \ldots, y_a, y_{a+1}, \ldots, y_{a+b})'$. Let us assume that it is a first order autocorrelation and a total of $a + b$ observations (a before interruption, b after). The simplest model is one where the observations are stationary or constant before and after the interruption. Thus, the model for observation Y_i at time i will be

$$Y_i = \mu + \beta Treatment_i + \varepsilon_i,\ i = 1, \ldots, a + b. \quad \varepsilon_i \sim \mathrm{NAR}(0, \sigma, \rho),$$

where $Treatment_i$ is equal to zero for all time points before the treatment and equal to one for all timepoints after the treatment, where NAR(0, σ, ρ) was defined above.

Statistical Analysis of the Interrupted Time Series Design

There are many different statistical analyses that could be performed, but a simple one is to use ordinary least squares. Matrix algebra is helpful to describe the statistical analysis of even this simple model. Define the $(a + b) \times 2$ design matrix $\mathbf{X}$ via

$$\mathbf{X} = \begin{pmatrix} \mathbf{1}_a & \mathbf{0}_a \\ \mathbf{1}_b & \mathbf{1}_b \end{pmatrix},$$

where $\mathbf{1}_a$, $\mathbf{1}_b$, and $\mathbf{0}_a$ are column vectors of a ones, b ones, and a zeroes, respectively. Let $\boldsymbol{\beta} = (\beta_1, \beta_2)'$, be the 2×1 vector of model coefficients so that $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, is the model for the observations and β_2 is the effect of the interruption.

To test the null hypothesis

$$H_0: \beta_2 = 0$$

of no interruption, use the statistic

$$F = \frac{\left(\bar{y}_b - \bar{y}_a\right)^2}{\left(\sum_{i=1}^{a}\left(y_i - \bar{y}\right)^2 + \sum_{i=a+1}^{a+b}\left(y_i - \bar{y}\right)^2\right)} \left(\frac{tr\left(\mathbf{I} - \mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X})\boldsymbol{\Sigma}\right)}{\mathbf{1}_b{}'\boldsymbol{\Sigma}_{bb}\mathbf{1}_b / b^2 + \mathbf{1}_a{}'\boldsymbol{\Sigma}_{aa}\mathbf{1}_a / a^2 - 2\mathbf{1}_a{}'\boldsymbol{\Sigma}_{ab}\mathbf{1}_b / ab} \right)$$

has the F-distribution with 1 degree of freedom in the numerator and h degrees of freedom in the denominator, where h is given by

$$h = \frac{\left[\left(tr\left(\mathbf{I} - \mathbf{I} - \mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\right)\boldsymbol{\Sigma}\right)\right]^2}{tr\left(\boldsymbol{\Sigma}^2\right) - 2tr\left(\mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\,\boldsymbol{\Sigma}^2\right) + tr\left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\,\boldsymbol{\Sigma}\,\mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\,\boldsymbol{\Sigma}\,\mathbf{X}'\right)},$$

and $\boldsymbol{\Sigma}_{aa}^{-1}$ is $a \times a$ and $\boldsymbol{\Sigma}_{bb}^{-1}$ is $b \times b$ are the submatrices defined by partitioning $\boldsymbol{\Sigma}$ into four submatrices

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}.$$

The analysis is performed with software that will do all the necessary calculations. When the null hypothesis is not true, then the test statistic has the noncentral F-distribution with noncentrality parameter

$$\lambda = \left(\frac{\beta_2^2}{\mathbf{1}_b{}'\boldsymbol{\Sigma}_{bb}\mathbf{1}_b / b^2 + \mathbf{1}_a{}'\boldsymbol{\Sigma}_{aa}\mathbf{1}_a / a^2 - 2\mathbf{1}_a{}'\boldsymbol{\Sigma}_{ab}\mathbf{1}_b / ab} \right)$$

with 1 degree of freedom in the numerator and h degrees of freedom in the denominator,

Effect Size

The data in interrupted time series models is almost always aggregate data, for example the average achievement of students in a lass or school at a particular timepoint. The treatment effect β2 can be interpreted as a mean difference: the difference between the mean of post interruption observations minus the man of the pre-interruption observations. The effect size is the ration of this mean difference to the variance of the observation that are used to compute that mean, that is

$$\frac{\beta_2}{\sigma_M / \sqrt{1-\rho^2}} = \frac{\beta_2\sqrt{1-\rho^2}}{\sigma_M},$$

where the symbol σ_M is used here to emphasize that the standard deviation is the standard deviation of he means of observations that are the observations. Effect sizes are conventionally expressed using the standard deviation of the observations of individuals in the denominator. If each observation is the mean of a sample of n observations, then

$$\sigma_M = \sigma / \sqrt{n}$$

where σ is the standard deviation of the individual observations. To express the effect size in interrupted times series designs in a way that is comparable to the effect size used in experiments it is helpful to define the effect size in interrupted time series designs as

$$\delta = \frac{\beta_2}{\sigma} = \frac{\beta_2\sqrt{n\left(1-\rho^2\right)}}{\sigma}.$$

Planning Interrupted Time Series Designs

The object planning a research design is to ensure that if the data is collected according to the design, the analysis of that data will yield unambiguous results. To ensure results will be unambiguous, the design has to be sufficiently sensitive. There are three different, but complementary, ways to describe design sensitivity, all of which can be useful in planning designs.

We discussed previously three ways to describe design sensitivity: Statistical power, minimum detectable effect size and precision of he estimated treatment effect. In the next section we describe how to use these characterizations to plan interrupted time series designs so that they will yield unambiguous conclusions about a treatment effect.

General Principles for Planning Interrupted Time Series Designs

Planning an interrupted time series design to evaluate an educational intervention primarily involves choosing a design that will have adequate sensitivity (high statistical power and precision, or small minimal detectable effect size). In principle, design sensitivity depends on the significance level chosen, the effect size expected, the autocorrelation, and the number of observations before (a) and after (b) the interruption. Because eh effect size implicitly includes the number of subjects (n) whose average scores comprise each intervention, design sensitivity also depends on n. Although significance level is in principle an arbitrary choice of the investigator, the 0.05 or 5% significance level has been so reified by convention that it is, for all practical purposes, not a choice but the de facto standard. The autocorrelation ρ is not under the control of the investigator. The effect size expected is also not completely a choice but is dictated by the prior experience of the investigator about what size effect is large enough to be important or is the effect size to be expected from the intervention being evaluated. Thus, planning the design amounts to deciding on how timepoints before and after the interruption (a and b) and many subjects (usually students) will contribute to the mean at each timepoint (n). That is, planning the design involves choosing n to achieve the desired design sensitivity.

Statistical tables that are not widely available for computing statistical power in interrupted time series designs. Specialized statistical software that can compute design sensitivity for this design and it is probably easiest to use software that is designed to do the computations directly (e.g., **Website reference**). While exact power computations using software are recommended, we offer a few general principles that are helpful in the sections that follow and illustrate them with computations of design sensitivity.

The most sensitive design given a fixed total sample size is a balanced design in which the same number of individuals contribute to the means at each timepoint and there are equal numbers of timepoints before and after the interruption ($a = b$). Table 2 shows the statistical power, minimum detectable effect size, and standard error of the estimated treatment effect for total sample sizes from $n = 50$, autocorrelations ranging from $\rho = 0.0$ to $\rho = 0.9$, and effect size values of $\delta = 0.20$ and 0.50. The table shows that the minimum detectable effect size and standard error of the treatment effect estimate are both increasing functions of ρ, but do not depend on the effect size. However, the statistical power does depend on effect size. For example, and the statistical power of designs with $\delta = 0.5$ is greater than that for designs with $\delta = 0.2$ (and the same ρ value). Because effect size δ depends on sample size n, design sensitivity also depends on the sample size n. The table also shows that design sensitivity depends strongly on the number of timepoints (a and b), with power being lower and MDES and SE being higher when $a = b = 3$ than when $a = b = 6$. Note also that the MDES is rather large (larger than 1.0) when $a = b = 3$ and the autocorrelation $\rho \geq 0.4$. Even when $a = b = 6$, the MDES is larger than 0.8 when the autocorrelation $\rho \geq 0.4$.

Insert Table 2 About Here

Unequal Numbers of Timepoints Before and After Interruption Can Increase Sensitivity

As in the case of experimental designs, imbalance can increase sensitivity of interrupted time series designs it permits the number of timepoints to increase. That is, making a and b unequal can increase sensitivity if it makes possible a larger value of $a + b$ in the design. This might happen, for example if the number of post-interruption timepoints was fixed (because additional timepoints would be in the future), but it was possible to obtain data at additional timepoints in the pre-interruption period (e.g., by reconstructing administrative records), albeit with increasing effort for points increasingly further in the past. In such a situation, gains in sensitivity would need to be weighed against the cost of collecting data at more timepoints.

The minimum detectable effect size as a function of *a* and *b* (which here can be unequal), for $n = 50$ and three values of ρ is shown in Table 3. This table further illustrates he importance of the autocorrelation in determining design sensitivity. The table shows that design sensitivity can be increased by adding additional timepoints to the time series. For example, if $\rho = 0.5$, the MDES when $a = b = 3$ is 0.61, but the MDES can be reduced to 0.51 if $a = 3$ and $b = 6$, and reduced further to 0.47 if $a = 3$ and $b = 9$. Because the MDES values depend on n, and table 3 is specific to $n = 50$, additional tables corresponding to Table 3, but based on $n = 25$, $n = 75$, and $n = 100$ are provided in the appendix to this chapter.

Examples

Conclusions

References

Amemiya, T. (1985). *Advanced econometrics.* Cambridge, MA: Harvard University Press.

Chapter 12
Nonequivalent Control Group Designs